EXTREMAL TORUS FIXED POINTS IN TYPE D: A CLASSIFICATION OF THE SIMPLE-SPECTRUM LOCUS

JAIDEEP SAI PADHI

ABSTRACT. Let $\Gamma = D_n$, let $\lambda = \omega_k$ be a fundamental weight with $2 \le k \le n-2$, let w be an element of the Weyl group and let V be the unique T-fixed representation on the Nakajima quiver variety $\mathfrak{M}(\lambda - w\lambda, \lambda)$. Writing $\mu = w\lambda = \sum_{i\in S} \eta_i e_i$ in Bourbaki coordinates, put $m = \#\{i \le n-2 : i \in S,\ \eta_i = -1\}$ and $r = |S \cap \{n-1, n\}|$. We prove, over an arbitrary ground field, that T acts on V with simple spectrum if and only if $r + 2m \le 2$. For the two spin weights every w is λ-minuscule and the spectrum is always simple. The proof identifies V with an iterated socle of the injective Π-module I_κ, decomposes I_κ into two multiplicity-free "strands" under the leaf-exchanging diagram automorphism, and reduces the statement to an explicit inequality between two integer sequences at $9(k-1)$ rank-uniform weights. The strand decomposition requires 2 to be invertible; it is used only to construct one witness submodule in the generic fibre, and a specialisation argument over $\mathbb{Z}_{(p)}$ carries the conclusion to every characteristic, $p = 2$ included. We also record an element of D_4 that is fully commutative but does not have simple spectrum. This answers, in type D for non-spin fundamental λ, the open problem of [5, §5.4.2]; §1.3 records what of that problem remains.

1. INTRODUCTION

1.1. The problem. Fix a simply-laced Dynkin diagram Γ with vertex set I, Cartan matrix $C = (a_{ij})$, simple roots α_i, fundamental weights ω_i and Weyl group W. Let $\mathfrak{M}(v, w)$ be the Nakajima quiver variety [8] with dimension vector v and framing w, and write $\lambda = \sum_i w_i\omega_i$ and $\mu = \lambda - \sum_i v_i\alpha_i$. When $\mu = w\lambda$ is extremal the variety is a single point, and that point is a representation $V = \bigoplus_{i\in I} V_i$ of the doubled quiver, acted on by a torus T. Dinkins, Karpov and Krylov [5, §5.4] attach to such a w a coloured graded poset, and their formulas for the vertex function require the T-action on V to have *simple spectrum*, i.e. pairwise distinct weights on each V_i. They ask [5, §5.4.2] for a combinatorial characterisation of the w for which this holds.

The present paper answers that question in type D_n for every non-spin fundamental λ, in the sharpest possible form: the answer depends only on the *signs* of the Bourbaki coordinates of $\mu = w\lambda$, and not on w, on any reduced word, or on the rank n.

Theorem 1.1 (type D)**.** *Let $\Gamma = D_n$ with the Bourbaki labelling $\alpha_i = e_i - e_{i+1}$ for $i \le n-1$ and $\alpha_n = e_{n-1} + e_n$, let $t = n-2$ be the trivalent node, and let $\lambda = \omega_k$ with $2 \le k \le n-2$. Let the ground field $\mathbb{K}$ be arbitrary. Write*

$$\mu = w\lambda = \sum_{i\in S} \eta_i e_i, \qquad |S| = k, \quad \eta_i = \pm 1,$$

and put

$$m = \#\{\, i \le n-2 :\ i \in S,\ \eta_i = -1 \,\}, \qquad r = |S \cap \{n-1, n\}|.$$

Then T acts on V with simple spectrum if and only if

$$\boxed{\ v_t \ \le\ 2 \qquad \Longleftrightarrow \qquad r + 2m \ \le\ 2\ }$$

— equivalently, iff $m = 0$, or $m = 1$ and $r = 0$.

Here v_t is the coordinate at the trivalent node of the dimension vector $v = \lambda - \mu$, so the left-hand form is a condition on the quiver data (v, w) that one starts from: it names no basis, no reduced word and no rank. The two forms agree because $v_t = r + 2m$ (Lemma 4.4), and it is the left-hand one that the proof establishes — the colour-t profile of I_κ is $(1, 2^{k-1}, 1)$ and its greedy top truncation to total v_t is multiplicity-free exactly when $v_t \le 2$. The right-hand form is the translation into the signs of μ, and is what makes the criterion checkable by inspection.

Theorem 1.2 (the spin weights)**.** *Let $\Gamma = D_n$ with $n \ge 4$, let $\lambda = \omega_\kappa$ with $\kappa \in \{1, n-1, n\}$, and let the ground field $\mathbb{K}$ be arbitrary. Then every w of minimal length in wW_λ is λ-minuscule, and the module $V = \mathrm{soc}_\sigma(I_\kappa)$ of Theorem 3.6 satisfies*

$$\dim_{\mathbb{K}} V_i[d] \le 1 \qquad \text{for every vertex } i \text{ and every degree } d,$$

i.e. V has simple spectrum in the sense fixed after Proposition 2.16. Over $\mathbb{C}$, where V is the T-fixed representation on $\mathfrak{M}(\lambda - w\lambda, \lambda)$ and the path-length grading is the T-weight grading, this says that T acts with simple spectrum, for every w. No hypothesis on the characteristic is needed, and the characteristic-free half of the statement uses neither Proposition 2.9 nor Proposition 2.16.

The excluded case $k = 1$ of Theorem 1.1 is contained in Theorem 1.2; the criterion $r + 2m \leq 2$ is in any case automatic there, since $|S| = 1$.

1.2. What is proved, and over which fields. Theorem 1.1 holds over every field. It is worth being precise about where char $\neq 2$ does and does not appear, because the role of that hypothesis is easy to misdescribe: it is *not* a standing hypothesis on the ambient field of the whole argument. The proof of Theorem 1.1 funnels through a single existential statement, (B) of Definition 7.9: at each of the $9(k-1)$ representatives μ of Proposition 7.7, *some* submodule of I_κ has the greedy top truncation as its graded profile. Everything else — the identification $V \cong \mathrm{soc}_\sigma(I_\kappa)$, the reduction to the trivalent node, the profile of I_κ, the reduction of $(*^\circ)$ to the representatives, the rigidity that turns "some submodule" into "the submodule", and the induction along σ — is proved directly over an arbitrary field. So char $\neq 2$ is a hypothesis on the field over which one *constructs the witness submodule* for (B), not on the field over which the theorem is true. The construction is carried out once and for all in the generic fibre, and it spreads by specialisation: Π is presented over $\mathbb{Z}$, the witness is saturated over $\mathbb{Z}_{(p)}$, and its reduction mod p is again a submodule with the same graded dimensions. That is Theorem 11.1 of §11, and it is what removes the hypothesis.

Explicitly:

- *Proved directly in every characteristic*: §3 entire, including Lemma 3.1 and Theorem 3.6; Corollary 2.13; Proposition 2.1, whose part (a) — the Ext^1 formula, which carries everything else — is *proved* in §2.2 rather than quoted; Propositions 2.6, 8.10; Lemmas 4.5, 4.2, 2.8, 4.4, 4.9; Theorems 4.6, 1.2; the whole of §7, together with §7.2; and the induction of §10, given (B). Lemma 4.9 is what the induction reads at the node, and is deliberately independent of the char $\neq 2$ Lemma 8.9. Since §7 lies outside the standing hypothesis char $\neq 2$ of §§5–6 and §§8–9, it is quoted at $p = 2$.
- *Proved only for* char $\neq 2$*, and then transported*: the strand decomposition and everything built on it — Proposition 5.1 and (W1)–(W3), Propositions 5.5, 5.6, 5.7, Corollary 5.8, Proposition 6.1 and (T0)–(T2), the canonical truncation $T(v)$ at a partially taken multiplicity-2 slot, and Theorem 9.4, i.e. (B) itself. At $p = 2$ these are used only through (B), which §11 transports. None is claimed at $p = 2$; Proposition 5.1 is in fact *false* there, in every formulation we know (Remark 11.8).
- *Off the logical path*: §12 and §13, neither of which is used by anything else, and both of which hold over an arbitrary field. Theorem 1.1 does not care either way.

Section 14 separates what is proved from what is merely checked. Its first part, §14.1, records item by item everything in this paper that is *asserted* on the strength of a computation rather than a proof. That list is *empty*: every assertion used in the proof of Theorem 1.1 is proved here or cited; the sharpness of the side condition in $(*^\circ)$, the last statement to have been offered on computational evidence, is Proposition 7.5. The computations formerly offered as evidence are retained in §14.2, which reports what they found: counts of weights, of obligations and of slots, over the ranges stated there. Those counts are assertions of fact resting on computation, and we do not claim otherwise; but each is a *check* on a statement proved elsewhere in this paper, not a step in any proof, and no proof here cites §14.

1.3. What remains of the problem of [5, §5.4.2]. The question of [5, §5.4.2] restricts neither the Dynkin type nor the weight: λ in [5, Def. 5.1] is arbitrary. So it is worth saying which part of it is settled here. Type A_n is vacuous at every fundamental $\lambda = \omega_k$, all of which are minuscule, so every w minimal in wW_λ is λ-minuscule [10, §2] and the spectrum is always simple; the same disposes of the minuscule weights ω_1, ω_6 of E_6 and ω_7 of E_7. Type D_n is settled at every fundamental λ, by Theorem 1.1 for the non-spin weights and Theorem 1.2 for $\omega_1, \omega_{n-1}, \omega_n$.

Two regions are untouched. The first is type E at a non-minuscule fundamental weight: $\omega_2, \omega_3, \omega_4, \omega_5$ in E_6, $\omega_1, \ldots, \omega_6$ in E_7, and every fundamental weight of E_8, which has no minuscule weight at all. We know of no criterion there and offer none. What transfers is the machinery and not the answer: Theorem 3.6 is proved for an arbitrary simply-laced Γ and is characteristic-free, so it applies verbatim; what is missing is the passage from it to a closed condition on μ, the analogue of $v_t \leq 2$, whose proof here uses the two-leaf symmetry of D_n throughout §§5–9. The second is λ not fundamental, in any type: every statement here assumes $\lambda = \omega_\kappa$, Theorem 3.6 included. Part of that case, however, is not open at all. When the framing is supported at

minuscule vertices — the setting of [5, Thm. B] — the fixed points of the framing torus F_w are the direct sums $\bigoplus_{i,s} V^{(i,s)}$ of point representatives indexed by the decompositions $v = \sum_i \sum_{s \le w_i} v_i^{(s)}$ [5, Ex. 4.29]. Each summand is the extremal module for a minuscule fundamental weight, so it has simple spectrum by Theorem 1.2; and distinct summands carry distinct characters of F_w, one per framing direction, so no weight of V_i is repeated across summands. Hence for $T \supseteq F_w \times \mathbb{C}_\hbar^\times$ the spectrum is simple for every such λ and every extremal μ, with nothing to characterise.

What remains open in the non-fundamental case is therefore narrower than it first appears, and is of two kinds: a λ that is not a sum of minuscule weights, such as $2\omega_2$ or $\omega_2 + \omega_3$ in D_n, where the fixed point is not a direct sum of point modules and the question is genuinely untouched; and a flavour torus too small to separate the summands above — for $T = \mathbb{C}_\hbar^\times$ alone the argument of the previous paragraph gives nothing, and [5, §5.4.2] allows any F. We address neither.

1.4. The criterion against the examples of [5]. The first thing to check is that the criterion reproduces the three worked examples of [5, §5.4]. It does. In each row below $\mu = w\lambda$ is computed in Bourbaki coordinates, and the last column is the list of colours occurring in each graded piece of V, read from the top degree down, as in the Hasse diagrams of [5, pp. 82–83].

[5]	Γ	λ	w	μ	(m, r)	$r + 2m$	rows of V
Ex. 5.10	D_4	ω_2	$s_1s_2s_3s_4s_2$	$(-1, 1, 0, 0)$	$(1, 0)$	2	$[2], [1, 3, 4], [2], [1]$
Ex. 5.11	D_5	ω_2	$s_2s_1s_3s_4s_5s_3s_2$	$(0, -1, 1, 0, 0)$	$(1, 0)$	2	$[2], [1, 3], [2, 4, 5], [3], [2]$
Ex. 5.12	D_5	ω_2	$s_2s_1s_2s_3s_4s_5s_3s_2$	$(-1, 0, 1, 0, 0)$	$(1, 0)$	2	$[2], [1, 3], [2, 4, 5], [3], [2], [1]$

All three have $r + 2m = 2 \le 2$, so Theorem 1.1 predicts simple spectrum; and indeed no colour is repeated inside any row, which by Proposition 2.16 is exactly simplicity of the spectrum. The dimension vectors agree as well: for Ex. 5.10 one gets $v = (2, 2, 1, 1)$, and $v_t = v_2 = 2 = r + 2m$ in accordance with Lemma 4.4. Note that Ex. 5.12 is *not* fully commutative, so full commutativity is not necessary for simple spectrum; the next example shows it is not sufficient either.

Example 1.3 (full commutativity is neither necessary nor sufficient)**.** In D_4 take $w = s_2s_4s_3s_1s_2$, so that $\mu = (0, -1, 1, 0)$ and $v = (1, 3, 1, 1)$. Here $m = 1$ and $r = 1$, so $r + 2m = 3$ and Theorem 1.1 predicts failure. Indeed the graded dimensions of $V \subseteq I_2$ are $\{2 : 1\}$ in degree 4, $\{1 : 1, 3 : 1, 4 : 1\}$ in degree 3 and $\{2 : 2\}$ in degree 2: the rows are $[2]$, $[1, 3, 4]$, $[2, 2]$, and the colour 2 is repeated. This w *is* fully commutative. So full commutativity does not imply simple spectrum; and [5, Ex. 5.12], whose $w = s_2s_1s_2s_3s_4s_5s_3s_2$ in D_5 is recorded there as *not* fully commutative yet does have simple spectrum, shows the converse fails too. Hence the condition fails in both directions.

1.5. The shape of the proof. The argument has five movements.

(1) From the quiver variety to a module. Let Π be the preprojective algebra of Γ and I_κ the injective envelope of the simple S_κ. Reading a reduced word for w from the right and recording at each step $c = \langle \alpha_i^\vee, \mu_{\mathrm{cur}} \rangle$ produces a sequence σ of colours with multiplicities, and $V \cong \mathrm{soc}_\sigma(I_\kappa)$, the iterated socle of Geiss–Leclerc–Schröer taken along σ (Theorem 3.6). Simple spectrum becomes $\dim V_i[d] \le 1$ for all i and d, where $[d]$ is the path-length grading.

(2) One vertex suffices. Inward injectivity of the arrows of I_κ (Lemma 4.5) transports a repeat at any vertex to a repeat at the trivalent node $t = n - 2$. So only V_t needs to be inspected (Theorem 4.6).

(3) The profile at t, and the top truncation. The colour-t multiplicity sequence of I_κ is $(1, 2, \dots, 2, 1)$ with $k - 1$ twos (Lemma 4.2), and $v_t = r + 2m$ (Lemma 4.4). The content of the theorem is that the colour-t multiplicities of V are the greedy *top* truncation of that sequence to total v_t (Proposition 4.11); then every multiplicity is ≤ 1 precisely when $v_t \le 2$.

(4) Proposition 4.11 by induction along σ. The hypothesis "X is a top truncation at t and an up-set at each leaf" is carried along the socle construction. Leaf steps are elementary (§10.2); colour-t steps need Claim (G) of §10.3, which in turn needs the statement

$$(*^\circ) \qquad \text{every extremal } M \subseteq I_\kappa \text{ with } \langle \alpha_t^\vee, \lambda - \underline{\dim}\, M \rangle \ge 1 \text{ is an up-set at } t - 1.$$

(5) $(*^\circ)$. This mentions no word and no order of construction, and it is invariant under the parabolic $W_{J'}$, $J' = \{1, \dots, t - 2\}$. The $W_{J'}$-minimal weights satisfying its side condition are $9(k - 1)$ explicit weights, *independently of* n (Proposition 7.7). At each of them one exhibits a submodule of I_κ of the right dimension

vector; by rigidity (Corollary 2.13) that submodule *is* V, and it is visibly an up-set at every vertex. The submodule is built from the two "strands" into which the leaf-exchanging automorphism τ splits I_κ (§5), and its closure reduces to two increment inequalities between integer sequences (§9).

Acknowledgements. I thank Vasily Krylov for the question that prompted this work and for correspondence about [5].

2. Setup

2.1. The preprojective algebra. Let Γ be a simply-laced Dynkin diagram with vertex set I, and let $\overline{Q}$ be the doubled quiver of an orientation Q of Γ: for each edge $\{i,j\}$ there are arrows $a_{i\to j}$ and $a_{j\to i}$. The preprojective algebra is

$$\Pi = \Pi_\Gamma = \mathbb{K}\overline{Q}\Big/\Big(\sum_{a\in Q}(aa^* - a^*a)\Big),$$

over a ground field $\mathbb{K}$. We fix once and for all the sign convention $\varsigma(i,j) = \operatorname{sign}(j-i)$ on arrows, so that the relation at a vertex j reads $\sum_{i\sim j}\varsigma(j,i)\,a_{i\to j}a_{j\to i} = 0$.

The orientation is fixed accordingly: we take for Q the orientation in which every arrow points to the *smaller* label, i.e. $a_{i\to j}\in Q$ exactly when $j<i$. With that choice $\varsigma(a) := \varsigma(t(a),s(a)) = +1$ precisely for the arrows a of Q and -1 for their reverses, which is what makes the sign convention agree with the presentation $\sum_{a\in Q}(aa^*-a^*a)$: cutting that element with e_j on both sides gives $+a\,a^*$ for each arrow $a\in Q$ with head j and $-b\,b^*$ for each reverse $b=a^*$ with head j ($a\in Q$ with tail j), i.e. the displayed relation, and (2) below. *For any other orientation the two would differ by signs*, so the reader checking (2) should use this one. (Nothing depends on the choice: different orientations give isomorphic Π, by rescaling the arrows.)

All modules in this paper are left Π*-modules*, and all submodules are $\mathbb{K}$-subspaces closed under every arrow. (The involution of Π fixing the vertices and sending $a\mapsto\varsigma\,a^*$ identifies Π with Π^{op}, so nothing is lost by this choice.) We use the following two standard facts about Π in Dynkin type, and we state both in full because they are used constantly; their provenance is discussed in Remark 2.2 immediately below.

Proposition 2.1 (standard)**.** *Let Γ be a simply-laced Dynkin diagram and $\Pi=\Pi_\Gamma$, over an arbitrary ground field $\mathbb{K}$.*

(a) (the Ext^1 formula; Crawley-Boevey [4, Lem. 1]) *For all finite-dimensional Π-modules M,N,*

$$\dim\operatorname{Ext}^1_\Pi(M,N) = \dim\operatorname{Hom}(M,N) + \dim\operatorname{Hom}(N,M) - (\underline{\dim}\,M, \underline{\dim}\,N),$$

where $(x,y) = x^{\mathsf{T}}Cy$. This part is proved below (Lemma 2.3 and Proposition 2.4) over an arbitrary field, and [4] is cited for the attribution only.

(b) *Π is finite dimensional and self-injective. Each indecomposable projective Πe_j is therefore also injective, and it has simple socle $S_{\nu(j)}$, where ν is the Nakayama permutation of Π, induced by the diagram automorphism $-w_0$. Consequently the injective envelope of S_κ is*

$$I_\kappa \;\cong\; \Pi e_{\nu^{-1}(\kappa)}. \tag{1}$$

Remark 2.2 (provenance of Proposition 2.1)**.** We are precise about attributions here, and about exactly how much of each part the paper consumes.

- (a) is Crawley-Boevey's Ext^1 formula for modules over a deformed preprojective algebra [4, Lem. 1], which reads: *"If M and N are finite dimensional $\Pi(Q)$-modules, then* $\dim\operatorname{Ext}^1(M,N) = \dim\operatorname{Hom}(M,N) + \dim\operatorname{Hom}(N,M) - (\dim M, \dim N)$*."* We do not, however, rely on it as an authority. The reason is a mismatch of hypotheses that matters for this paper: [4] is a paper about Kleinian singularities and works throughout over an algebraically closed field of characteristic 0, whereas part (a) is the one input that carries *everything* that is characteristic-free here — Theorem 3.6 through Lemma 3.1, and the inequality of Corollary 2.13 through which §11 runs. The formula is nevertheless true over any field, for any quiver, by the standard four-term sequence; we write that proof out in Lemma 2.3 and Proposition 2.4 below, so that (a) is a proved statement of this paper and [4] an attribution.
- (b) is sometimes attributed to Crawley-Boevey; it is not his. Self-injectivity of Π_Γ in Dynkin type, and the identification of the Nakayama permutation with the diagram automorphism induced by $-w_0$, are due to Ringel and Schofield (unpublished manuscript *Wild algebras with periodic Auslander–Reiten translate*); the first published proof, with the Nakayama automorphism written out explicitly in each *ADE* case, is Brenner–Butler–King [1]. Neither [2, §2] nor [7] contains this statement.

Part (b) is the only statement quoted rather than proved in this paper, so we record precisely which of its assertions are consumed and which are not. Since I_κ is *defined* as the injective envelope of S_κ inside the finite-dimensional Π-modules, two of the assertions that (b) would supply cost nothing: I_κ is injective by definition, and $\operatorname{soc} I_\kappa = S_\kappa$ because an injective envelope is an essential extension, so every nonzero submodule meets S_κ. Neither needs (b), and below we appeal to the definition rather than to [1] for them. What (b) is genuinely used for is the following three items, and nothing else:

(b1) Π_Γ is finite dimensional, so that the injective envelope exists inside the finite-dimensional modules and is itself finite dimensional;

(b2) $I_\kappa \cong \Pi e_{\nu^{-1}(\kappa)}$ as in (1). This is what supplies the path-length grading used throughout, the fact that I_κ is generated in degree 0 (fact (G1) in the proof of Proposition 3.12), and the presentation $F/\mathcal{R}$ of I_κ from which Lemma 2.8 is proved;

(b3) ν fixes every chain vertex of D_n, i.e. $I_k \cong \Pi e_k$ for $1 \le k \le n-2$. This enters once, as the initial datum $\hat{a}_0 = \delta_{\cdot k}$ of Proposition 3.12(a) (together with fact (G3) there, which is the same statement read at the top degree).

What is *not* consumed is the identification of ν with the automorphism induced by $-w_0$ beyond (b3). In particular the explicit type-D form of ν on the two leaves — the identity for n even, the leaf exchange for n odd — is never used: at a leaf weight the paper carries $\nu^{-1}(\kappa)$ as an unspecified leaf throughout, and Proposition 3.12(b) is stated in the folded coordinates, in which the two leaves are indistinguishable by construction. So the residue of the citation is: Π_{D_n} *is finite dimensional and self-injective, and its Nakayama permutation fixes the chain vertices.*

Nor is any dimension count imported. The type-D profile and dimension of I_κ are computed outright: Proposition 3.12 determines the whole graded profile of I_κ from Lemma 2.8 (the recursion), Lemma 2.7 (the socle sits in the top degree), part (a), and integer combinatorics, with no hypothesis on the ground field; the resulting Corollary 3.13 is Proposition 2.6. The numbers, the simplicity of $\operatorname{soc}\Pi e_j$ and the vertex at which it sits are independently confirmed over $\mathbb{Q}$ and over $\mathbb{F}_p$ for $p \in \{2,3,5,7\}$, for D_4 through D_{10} and every vertex j, by the computations of §14.

2.2. Proof of the Ext^1 formula over an arbitrary field. We now prove Proposition 2.1(a). The argument is the standard four-term sequence for Π; it is elementary, it is uniform in the ground field, and — as will be clear — it uses neither the Dynkin hypothesis nor any hypothesis on $\mathbb{K}$. We record it because everything in this paper that is asserted over an arbitrary field passes through part (a).

Fix the notation, used in this subsection only. For an arrow $a = a_{i\to j}$ of $\overline{Q}$ put $s(a) = i$ (source), $t(a) = j$ (target) and $a^* = a_{j\to i}$, and write $M_a : M_i \to M_j$ for the action of a on a Π-module M. (The symbol t for the trivalent node of D_n is introduced only in Convention 4.1, well after this subsection ends, and $s(\cdot), t(\cdot)$ occur nowhere else.) Write also, extending the sign convention of §2 to a one-argument form,

$$\varsigma(a) := \varsigma\big(t(a), s(a)\big) = \operatorname{sign}\big(s(a) - t(a)\big), \qquad \text{so that} \qquad \varsigma(a^*) = -\varsigma(a),$$

and the preprojective relation at a vertex j reads

$$\sum_{a\in\overline{Q}:\ t(a)=j} \varsigma(a)\, M_a M_{a^*} = 0. \tag{2}$$

Lemma 2.3 (the four-term sequence)**.** *Let Γ be any graph without loops, $\overline{Q}$ the doubled quiver of an orientation, Π the preprojective algebra over an arbitrary field $\mathbb{K}$, and M, N finite-dimensional Π-modules. Put*

$$C^0 = \bigoplus_{i\in I} \operatorname{Hom}_{\mathbb{K}}(M_i, N_i), \quad C^1 = \bigoplus_{a\in\overline{Q}} \operatorname{Hom}_{\mathbb{K}}(M_{s(a)}, N_{t(a)}), \quad C^2 = \bigoplus_{i\in I} \operatorname{Hom}_{\mathbb{K}}(M_i, N_i),$$

and define $\alpha : C^0 \to C^1$ and $\beta : C^1 \to C^2$ by

$$\alpha(\varphi)_a = N_a\varphi_{s(a)} - \varphi_{t(a)}M_a, \qquad \beta(\psi)_j = \sum_{t(a)=j} \varsigma(a)\big(N_a\psi_{a^*} + \psi_a M_{a^*}\big).$$

Then $\beta\alpha = 0$, and

(i) $\ker\alpha = \operatorname{Hom}_\Pi(M,N)$;

(ii) $\ker\beta/\operatorname{im}\alpha \cong \operatorname{Ext}^1_\Pi(M,N)$;

(iii) $\operatorname{coker}\beta \cong \operatorname{Hom}_\Pi(N,M)^*$.

Proof. $\beta\alpha = 0$. Substituting $\alpha(\varphi)_{a^*} = N_{a^*}\varphi_j - \varphi_{s(a)}M_{a^*}$ and $\alpha(\varphi)_a = N_a\varphi_{s(a)} - \varphi_j M_a$ (for $t(a) = j$) into β, the two cross terms $\pm N_a\varphi_{s(a)}M_{a^*}$ cancel and what is left is

$$\beta\big(\alpha(\varphi)\big)_j = \Big(\sum_{t(a)=j} \varsigma(a) N_a N_{a^*}\Big)\varphi_j - \varphi_j\Big(\sum_{t(a)=j} \varsigma(a) M_a M_{a^*}\Big) = 0$$

by (2) for N and for M.

(i). $\alpha(\varphi) = 0$ says exactly $N_a\varphi_{s(a)} = \varphi_{t(a)}M_a$ for every arrow, i.e. that φ is a morphism of Π-modules.

(ii). Let $\psi \in C^1$ and define $E(\psi)$ by $E(\psi)_i = M_i \oplus N_i$ and $E(\psi)_a(m,n) = \big(M_a m,\ \psi_a m + N_a n\big)$. Then

$$\sum_{t(a)=j} \varsigma(a) E(\psi)_a E(\psi)_{a^*}(m,n) = \big(0,\ \beta(\psi)_j\, m\big),$$

again by (2) for M and N; so $E(\psi)$ is a Π-module if and only if $\beta(\psi) = 0$, and then $0 \to N \to E(\psi) \to M \to 0$ is an extension of Π-modules. Conversely every extension of M by N arises this way: choose a $\mathbb{K}$-linear splitting at each vertex, which identifies the underlying graded vector space with $M \oplus N$ and writes each arrow in the displayed lower-triangular form. Finally, an isomorphism $E(\psi) \to E(\psi')$ inducing the identity on N and on M is precisely a map f with $f_i(m,n) = (m, n + \varphi_i m)$, and imposing $f_{t(a)}E(\psi)_a = E(\psi')_a f_{s(a)}$ gives

$$\psi_a + \varphi_{t(a)}M_a = \psi'_a + N_a\varphi_{s(a)}, \qquad \text{i.e.} \qquad \psi'_a - \psi_a = \varphi_{t(a)}M_a - N_a\varphi_{s(a)} = -\alpha(\varphi)_a.$$

The fibres of $\psi \mapsto [E(\psi)]$ are therefore the cosets of $\operatorname{im}(-\alpha) = \operatorname{im}\alpha$ (the sign is immaterial, α being linear, but we record it correctly): that map is a surjection $\ker\beta \to \operatorname{Ext}^1_\Pi(M,N)$ with kernel $\operatorname{im}\alpha$, and it is additive because the Baer sum of $E(\psi)$ and $E(\psi')$ is $E(\psi+\psi')$.

(iii). Use the trace pairing to identify $(C^2)^*$ with $\bigoplus_i \operatorname{Hom}(N_i, M_i)$ and $(C^1)^*$ with $\bigoplus_a \operatorname{Hom}(N_{t(a)}, M_{s(a)})$. For $\theta \in \bigoplus_i \operatorname{Hom}(N_i, M_i)$,

$$\langle \beta(\psi), \theta\rangle = \sum_{a\in\overline{Q}} \varsigma(a)\Big[\operatorname{tr}\big(\psi_{a^*}\theta_{t(a)}N_a\big) + \operatorname{tr}\big(\psi_a M_{a^*}\theta_{t(a)}\big)\Big].$$

Re-indexing the first sum by $a \mapsto a^*$ and using $\varsigma(a^*) = -\varsigma(a)$ and $t(a^*) = s(a)$, it becomes

$$-\sum_{a\in\overline{Q}} \varsigma(a)\ \operatorname{tr}\big(\psi_a\theta_{s(a)}N_{a^*}\big),$$

whence

$$\langle \beta(\psi), \theta\rangle = \sum_{a\in\overline{Q}} \varsigma(a)\ \operatorname{tr}\Big(\psi_a\big(M_{a^*}\theta_{t(a)} - \theta_{s(a)}N_{a^*}\big)\Big).$$

This vanishes for every ψ if and only if $M_{a^*}\theta_{t(a)} = \theta_{s(a)}N_{a^*}$ for every a, i.e. (replacing a by a^*) $M_b\theta_{s(b)} = \theta_{t(b)}N_b$ for every arrow b, i.e. $\theta \in \operatorname{Hom}_\Pi(N,M)$. So $(\operatorname{coker}\beta)^* = \ker\beta^* = \operatorname{Hom}_\Pi(N,M)$. □

Proposition 2.4. *Proposition 2.1*(a) *holds over an arbitrary field* $\mathbb{K}$*: for all finite-dimensional* Π*-modules* M, N*, with* $m = \underline{\dim}\, M$ *and* $n = \underline{\dim}\, N$*,*

$$\dim\operatorname{Ext}^1_\Pi(M,N) = \dim\operatorname{Hom}(M,N) + \dim\operatorname{Hom}(N,M) - (m,n), \qquad (m,n) = m^{\mathsf{T}}Cn.$$

Proof. Count dimensions in Lemma 2.3. Writing $h = \dim\operatorname{Hom}_\Pi(M,N)$, $h' = \dim\operatorname{Hom}_\Pi(N,M)$ and $e = \dim\operatorname{Ext}^1_\Pi(M,N)$, parts (i)–(iii) give $\dim C^0 = h + \dim\operatorname{im}\alpha$, $\dim\ker\beta = \dim\operatorname{im}\alpha + e$, $\dim C^1 = \dim\ker\beta + \dim\operatorname{im}\beta$ and $\dim C^2 = \dim\operatorname{im}\beta + h'$, whence

$$\dim C^0 - \dim C^1 + \dim C^2 = h + h' - e.$$

On the other hand $\dim C^0 = \dim C^2 = \sum_i m_i n_i$ and, since the arrows of $\overline{Q}$ are the ordered pairs (i,j) with $i \sim j$, $\dim C^1 = \sum_{\{i,j\}\in\Gamma}(m_i n_j + m_j n_i)$. As $C = 2\,\mathrm{Id} - A$ with A the adjacency matrix of Γ,

$$\dim C^0 - \dim C^1 + \dim C^2 = 2\sum_i m_i n_i - \sum_{i\sim j} m_i n_j = m^{\mathsf{T}}Cn = (m,n).$$

Comparing the two displays gives $e = h + h' - (m,n)$. □

Remark 2.5. Neither Lemma 2.3 nor Proposition 2.4 uses that Γ is Dynkin, or that $\mathbb{K}$ is algebraically closed, or anything about $\operatorname{char}\mathbb{K}$. Part (b) of Proposition 2.1 does use the Dynkin hypothesis and remains an input, cited to [1]; it is used only to identify I_κ with a graded Πe_j, never in a characteristic-sensitive way, and only to the extent itemised as (b1)–(b3) in Remark 2.2.

We label (1) as Proposition 2.6 for reference:

Proposition 2.6. *$\underline{\dim} I_\kappa = \omega_\kappa - w_0\omega_\kappa$ in root coordinates. In type D_n this gives*

$$\dim I_\kappa = k(2n-1-k) \quad \text{for } \kappa = k \le n-2, \qquad \dim I_\kappa = \tfrac{n(n-1)}{2} \quad \text{for } \kappa \in \{n-1, n\}.$$

Proof. *This proposition is proved outright, over an arbitrary field, in Corollary 3.13 below*, as a consequence of Proposition 3.12; nothing between here and there uses it in a proof — the sole reference, a parenthetical in the proof of Theorem 3.6, is flagged there as not used below — so there is no circularity. What follows is only the root-coordinate bookkeeping that identifies the two displayed integers with the coordinate sum of $\omega_\kappa - w_0\omega_\kappa$, i.e. that shows the two formulations of the proposition to be the same statement. Now $-w_0$ is the identity on chain vertices of D_n and either the identity or the leaf exchange on $\{n-1, n\}$; in both cases $\omega_\kappa - w_0\omega_\kappa = \omega_\kappa + \omega_{\kappa^*}$ has the same coordinate sum as $2\omega_\kappa$. Writing $\omega_k = e_1 + \cdots + e_k = \sum_j c_j\alpha_j$ and comparing coefficients of e_p gives $c_j = \min(j,k)$ for $j \le n-2$ and $c_{n-1} = c_n = k/2$, whence

$$\langle \omega_k - w_0\omega_k \rangle := 2\Big(\sum_{j \le n-2} \min(j,k) + k \Big) = k(k+1) + 2k(n-1-k) = k(2n-1-k),$$

where $\langle\,\cdot\,\rangle$ denotes the coordinate sum. For $\kappa = n$ one has $\omega_n = \frac{1}{2}(e_1 + \cdots + e_n)$ and the same comparison gives $c_j = j/2$ for $j \le n-2$, $c_{n-1} = (n-2)/4$, $c_n = n/4$; the coordinate sum is $\frac{(n-2)(n-1)}{4} + \frac{n-2}{4} + \frac{n}{4} = \frac{n(n-1)}{4}$, and doubling gives $n(n-1)/2$. The case $\kappa = n-1$ is the same by τ. □

We emphasise that we work exclusively with *submodules of the injective I_κ* — that is what Proposition 2.9 below describes — and never with quotients of the projective Πe_k; the identification (1) is used only to know that I_κ has the path-length grading of $\Pi e_{\nu^{-1}(\kappa)}$. (That $\operatorname{soc} I_\kappa = S_\kappa$ is simple needs no identification at all: it is the definition of an injective envelope. See Remark 2.2.)

For $\kappa = k \le n-2$, a chain vertex, ν fixes κ and $I_\kappa \cong \Pi e_k$; that is the case of Theorem 1.1 and of §§5–9. The leaf weights $\kappa \in \{n-1, n\}$ occur too, throughout §§2–3: Lemma 2.7, Lemma 2.8 and Lemma 3.10 are stated for an arbitrary vertex κ; Proposition 2.6 and Corollary 3.13 state the leaf case explicitly, and the latter is what Step 1 of Theorem 11.1 uses at $p = 2$; Lemma 3.11(b) and Proposition 3.12(b) exist only for the leaf case; and Corollary 3.14 and Theorem 1.2 are about it. There $\nu^{-1}(\kappa)$ may be the other leaf, so below we write $I_\kappa \cong \Pi e_{\nu^{-1}(\kappa)}$ and keep track of which vertex carries degree 0. We never need to decide which leaf it is — for the record, ν is the identity when n is even and the leaf exchange when n is odd, but that refinement is not used anywhere, and Proposition 3.12(b) is stated in folded coordinates precisely because it does not distinguish the two leaves.

Every Πe_j is graded by path length, *and this is the grading used throughout.* We write $M_i[d]$ for the degree-d part of a graded module M at vertex i.

Lemma 2.7 (the socle is the top degree)**.** *Let d^* be the largest degree with $(I_\kappa)[d^*] \neq 0$. Then $\operatorname{soc}(I_\kappa) = (I_\kappa)[d^*]$; it is one dimensional and sits at the vertex κ. In particular $\operatorname{soc}(I_\kappa) = S_\kappa$ is concentrated in the top degree, and the path-length grading is the socle-layer grading read backwards.*

Proof. Every arrow raises the path-length degree by 1, so every element of $(I_\kappa)[d^*]$ is killed by every arrow and hence lies in $\operatorname{soc}(I_\kappa)$; thus $0 \neq (I_\kappa)[d^*] \subseteq \operatorname{soc}(I_\kappa)$. The socle is the simple module S_κ, of dimension 1 and supported at the vertex κ: this is not an appeal to Proposition 2.1(b) but the definition of I_κ, an injective envelope being an essential extension of S_κ, so that every nonzero submodule of I_κ meets S_κ and hence $\operatorname{soc}(I_\kappa) = S_\kappa$. (Only the path-length grading, which is part (b2) of Remark 2.2, is quoted here.) A one-dimensional space containing a nonzero subspace equals it, so $\operatorname{soc}(I_\kappa) = (I_\kappa)[d^*] = \mathbb{K}$, at vertex κ. □

Lemma 2.8 (the recursion)**.** *Let Γ be simply-laced Dynkin, κ any vertex, and $a_d(i) = \dim(I_\kappa)_i[d]$. Then for every i and every d strictly below the top degree d^*,*

$$a_{d+1}(i) = \sum_{j \sim i} a_d(j) - a_{d-1}(i), \qquad a_0(i) = \delta_{i,\nu^{-1}(\kappa)}, \quad a_{-1} = 0.$$

More precisely, the map $Y : (I_\kappa)_i[d-1] \to \bigoplus_{j\sim i}(I_\kappa)_j[d]$, $u \mapsto (a_{i\to j}u)_j$, is injective for every i and every $d-1$ strictly below d^*.

Proof. Write $c = \nu^{-1}(\kappa)$, so that $I_\kappa = F/\mathcal{R}$ with $F = \mathbb{K}\overline{Q}e_c$ the free module and $\mathcal{R}$ the submodule spanned by the elements $p\, r_l\, q$, where l runs over the vertices, $r_l = \sum_{j\sim l} \varsigma(l,j)\, a_{j\to l}a_{l\to j}$ is the preprojective relation at l, and p, q are paths $l \to i$ and $c \to l$ respectively. All of F, $\mathcal{R}$ and I_κ are graded by path length.

Every path $c \to i$ of length $d+1$ factors uniquely as $a_{j\to i}q'$ with $j \sim i$ and q' a path $c \to j$ of length d, so

$$\Phi\colon \bigoplus_{j\sim i} F_j[d] \xrightarrow{\sim} F_i[d+1], \qquad (u_j)_{j\sim i} \longmapsto \sum_{j\sim i} a_{j\to i}u_j,$$

is an isomorphism. We compute $\Phi^{-1}\big(\mathcal{R}_i[d+1]\big)$. A spanning element $p\, r_l\, q$ of $\mathcal{R}_i[d+1]$ has $|p|+|q| = d-1$, where $|\cdot|$ denotes path length, and there are two cases. If $|p| \geq 1$, write $p = a_{j\to i}p'$ with $j \sim i$; then $p\, r_l\, q = \Phi$ of the tuple whose j-entry is $p' r_l q \in \mathcal{R}_j[d]$ and whose other entries vanish. If $|p| = 0$ then $p = e_i$, hence $l = i$ and q is a path $c \to i$ of length $d-1$, and

$$r_i\, q = \sum_{j\sim i} \varsigma(i,j)\, a_{j\to i}\big(a_{i\to j}q\big) = \Phi\Big(\big(\varsigma(i,j)\, a_{i\to j}q\big)_{j\sim i}\Big).$$

So $\Phi^{-1}(\mathcal{R}_i[d+1]) = \bigoplus_{j\sim i}\mathcal{R}_j[d] + \{(\varsigma(i,j)a_{i\to j}q)_j : q \in F_i[d-1]\}$: the first case contributes only elements that are *already zero one degree down*, in $(I_\kappa)_j[d]$, and the second contributes the image of Y up to the signs $\varsigma(i,j)$, which rescale the coordinates by ± 1 and so change neither the dimension of the image nor the injectivity of Y. Passing to the quotient, the induced surjection

$$\overline{\Phi}\colon \bigoplus_{j\sim i}(I_\kappa)_j[d] \longrightarrow (I_\kappa)_i[d+1]$$

has kernel exactly $\operatorname{im} Y$. This is the point at which "the preprojective relations impose one condition for each basis vector of $(I_\kappa)_i[d-1]$, and no others" is proved rather than assumed. Hence $a_{d+1}(i) = \sum_{j\sim i} a_d(j) - \dim\operatorname{im} Y$, and the stated recursion is equivalent to the injectivity of Y.

That injectivity holds below the top degree: a nonzero element of $\ker Y$ is killed by every arrow out of i, hence spans a copy of S_i inside $\operatorname{soc}(I_\kappa)$, which by Lemma 2.7 is a one-dimensional space in the top degree d^*. So $\ker Y = 0$ whenever $d-1 < d^*$. □

This lemma and Lemma 2.7 are used without further comment throughout.

2.3. Iterated socles. For $j \in I$ and a Π-module M let $\operatorname{soc}_{(j)}(M)$ be the largest submodule of M that is a direct sum of copies of S_j. Given a submodule $X \subseteq M$, the *socle step of colour j* replaces X by the preimage in M of $\operatorname{soc}_{(j)}(M/X)$; concretely,

$$X \;\longmapsto\; \{\, x \in M :\; a_{j\to i}\, x \in X \text{ for every } i \sim j \,\} \;\cap\; (X + M_j),$$

which enlarges X only at vertex j. For a sequence σ of colours, $\operatorname{soc}_\sigma(M)$ denotes the result of applying the corresponding steps to $X = 0$ in order. This is the iterated socle $\operatorname{soc}_{(j_1,\dots,j_\ell)}$ of Geiss–Leclerc–Schröer [7, §2.4].

2.4. The fixed point, and why it is a submodule. We use two inputs, which we state in full rather than by number, and which we keep strictly apart: a *geometric* identification, which is available over $\mathbb{C}$, and an *algebraic* rigidity property, which needs no hypothesis on the ground field at all. The separation matters: the characteristic-p argument of §11 uses only the second.

Proposition 2.9 (Savage–Tingley [9, Thm. 4.4, Prop. 4.9])**.** *Work over $\mathbb{C}$, and let $\mathfrak{L}(v,w)$ be the Lagrangian subvariety of $\mathfrak{M}(v,w)$. Then $\mathfrak{L}(v,w)$ is in natural bijection with the quiver Grassmannian of submodules of dimension vector v inside $\bigoplus_i I_i^{w_i}$; the bijection is a bijective algebraic map, hence a homeomorphism. Moreover the following are equivalent: v is w-extremal; $\mathfrak{L}(v,w)$ is a single point; that quiver Grassmannian is a single point; $\bigoplus_i I_i^{w_i}$ has a unique submodule of graded dimension v.*

Remark 2.10. Two points of bookkeeping. First, [9] states Theorem 4.4 as a bijective algebraic map and a homeomorphism, not as an isomorphism of varieties, and we have quoted it in that form; nothing below uses more. Second, the quiver Grassmannian $\mathrm{Gr}_P(v, q_w)$ of [9] is *defined* as a space of submodules of the injective module q_w, which is the convention of this paper (see the paragraph after Proposition 2.6); no translation between submodules of an injective and quotients of a projective is needed anywhere. Finally, [9] works throughout over $\mathbb{C}$, and Proposition 2.9 is stated here with that hypothesis.

Proposition 2.11 (rigidity; Baumann–Kamnitzer–Tingley [3, §3])**.** *Let T be an object of an essentially small abelian category in which every object has finite length, and let $\mathrm{Pol}(T)$ be its Harder–Narasimhan polytope. If x is a vertex of $\mathrm{Pol}(T)$, then T has a unique subobject X with $[X] = x$.*

Remark 2.12 (attribution, and the absence of a field hypothesis)**.** This is stated in [3, §3] as an unnumbered consequence of Corollary 3.3 there: *"Another noteworthy consequence is the following rigidity property: if x is a vertex of* $\mathrm{Pol}(T)$*, then T has a unique subobject X such that $[X] = x$."* Two points deserve emphasis, because both are easy to get wrong.

(i) The result is in [3]*, not in* [2]. What [2, Rem. 4.5] proves is the inclusion $\underline{\dim} U \in \mathrm{Pol}(T)$ for every submodule $U \subseteq T$; both the convex-hull description of $\mathrm{Pol}(T)$ and the description of the submodules realising its extremal points are there explicitly *deferred* to a paper in preparation of Baumann, Kamnitzer and Sadanand. So [3] is the load-bearing reference and [2] the incidental one.

(ii) There is no hypothesis on the ground field. Section 3 of [3] opens *"We consider an essentially small abelian category $\mathcal{A}$ such that all objects have finite length"*, and nothing else is assumed there; the convention *"We fix a base field K, which we assume for convenience to be algebraically closed of characteristic 0"* appears only in [3, §4.1], where preprojective algebras enter, and does not govern §3. Over any field $\mathbb{K}$ the algebra Π_Γ of a Dynkin Γ is finite dimensional, so its category of finite-dimensional modules satisfies the hypothesis of Proposition 2.11. This is what makes §11 possible.

For the one case this paper actually needs — submodules of I_κ with Γ Dynkin — we do not even have to rely on [3]: Corollary 2.13 below is given a short self-contained proof, valid over every field.

Corollary 2.13. *Let Γ be simply-laced Dynkin, $\mathbb{K}$ any field, and $U \subseteq I_\kappa$ a submodule with $v = \underline{\dim} U$. Then*

$$\delta(v) := 2v_\kappa - v^{\mathsf{T}} C v \;\geq\; 0,$$

and $\delta(v) = (\lambda,\lambda) - (\mu,\mu)$ where $\mu = \lambda - \sum_i v_i\alpha_i$; in particular $\delta(v) = 0$ whenever v is w-extremal. If $\delta(v) = 0$ then I_κ has at most one *submodule of dimension vector v. Consequently, if $v = \lambda - w\lambda$ then I_κ has at most one submodule of dimension vector v; over $\mathbb{C}$ there is exactly one, and it is the T-fixed representation V.*

Proof. Everything here is characteristic-free.

The identity. With $\lambda = \omega_\kappa$ and $\mu = \lambda - \sum_i v_i\alpha_i$ one has $(\lambda,\lambda) - (\mu,\mu) = 2(\lambda, \sum_i v_i\alpha_i) - (\sum_i v_i\alpha_i, \sum_i v_i\alpha_i) = 2v_\kappa - v^{\mathsf{T}} C v$, since $(\omega_\kappa, \alpha_i) = \delta_{i\kappa}$ in the normalisation $(\alpha_i,\alpha_i) = 2$. If $\mu = w\lambda$ then $(\mu,\mu) = (\lambda,\lambda)$, so $\delta(v) = 0$.

The inequality. Since I_κ is injective with $\operatorname{soc} I_\kappa = S_\kappa$ — both by the definition of an injective envelope, not by Proposition 2.1(b) — the functor $\mathrm{Hom}_\Pi(-, I_\kappa)$ is exact and $\dim \mathrm{Hom}(S_i, I_\kappa) = \delta_{i\kappa}$, so counting along a composition series of U gives $\dim \mathrm{Hom}(U, I_\kappa) = v_\kappa$. Composition with the inclusion $U \hookrightarrow I_\kappa$ embeds $\mathrm{End}_\Pi(U)$ into $\mathrm{Hom}(U, I_\kappa)$, whence $\dim \mathrm{End}(U) \leq v_\kappa$. By Proposition 2.1(a) with $M = N = U$,

$$v^{\mathsf{T}} C v = (v,v) = 2\dim \mathrm{End}(U) - \dim \mathrm{Ext}^1_\Pi(U,U) \;\leq\; 2v_\kappa,$$

which is $\delta(v) \geq 0$.

Uniqueness. Let $U \neq U'$ be submodules of I_κ with $\underline{\dim} U = \underline{\dim} U' = v$ and $\delta(v) = 0$. Both $U \cap U'$ and $U + U'$ are submodules, and $\underline{\dim}\,(U \cap U') + \underline{\dim}\,(U + U') = 2v$; write $\underline{\dim}\,(U \cap U') = v - e$ and $\underline{\dim}\,(U + U') = v + e$. Since δ is the quadratic polynomial $x \mapsto 2x_\kappa - (x,x)$,

$$\delta(v - e) + \delta(v + e) = 2\delta(v) - 2(e,e) = -2(e,e).$$

The Cartan matrix of a Dynkin diagram is positive definite, so $(e,e) > 0$ unless $e = 0$; and $e \neq 0$ would make one of $\delta(v \pm e)$ negative, contradicting the inequality just proved. So $e = 0$, i.e. $U \cap U'$ has dimension vector v, and therefore $U \cap U' = U = U'$.

Existence over $\mathbb{C}$, and the identification with V, is Proposition 2.9: for v w-extremal that proposition gives a unique submodule of I_κ of dimension vector v and a single point of $\mathfrak{L}(v, \omega_\kappa)$, which is the T-fixed representation V. (Over an arbitrary field existence is not quoted at all: Theorem 3.6 constructs such a submodule directly, by an iterated socle.) □

Remark 2.14. Corollary 2.13 is exactly the rigidity property of Proposition 2.11 in the case at hand, and we could have quoted [3, §3] for it, that section carrying no hypothesis on the field (Remark 2.12). We prefer the proof above because it is three lines long, because it makes the absence of a characteristic hypothesis visible rather than inherited, and because §11 rests on precisely this point.

We refer to a submodule U with $\underline{\dim} U$ extremal as an *extremal module*, and write $V(\mu)$ for the unique submodule with $\underline{\dim} V(\mu) = \lambda - \mu$. Over an arbitrary field its *existence* is not an input: Theorem 3.6 below constructs a submodule of I_κ of dimension vector $\lambda - w\lambda$ by an iterated socle, over any field and with no appeal to geometry. So $V(\mu)$ is well defined over every $\mathbb{K}$, and over $\mathbb{C}$ it is the T-fixed point of $\mathfrak{M}(\lambda - \mu, \lambda)$.

The last sentence of Corollary 2.13 is what we use constantly: to identify $V(\mu)$ it suffices to *exhibit any submodule* of I_κ of the correct dimension vector.

2.5. The two gradings agree.

Lemma 2.15 (the bidegree is determined)**.** Π *is bigraded by the numbers of* a*- and* a^**-arrows separately, the relations being bihomogeneous of bidegree* $(1,1)$*. For* Γ *a tree, the piece* $(\Pi e_j)_i[d]$ *is pure of a single bidegree, namely* $p = \frac{1}{2}(d + f(i) - f(j))$*,* $q = \frac{1}{2}(d - f(i) + f(j))$*, where* f *is any function on vertices with* $f(\mathrm{head}) - f(\mathrm{tail}) = 1$ *on every arrow of* Q*.*

Proof. Such an f exists because Γ is a tree. An a-arrow raises f by 1 and its reverse lowers it by 1, so along any path from j to i one has $p - q = f(i) - f(j)$; with $p + q = d$ this determines p, q. □

Proposition 2.16. *The* T*-weight decomposition of* V_i *coincides with its path-length decomposition, up to an affine normalisation depending only on* i*. In particular*

$$T \text{ acts on every } V_i \text{ with simple spectrum} \iff \dim V_i[d] \le 1 \quad \text{for all } i \in I \text{ and all } d.$$

Proof. Since $\lambda = \omega_\kappa$ is fundamental, the framing space is one dimensional and the framing torus acts on all of V through one character, contributing only a global shift. The rest of T acts through the bigrading of Π, assigning to $(\Pi e_{\kappa'})_i[d]$ a weight $\alpha p + \beta q$ for fixed integers α, β. By Lemma 2.15 both p and q are affine in d of slope $\pm\frac{1}{2}$, so $\alpha p + \beta q$ is affine in d of slope $\frac{1}{2}(\alpha + \beta)$, injective in d unless $\alpha + \beta = 0$. The latter is excluded because the symplectic form has nonzero weight — that being what makes the T-fixed locus isolated. So the weight determines and is determined by d. Finally V is a *graded* submodule: by Corollary 2.13 it is the unique submodule of its dimension vector, and uniqueness forces stability under the whole torus. □

From now on "simple spectrum" means $\dim V_i[d] \le 1$ for all i, d.

3. The socle-step identity and the main theorem

This section is uniform in type and characteristic-free.

Lemma 3.1. *Let* $X \subseteq I_\kappa$ *be a submodule,* $v' = \underline{\dim}\, X$ *and* $\mu' = \lambda - \sum_i v'_i \alpha_i$*. Then for every* $i \in I$

$$\dim \mathrm{soc}_{(i)}(I_\kappa / X) = \langle \alpha_i^\vee, \mu' \rangle + \dim \mathrm{top}_i(X).$$

Proof. Write $I = I_\kappa$, $Q = I/X$. Applying $\mathrm{Hom}_\Pi(S_i, -)$ to $0 \to X \to I \to Q \to 0$ gives

$$0 \to \mathrm{Hom}(S_i, X) \to \mathrm{Hom}(S_i, I) \to \mathrm{Hom}(S_i, Q) \to \mathrm{Ext}^1(S_i, X) \to \mathrm{Ext}^1(S_i, I),$$

and the last term vanishes as I is injective. Since $\mathrm{soc}(I_\kappa) = S_\kappa$ we have $\dim \mathrm{Hom}(S_i, I) = \delta_{i\kappa}$, so

$$\dim \mathrm{soc}_{(i)}(Q) = \delta_{i\kappa} - \dim \mathrm{soc}_{(i)}(X) + \dim \mathrm{Ext}^1(S_i, X). \tag{3}$$

By the Ext^1 formula, Proposition 2.1(a) — proved over an arbitrary field in Proposition 2.4 —

$$\dim \mathrm{Ext}^1_\Pi(M, N) = \dim \mathrm{Hom}(M, N) + \dim \mathrm{Hom}(N, M) - (\underline{\dim}\, M, \underline{\dim}\, N),$$

where $(x, y) = x^{\mathsf{T}} C y$. With $M = S_i$, $N = X$, and using $\mathrm{Hom}(S_i, X) = \mathrm{soc}_{(i)}(X)$, $\mathrm{Hom}(X, S_i) = \mathrm{top}_i(X)$, this reads $\dim \mathrm{Ext}^1(S_i, X) = \dim \mathrm{soc}_{(i)}(X) + \dim \mathrm{top}_i(X) - (\alpha_i, v')$. Substituting into (3), the socle terms cancel: $\dim \mathrm{soc}_{(i)}(Q) = \delta_{i\kappa} - (\alpha_i, v') + \dim \mathrm{top}_i(X)$. Finally $(\alpha_i, v') = \langle \alpha_i^\vee, \sum_j v'_j \alpha_j \rangle$ and $\langle \alpha_i^\vee, \omega_\kappa \rangle = \delta_{i\kappa}$, so $\delta_{i\kappa} - (\alpha_i, v') = \langle \alpha_i^\vee, \mu' \rangle$. □

Corollary 3.2. *A socle step of colour* i *applied to* $X \subseteq I_\kappa$ *enlarges* X *by exactly* $\langle \alpha_i^\vee, \mu' \rangle + \dim \mathrm{top}_i(X)$ *dimensions, all at vertex* i*; in particular by at least* $\langle \alpha_i^\vee, \mu' \rangle$*, with equality iff* S_i *does not occur in the head of* X*.*

Remark 3.3. Repeating a colour is harmless. If a step of colour i with $c = \langle \alpha_i^\vee, \mu' \rangle$ has just added exactly c dimensions, producing X', those c elements lie in the head of X', so $\dim \mathrm{top}_i(X') = c$, while the weight has become $s_i \mu'$ with $\langle \alpha_i^\vee, s_i \mu' \rangle = -c$. Lemma 3.1 gives $\dim \mathrm{soc}_{(i)}(I_\kappa / X') = 0$. Thus soc_σ is unchanged if a block i^c is replaced by $i^{c'}$ for any $c' \ge c$, and "apply $\mathrm{soc}_{(i)}$ until it stabilises" is an equivalent description.

Definition 3.4 (the c-trace)**.** Let $w = s_{i_1} \cdots s_{i_\ell}$ be reduced. Set $\mu^{(\ell)} = \lambda$ and $\mu^{(p-1)} = s_{i_p} \mu^{(p)}$, and put $c_p = \langle \alpha_{i_p}^\vee, \mu^{(p)} \rangle$. The c-weighted sequence is $\sigma = (i_\ell^{c_\ell}, \ldots, i_1^{c_1})$, read from the right end of the word inwards.

Lemma 3.5. *Let* w *be of minimal length in* wW_λ *and* $\mu = w\lambda$*.*

(i) Every c*-trace value of every reduced word for* w *satisfies* $c_p \ge 1$*.*

(ii) If $\mu \ne w_0 \lambda$ *there is* i *with* $\langle \alpha_i^\vee, \mu \rangle \ge 1$*, and for any such* i *one has* $\ell(s_i w) = \ell(w) + 1$*, with* $s_i w$ *again minimal in* $s_i w W_\lambda$*.*

Consequently every reduced word for w extends on the left to a reduced word for the longest minimal-length representative w_0^λ, with all c-trace values ≥ 1.

Proof. (i) Write $u = s_{i_{p+1}} \cdots s_{i_\ell}$, so $\mu^{(p)} = u\lambda$. The word being reduced gives $u^{-1}\alpha_{i_p} > 0$, and λ dominant gives $c_p = \langle (u^{-1}\alpha_{i_p})^\vee, \lambda \rangle \geq 0$. If $c_p = 0$ then $s_{i_p} u\lambda = u\lambda$, so $w\lambda = s_{i_1} \cdots s_{i_{p-1}} u\lambda$ with an element of length $\leq \ell - 1$, contradicting minimality.

(ii) If $\langle \alpha_i^\vee, \mu \rangle \leq 0$ for all i then μ is antidominant, i.e. $\mu = w_0\lambda$. Otherwise pick i with $\langle \alpha_i^\vee, w\lambda \rangle > 0$; then $\langle \lambda, w^{-1}\alpha_i^\vee \rangle > 0$, and λ dominant forces $w^{-1}\alpha_i > 0$, whence $\ell(s_i w) = \ell(w) + 1$. With $J = \{j : \langle \alpha_j^\vee, \lambda \rangle = 0\}$ one has $W_\lambda = W_J$, and w minimal in wW_J iff $w\alpha_j > 0$ for $j \in J$. For such j, $s_i(w\alpha_j) < 0$ would force $w\alpha_j = \alpha_i$, giving $\langle \alpha_i^\vee, w\lambda \rangle = \langle \alpha_j^\vee, \lambda \rangle = 0$, a contradiction. □

Theorem 3.6 (Main Theorem)**.** *Let Γ be simply-laced Dynkin, $\lambda = \omega_\kappa$, $w \in W$ of minimal length in wW_λ, and σ the c-weighted sequence of any reduced word for w. Then*

$$\underline{\dim}\, \mathrm{soc}_\sigma(I_\kappa) \;=\; v \;=\; \lambda - w\lambda, \qquad \textit{and consequently} \qquad V \;\cong\; \mathrm{soc}_\sigma(I_\kappa)$$

as graded representations of the doubled quiver. In particular the isomorphism class of $\mathrm{soc}_\sigma(I_\kappa)$ is independent of the reduced word.

Proof. Extend the word, by Lemma 3.5, to a reduced word for w_0^λ; let σ_{full} be its c-weighted sequence and σ the prefix belonging to w. *Index the blocks in the order in which they are applied*: $X_0 = 0$, and X_p is the result of applying the first p blocks, so that $X_0 \subseteq X_1 \subseteq \cdots \subseteq X_{\ell_{\mathrm{full}}} = \mathrm{soc}_{\sigma_{\mathrm{full}}}(I_\kappa)$, where ℓ_{full} is the number of blocks of σ_{full}, i.e. the length of the word. Write $i(p)$ for the colour of the p-th block applied and $c(p)$ for its multiplicity, and put $\mu[0] = \lambda$, $\mu[p] = s_{i(p)}\mu[p-1]$; reading Definition 3.4 from the right end of the word inwards, this is exactly the c-trace, so

$$c(p) = \langle \alpha_{i(p)}^\vee, \mu[p-1] \rangle \;\geq\; 1$$

by Lemma 3.5(i), and $\mu[p] = \mu[p-1] - c(p)\alpha_{i(p)} \in W\lambda$.

We prove by induction on p that

$$\underline{\dim}\, X_p = \lambda - \mu[p] \qquad (0 \leq p \leq \ell_{\mathrm{full}}), \tag{4}$$

which for $p = 0$ is $X_0 = 0$. So let $p \geq 1$ and assume $\underline{\dim}\, X_{p-1} = \lambda - \mu[p-1] =: v$. Write $i = i(p)$, $c = c(p)$, and let $\delta_p := \dim X_p - \dim X_{p-1}$; a block of colour i enlarges X only at the vertex i, so $\underline{\dim}\, X_p = v + \delta_p\, \epsilon_i$ with ϵ_i the i-th coordinate vector.

Lower bound: $\delta_p \geq c$. The inductive hypothesis is what makes Corollary 3.2 quotable with the *intended* weight: applied to X_{p-1}, whose weight $\mu' = \lambda - \sum_j v_j\alpha_j$ *is* $\mu[p-1]$, the first letter of the block already adds $\langle \alpha_i^\vee, \mu[p-1] \rangle + \dim \mathrm{top}_i(X_{p-1}) \geq c$ dimensions, and the remaining letters of the block only enlarge X further.

Upper bound: $\delta_p \leq c$. Here we use the inequality of Corollary 2.13, applied to the submodule $X_p \subseteq I_\kappa$. Since $\mu[p-1] \in W\lambda$, the same corollary gives $\delta(v) = 0$. With $\delta(x) = 2x_\kappa - (x,x)$ and $(\epsilon_i, \epsilon_i) = 2$, $(\epsilon_i, v) = (Cv)_i$,

$$\delta\big(v + \delta_p\epsilon_i\big) = \delta(v) + 2\delta_p\big(\delta_{i\kappa} - (Cv)_i\big) - 2\delta_p^2 = 2\delta_p\big(c - \delta_p\big),$$

because $\delta_{i\kappa} - (Cv)_i = \langle \alpha_i^\vee, \lambda - \sum_j v_j\alpha_j \rangle = \langle \alpha_i^\vee, \mu[p-1] \rangle = c$. Corollary 2.13 says this is ≥ 0, i.e. $2\delta_p(\delta_p - c) \leq 0$; and $\delta_p \geq c \geq 1 > 0$, so $\delta_p \leq c$.

Hence $\delta_p = c$ and $\underline{\dim}\, X_p = v + c\,\epsilon_i = \lambda - \mu[p-1] + c(p)\alpha_{i(p)} = \lambda - \mu[p]$, closing the induction. (Note that the two bounds are established *at the same step*, from the inductive hypothesis alone; no later block is invoked.)

Truncating at σ, i.e. stopping at the value of p at which the blocks of σ are exhausted, gives $\underline{\dim}\, \mathrm{soc}_\sigma(I_\kappa) = \lambda - w\lambda$, which is the v of the statement. By Corollary 2.13 the submodule of I_κ with this dimension vector is unique and equals V. (Taking $p = \ell_{\mathrm{full}}$ gives $\dim \mathrm{soc}_{\sigma_{\mathrm{full}}}(I_\kappa) = \dim(\lambda - w_0\lambda) = \dim I_\kappa$ by Proposition 2.6, so the full sequence exhausts I_κ; that consequence is not used below.) □

Corollary 3.7. *T acts with simple spectrum if and only if $\dim \mathrm{soc}_\sigma(I_\kappa)_i[d] \leq 1$ for all $i \in I$, d. The criterion is decidable in time polynomial in $|I|$ and $\ell(w)$ from any reduced word.*

Corollary 3.8. *Every X_p of the filtration of Theorem 3.6 is itself extremal: $\underline{\dim}\, X_p = \lambda - \mu[p]$ with $\mu[p] \in W\lambda$, by (4). In the word, $\mu[p] = w'\lambda$ where w' is the product of the last p letters of the reduced word, read as an element of W.*

3.1. The half-line wave, and the spin weights. The whole of the multiplicity bookkeeping in this paper rests on one elementary family of integer sequences, which we isolate now because it is used twice: here for the spin weights, and in Lemma 4.2 for the profile of I_κ. Nothing in this subsection uses the characteristic.

Lemma 3.9 (the half-line wave)**.** *Let $\rho \geq 1$ be an integer. For $j \geq 0$ and $d \in \mathbb{Z}$ put*

$$A_d^\rho(j) = \begin{cases} 1 & |j-\rho| \leq d \leq j+\rho-2 \text{ and } d \equiv j+\rho \ (2), \\ 0 & \text{otherwise.} \end{cases}$$

Then:
(i) $A_{d+1}^\rho(j) + A_{d-1}^\rho(j) = A_d^\rho(j-1) + A_d^\rho(j+1)$ *for every $j \geq 1$ and every d;*
(ii) $A_d^\rho(0) = 0$, $A_0^\rho(j) = \delta_{j\rho}$ *and* $A_d^\rho(j) = 0$ *for $d < 0$;*
(iii) $A_d^\rho(j) = 0$ *whenever $d > j+\rho-2$; in particular $A_d^\rho(j) = 0$ for all $j \leq \rho+1$ once $d \geq 2\rho$;*
(iv) $\#\{d : A_d^\rho(j) = 1\} = \min(j, \rho)$.

Proof. Let $u_d(i) = \big[\, |i-\rho| \leq d \text{ and } i \equiv \rho + d \ (2) \,\big]$ for $i \in \mathbb{Z}$. An immediate induction on d shows that u is the solution of the free chain recursion $u_{d+1}(i) + u_{d-1}(i) = u_d(i-1) + u_d(i+1)$ on $\mathbb{Z}$ with $u_0 = \delta_\rho$ and $u_{-1} = 0$. Set $A_d^\rho(j) = u_d(j) - u_d(-j)$; being a difference of two solutions of that recursion (the second reflected), A^ρ satisfies it for every j, which is (i). Under the parity constraint $d \equiv j+\rho$ one has $u_d(-j) = [\, j+\rho \leq d \,]$, so $u_d(j) - u_d(-j) = [\, |j-\rho| \leq d \leq j+\rho-2 \,]$, which is the displayed formula. (ii) is clear ($A_d^\rho(0) = u_d(0) - u_d(0) = 0$, and $d \geq 0$ is forced by $d \geq |j-\rho|$ unless $j = \rho$, where $d = 0$ gives 1). (iii) is read off the formula, together with $j+\rho-2 \leq 2\rho-1$ for $j \leq \rho+1$. For (iv), the degrees of the correct parity in $[\, |j-\rho|,\ j+\rho-2 \,]$ number $\frac{(j+\rho-2)-|j-\rho|}{2} + 1 = \frac{j+\rho-|j-\rho|}{2} = \min(j,\rho)$. □

Lemma 3.10 (folding the two leaves)**.** *Let $\Gamma = D_n$ with $n \geq 4$, $t = n-2$, $L = n-1$, and let κ be any vertex. Put*

$$\hat{a}_d(j) = \dim(I_\kappa)_j[d] \quad (1 \leq j \leq t), \qquad \hat{a}_d(L) = \dim(I_\kappa)_{n-1}[d] + \dim(I_\kappa)_n[d],$$

and let d^* be the top degree of I_κ. Then, with the boundary conventions $\hat{a}_d(0) := 0$ and $\hat{a}_d(L+1) := \hat{a}_d(L-1)$,

$$\hat{a}_{d+1}(j) + \hat{a}_{d-1}(j) = \hat{a}_d(j-1) + \hat{a}_d(j+1) \qquad (1 \leq j \leq L, \quad 0 \leq d \leq d^*-1),$$

with $\hat{a}_{-1} \equiv 0$ and $\hat{a}_0(j) = \delta_{j,\iota(\kappa)}$, where $\iota(\kappa) = \kappa$ for a chain vertex and $\iota(\kappa) = L$ for a leaf.

Proof. Lemma 2.8 gives $a_{d+1}(i) = \sum_{j \sim i} a_d(j) - a_{d-1}(i)$ at every vertex i and every $d < d^*$. At a chain vertex $j < t$ the neighbours are $j \pm 1$, which is the displayed equation. At $j = t$ the neighbours are $t-1$, $n-1$ and n, and $\hat{a}_d(t-1) + \hat{a}_d(L) = \hat{a}_d(t-1) + \hat{a}_d(t+1)$ since $L = t+1$. At a leaf the only neighbour is t; adding the two leaf equations gives $\hat{a}_{d+1}(L) + \hat{a}_{d-1}(L) = 2\hat{a}_d(t) = \hat{a}_d(L-1) + \hat{a}_d(L+1)$ under the mirror convention. The degree-0 part of $I_\kappa \cong \Pi e_{\nu^{-1}(\kappa)}$ is spanned by the idempotent at $\nu^{-1}(\kappa)$; after folding, that vertex is $\iota(\kappa)$ whether or not ν fixes κ, whence the initial datum $\hat{a}_0(j) = \delta_{j,\iota(\kappa)}$. □

Lemma 3.11 (rigidity of the folded profile)**.** *Let $n \geq 4$, $t = n-2$, $L = n-1$, and let A^ρ be as in Lemma 3.9.*
(a) *Let $1 \leq k \leq t$ and put $\hat{A}_d(j) = A_d^k(j) + A_{2t-d}^k(j)$. If $d \geq 0$ and $\hat{A}_d(j) = \delta_{jk}$ for every $1 \leq j \leq L$, then $d \in \{0, 2t\}$.*
(b) *If $d \geq 0$ and $A_d^L(j) = \delta_{jL}$ for every $1 \leq j \leq L$, then $d \in \{0, 2L-2\}$.*

Proof. Throughout we use the defining formula: $A_d^\rho(j) = 1$ exactly when $|j-\rho| \leq d \leq j+\rho-2$ and $d \equiv j+\rho$ (2).

(a). For $j \leq L$ one has $j+k-2 \leq L+t-2 = 2t-1$, so $A_e^k(j) = 0$ whenever $e > 2t-1$; hence $\hat{A}_d = 0$ for $d < 0$ (the first summand vanishes as $d < 0$, the second as $2t-d > 2t$). Since $\hat{A}_{2t-d} = \hat{A}_d$, it suffices to treat $0 \leq d \leq t$ and show $d = 0$. So suppose $1 \leq d \leq t$ and $\hat{A}_d = \delta_{\cdot k}$ on $\{1, \dots, L\}$; we derive a contradiction in five steps.

(1) $d \geq t+2-k$, and $k \geq 2$. If $k+d \leq L$ then $A_d^k(k+d) = 1$: indeed $|(k+d)-k| = d \leq d$, $d \leq (k+d)+k-2$ because $k \geq 1$, and $(k+d)+k \equiv d$ (2). As $k+d \neq k$ this contradicts $\hat{A}_d = \delta_{\cdot k}$. Hence $k+d > L = t+1$, i.e. $d \geq t+2-k$; with $d \leq t$ this forces $k \geq 2$.

(2) $d \equiv t+k$ (2). Take $j = L = t+1$, which is $\neq k$ since $k \leq t$. We have $|L-k| = t+1-k \leq d$ by (1), and $d \leq t \leq L+k-2 = t+k-1$; also $|L-k| \leq t \leq 2t-d$ and $2t-d \leq 2t-(t+2-k) = t+k-2 \leq L+k-2$. So both $A_d^k(L)$ and $A_{2t-d}^k(L)$ equal 1 as soon as the common parity condition $d \equiv 2t-d \equiv L+k$ (2) holds. If it held we would get $\hat{A}_d(L) = 2 \neq 0$; so $d \equiv L+k+1 = t+k$ (2).

(3) $k = t$. Take $j = t$. Then $|t-k| = t-k \leq t+2-k \leq d$, $d \leq t \leq t+k-2$ (using $k \geq 2$), and $d \equiv t+k$ (2) by (2); hence $A_d^k(t) = 1$ and so $\hat{A}_d(t) \geq 1$, forcing $t = k$.

(4) $d = 2$. By (2) and (3), d is even and $2 \le d \le t$. Put $e = 2t - d$ and $j_0 = e - t + 2 = t - d + 2$, so $2 \le j_0 \le t \le L$. Then $A^t_e(j_0) = 1$: $|j_0 - t| = d - 2 \le t \le e$; $e \le j_0 + t - 2 = e$; and $e \equiv j_0 + t = 2t - d \equiv e$ (2). So $\hat{A}_d(j_0) \ge 1$, forcing $j_0 = k = t$, i.e. $d = 2$.

(5) Contradiction. So $k = t$ and $d = 2$. If $t \ge 3$ then $A^t_2(t-2) = 1$ — $|-2| \le 2$, $2 \le (t-2) + t - 2 = 2t - 4$, and $2 \equiv (t-2) + t$ (2) — with $1 \le t - 2 \ne t$, contradicting $\hat{A}_2 = \delta_{\cdot t}$. If $t = 2$ then $d = 2 = 2t - d$, so $\hat{A}_2(2) = 2A^2_2(2) = 2 \ne 1$, again a contradiction.

(b). From $A^L_d(L) = 1$ we get $0 \le d \le 2L - 2$ and $d \equiv 2L$, i.e. d even. Suppose $2 \le d \le 2L - 4$. Since $n \ge 4$ we have $L \ge 3$, so $1 \le L - 2 < L$, and $A^L_d(L-2) = 1$: $|(L-2) - L| = 2 \le d$, $d \le (L-2) + L - 2 = 2L - 4$, and $d \equiv (L-2) + L = 2L - 2$ (2) as both are even. This contradicts $A^L_d = \delta_{\cdot L}$. Hence $d = 0$ or $d = 2L - 2$. □

Proposition 3.12 (the folded solution)**.** *Let* $\Gamma = D_n$, $n \ge 4$, $t = n - 2$, $L = n - 1$.
(a) *If* $\kappa = k$ *is a chain vertex with* $1 \le k \le t$, *then the top degree of* I_κ *is* $2t$ *and*

$$\hat{a}_d(j) = A^k_d(j) + A^k_{2t-d}(j) \qquad (1 \le j \le L,\ d \ge 0).$$

(b) *If* κ *is a leaf, then the top degree of* I_κ *is* $2L - 2$ *and* $\hat{a}_d(j) = A^L_d(j)$ *for* $1 \le j \le L$ *and* $d \ge 0$.

Proof. Write $\hat{A}$ for the proposed right-hand side and $\bar{d}$ for the asserted top degree. We first record three facts that hold in both cases.

(G1) I_κ *is generated in degree* 0. By (1), $I_\kappa \cong \Pi e_{\nu^{-1}(\kappa)}$ is spanned by the paths starting at $\nu^{-1}(\kappa)$, so $(I_\kappa)[d+1]$ is spanned by the elements $a\,x$ with $x \in (I_\kappa)[d]$ and a an arrow. Consequently $\hat{a}_d = 0$ forces $\hat{a}_{d'} = 0$ for every $d' \ge d$, and d^* is the unique degree with $\hat{a}_{d^*} \ne 0 = \hat{a}_{d^*+1}$.

(G2) $d^* \ge 1$. Pick $i \sim \kappa$. By Proposition 2.4, $\dim \mathrm{Ext}^1_\Pi(S_i, S_\kappa) = 0 + 0 - (\epsilon_i, \epsilon_\kappa) = -C_{i\kappa} = 1 \ne 0$, so S_κ is not injective and $I_\kappa \ne S_\kappa$. (This is the only place in this proof where the Ext^1 formula is used, and Proposition 2.4 supplies it over an arbitrary field.)

(G3) $\hat{a}_{d^*}(j) = \delta_{j,\iota(\kappa)}$ *for* $1 \le j \le L$. By Lemma 2.7, $(I_\kappa)[d^*] = \mathrm{soc}(I_\kappa) = S_\kappa$ is one dimensional and sits at the vertex κ. For a chain vertex $\kappa = k \le t$ this is the assertion; for a leaf κ the folded value $\hat{a}_{d^*}(L)$ is the sum of the two leaf dimensions, hence 1, and all other folded values vanish.

(a). Here $\bar{d} = 2t$ and $\iota(\kappa) = k$. Non-negativity is clear. Both $d \mapsto A^k_d(j)$ and $d \mapsto A^k_{2t-d}(j)$ satisfy the recursion of Lemma 3.9(i) for every $j \ge 1$ (the recursion is invariant under $d \mapsto 2t - d$), and both vanish at $j = 0$. The mirror condition $\hat{A}_d(L+1) = \hat{A}_d(L-1)$, i.e. $\hat{A}_d(t+2) = \hat{A}_d(t)$, follows for *every* d from

$$A^k_d(t+2) = A^k_{2t-d}(t) \qquad \text{for all } d, \tag{5}$$

applied both as it stands and with d replaced by $2t-d$. Indeed, $k \le t$ gives $|t+2-k| = t+2-k$ and $|t-k| = t-k$, so $A^k_d(t+2) = [\,t+2-k \le d \le t+k\,]$ while $A^k_{2t-d}(t) = [\,t-k \le 2t-d \le t+k-2\,] = [\,t-k+2 \le d \le t+k\,]$: the same interval, and both conditions carry the parity $d \equiv t+k$. Initial data: $A^k_{2t}(j) = 0$ and $A^k_{2t+1}(j) = 0$ for $j \le L$ by Lemma 3.9(iii), since $j + k - 2 \le L + t - 2 = 2t - 1$; hence $\hat{A}_0(j) = A^k_0(j) = \delta_{jk}$ and $\hat{A}_{-1} = 0$. Vanishing above $2t$: for $d > 2t$ both summands vanish, the first by 3.9(iii) and the second because $2t - d < 0$. Finally $\hat{A}_{2t}(j) = A^k_{2t}(j) + A^k_0(j) = \delta_{jk}$, by the same two facts.

So $\hat{A}$ is a non-negative solution of the recursion and the initial data of Lemma 3.10, *valid at every degree.* Since the recursion of Lemma 3.10 holds for I_κ at every degree $d \le d^* - 1$ and determines $\hat{a}_{d+1}$ from $\hat{a}_d, \hat{a}_{d-1}$, induction on d gives

$$\hat{a}_d = \hat{A}_d \qquad \text{for every } d \le d^*.$$

If $d^* > 2t$ this would give $\hat{a}_{d^*} = \hat{A}_{d^*} = 0$, absurd; so $d^* \le 2t$. By (G3), $\hat{A}_{d^*} = \hat{a}_{d^*} = \delta_{\cdot k}$ on $\{1, \dots, L\}$, so Lemma 3.11(a) gives $d^* \in \{0, 2t\}$, and $d^* \ne 0$ by (G2). Hence $d^* = 2t$, and $\hat{a}_d = \hat{A}_d$ for $d \le 2t$; for $d > 2t$ both sides vanish, by (G1) and by the previous paragraph. This is the assertion.

(b). Here $\bar{d} = 2L - 2$, $\iota(\kappa) = L$ and $\hat{A}_d(j) = A^L_d(j)$, which is non-negative, vanishes at $j = 0$ and satisfies the recursion for all $j \ge 1$. Its initial data are $\hat{A}_0(j) = \delta_{jL} = \delta_{j,\iota(\kappa)}$ and $\hat{A}_{-1} = 0$, as required. The mirror condition holds for $0 \le d \le 2L - 3$: $A^L_d(L+1) = [\,1 \le d \le 2L-1\,]$ and $A^L_d(L-1) = [\,1 \le d \le 2L-3\,]$ agree there (both with d odd). Hence induction on d, using Lemma 3.10 at degrees $d \le d^* - 1$ and the mirror condition at those degrees, gives

$$\hat{a}_d = \hat{A}_d \qquad \text{for every } d \le \min(d^*, 2L-2).$$

Suppose first $d^* < 2L - 2$. Then $\hat{A}_{d^*} = \hat{a}_{d^*} = \delta_{\cdot L}$ by (G3), so $d^* \in \{0, 2L-2\}$ by Lemma 3.11(b), hence $d^* = 0$, contradicting (G2). So $d^* \ge 2L - 2$ and $\hat{a}_d = \hat{A}_d$ for all $d \le 2L - 2$. Suppose next $d^* > 2L - 2$. Reading off the defining formula, $\hat{A}_{2L-2}(j) = \delta_{jL}$ (the condition $2L - 2 \le j + L - 2$ forces $j \ge L$) and $\hat{A}_{2L-3}(j) = \delta_{j,L-1}$

(the condition $2L-3 \le j+L-2$ forces $j \ge L-1$, and $j=L$ has the wrong parity). Since $2L-2 \le d^*-1$, Lemma 3.10 applies at degree $2L-2$ and gives, for every $1 \le j \le L$,

$$\hat{a}_{2L-1}(j) = \hat{A}_{2L-2}(j-1) + \hat{A}_{2L-2}(j+1) - \hat{A}_{2L-3}(j),$$

with the conventions $\hat{A}_d(0)=0$ and $\hat{A}_d(L+1) := \hat{A}_d(L-1)$. At $j=L$ this is $2\hat{A}_{2L-2}(L-1) - \hat{A}_{2L-3}(L) = 0-0=0$; at $j=L-1$ it is $\hat{A}_{2L-2}(L-2) + \hat{A}_{2L-2}(L) - \hat{A}_{2L-3}(L-1) = 0+1-1=0$; and at $j<L-1$ all three terms vanish. So $\hat{a}_{2L-1}=0$, whence $d^* \le 2L-2$ by (G1) — a contradiction. Therefore $d^*=2L-2$ and $\hat{a}=\hat{A}$. □

Corollary 3.13 (the dimension of I_κ in type D)**.** *Let $\Gamma = D_n$ with $n \ge 4$, and let $\mathbb{K}$ be an arbitrary field. Then*

$$\dim_{\mathbb{K}} I_\kappa = k(2n-1-k) \quad (\kappa = k \le n-2), \qquad \dim_{\mathbb{K}} I_\kappa = \tfrac{n(n-1)}{2} \quad (\kappa \in \{n-1, n\}).$$

Equivalently, $\underline{\dim}\, I_\kappa = \omega_\kappa - w_0\omega_\kappa$ in root coordinates. This is Proposition 2.6; it is proved here from Lemma 2.8, Lemma 2.7, Proposition 2.4 and integer combinatorics, with no hypothesis on $\mathbb{K}$.

Proof. Sum Proposition 3.12 over all $d \ge 0$ and all $1 \le j \le L$; the folded total is the total dimension, since folding only adds the two leaf dimensions. In case (a), $\hat{A}_d(j)=0$ for $d<0$ and $d \mapsto 2t-d$ carries the support of $A^k_\bullet(j)$ into that of $A^k_{2t-\bullet}(j)$ bijectively, so by Lemma 3.9(iv)

$$\dim I_\kappa = \sum_{d\ge 0}\sum_{j=1}^{L} \hat{A}_d(j) = 2\sum_{j=1}^{L} \min(j,k) = 2\Big(\sum_{j=1}^{k} j + \sum_{j=k+1}^{n-1} k\Big) = k(k+1) + 2k(n-1-k) = k(2n-1-k).$$

In case (b), Lemma 3.9(iv) gives $\sum_{j=1}^{L} \min(j,L) = \sum_{j=1}^{L} j = \frac{L(L+1)}{2} = \frac{n(n-1)}{2}$. The final sentence is the root-coordinate computation carried out in the proof of Proposition 2.6. □

Corollary 3.14 (multiplicity-freeness at the three minuscule vertices)**.** *Let $\Gamma = D_n$, $n \ge 4$, and $\kappa \in \{1, n-1, n\}$. Then $\dim(I_\kappa)_i[d] \le 1$ for every vertex i and every degree d.*

Proof. For $\kappa=1$ apply Proposition 3.12(a) with $k=1$ (legitimate, since $1 \le t$ for $n \ge 4$): $A^1_d(j) = [\,d = j-1\,]$, so $\hat{a}_d(j) = [\,d=j-1\,] + [\,d = 2t-j+1\,]$. For $j \le t$ the two degrees $j-1 \le t-1$ and $2t-j+1 \ge t+1$ are distinct, so $\hat{a}_d(j) \le 1$. At $j = L = t+1$ they coincide and $\hat{a}_d(L) = 2[\,d=t\,]$, so each of the two leaves has $\dim(I_1)_l[d] = b_d$ with $2b_d \le 2$, i.e. $b_d \le 1$.

For $\kappa \in \{n-1, n\}$ apply Proposition 3.12(b): $\hat{a}_d(j) = A^L_d(j) \le 1$ for every $j \le L$. For $j \le t$ this is the claim directly; for $j = L$ it says that the two leaf dimensions *sum* to at most 1, so each is at most 1. □

Proof of Theorem 1.2. Let $\lambda = \omega_\kappa$ with $\kappa \in \{1, n-1, n\}$, and let w be of minimal length in wW_λ. These three weights of D_n are minuscule, so $\langle \alpha^\vee, \nu\rangle \in \{0, \pm 1\}$ for every root α and every weight ν of the irreducible representation of highest weight λ; hence every c-trace value c_p equals 1, which is exactly Stembridge's condition that w be λ-minuscule [10, §2], and is [5, Def. 5.1] verbatim — note that [5, §5.1] is headed "dominant minuscule" while its Definition 5.1 defines "λ-minuscule", by the condition $\langle \alpha^\vee_{i_p}, s_{i_{p+1}} \cdots s_{i_\ell}(\lambda)\rangle = 1$ for $1 \le p \le \ell$. This is DKK's condition verbatim with their word index renamed to p; they write k for it, which in this paper is the fundamental-weight label. It is exactly our $c_p = 1$: Definition 3.4 sets $\mu^{(\ell)} = \lambda$ and $\mu^{(p-1)} = s_{i_p}\mu^{(p)}$, so $\mu^{(p)} = s_{i_{p+1}} \cdots s_{i_\ell}\lambda$ — an empty product, hence $\mu^{(\ell)} = \lambda$, at the endpoint $p = \ell$ — and therefore $c_p = \langle \alpha^\vee_{i_p}, \mu^{(p)}\rangle = \langle \alpha^\vee_{i_p}, s_{i_{p+1}} \cdots s_{i_\ell}\lambda\rangle$, character for character. By Stembridge [10, Prop. 2.1] such a w is in particular fully commutative; we do not use this.

For the spectrum there is nothing more to do, and in particular no geometry is needed. By Theorem 3.6, which holds over an arbitrary field, $V = \mathrm{soc}_\sigma(I_\kappa)$ is a *submodule* of I_κ; and it is a *graded* submodule, because soc_σ is built from socle steps and a socle step is degree-homogeneous — $\mathrm{soc}_{(j)}(I_\kappa/X)$ is cut out of the graded module I_κ/X by the degree-homogeneous conditions $a_{j\to i}x \in X$, so it is a graded subspace, and $X=0$ is graded to begin with. Therefore

$$\dim V_i[d] \;\le\; \dim(I_\kappa)_i[d] \;\le\; 1 \qquad \text{for every } i \text{ and } d$$

by Corollary 3.14, which is simplicity of the spectrum in the sense fixed after Proposition 2.16.

No hypothesis on the characteristic was used, and nothing here is quoted from a source that carries one: Theorem 3.6 rests on Lemma 3.1 and Corollary 2.13, both proved over an arbitrary field (the former through Proposition 2.4), and Corollary 3.14 rests on Proposition 3.12, a statement about the dimensions of I_κ proved from Lemma 2.8, Lemma 2.7, Proposition 2.4 and Lemma 3.11 alone. Over $\mathbb{C}$, Proposition 2.9 together with

Corollary 2.13 identifies this V with the T-fixed representation and Proposition 2.16 identifies the path-length grading with the T-weight grading, giving the geometric reading of the statement; neither is used in the argument above. □

Remark 3.15. The route through the heap module of [6, §3] and the claim that the rank function of $H(w)$ is the path-length degree — is avoided entirely. Corollary 3.14 is stronger than what the heap argument would give: it bounds *every* submodule of I_κ, not only the extremal ones.

4. The injective I_κ in type D

Convention 4.1. From now on $\Gamma = D_n$ with $n \geq 4$, Bourbaki labelling $\alpha_i = e_i - e_{i+1}$ $(i \leq n-1)$, $\alpha_n = e_{n-1} + e_n$. The trivalent node is $t = n-2$, the two *leaves* are $n-1$ and n, and the *chain* is $\{1, \dots, t\}$. We fix $\lambda = \omega_k$ with $2 \leq k \leq n-2$ and $\kappa = k$, so that $I_\kappa = \Pi e_k$. The nontrivial diagram automorphism exchanging $n-1$ and n is denoted τ; it fixes every chain vertex, hence fixes κ, hence acts on I_κ. For $\mu \in W\lambda$ we write $v = \lambda - \mu$ in root coordinates, so $\mu = \lambda - \sum_i v_i \alpha_i$.

We also fix the direction words once and for all: along the chain, the arrow $a_{j\to j+1}$ points *inward* (towards the node t) and $a_{j\to j-1}$ points *outward*; the arrows $a_{t\to l}$ to the leaves and $a_{l\to t}$ from them are called the *node–leaf* arrows.

Finally we fix the symbols that would otherwise collide. Throughout, ς is the sign convention on arrows, $\varepsilon \in \{+,-\}$ is a strand label (§5), η_i denotes a Bourbaki coordinate of a weight, and ℓ is the length of a reduced word or of a colour sequence. The letter a is reserved for the quiver: the arrows $a_{i\to j}$ and the multiplicities $a_d(j)$, $a_d^\varepsilon(j)$, $\hat{a}_d(j)$ of I_κ; the integer parameter of the nine families of §8, and the corresponding parameter in $v'_t = 2h$ or $2h+1$, is called h instead. The letter M is reserved for modules, and $\mathrm{m}_j := \min(j,k)$ — the number of slots of one strand at a chain vertex j — is set in upright type to keep it apart. Two *different* degrees occur and are named differently: $D_j(X)$ is always the threshold of Definition 4.8 (the largest degree at which X_j is a *proper* subspace of $(I_\kappa)_j$), while E_j is always the lowest *taken* degree of a greedy top truncation at j, with E_j^ε its refinement to one strand (Lemma 8.2); the two are compared in Remark 4.10. The letter L is used only for the folded leaf vertex $L = n-1$ of Lemma 3.10; the Lagrangian subvariety $\mathfrak{L}(v,w)$ of §2 is in fraktur.

Two reuses are left standing because they are separated both by section and by shape, and no formula contains both. In §§8–9 the constant $B := n-1-k$ never takes an argument, whereas in §13 the double-path operators $A(x), B(x), C(x)$ always do; and the half-line wave $A_d(j)$ of §3.1 always carries a degree subscript. Likewise m and r of Theorem 1.1, and the scalars $m_\pm$ of §13, are lower-case italic and never carry a vertex subscript; the operators $P(x) = B(x) + C(x)$ and $Q(x) = B(x) - C(x)$ of §13 always carry their argument, as does the extension module $E(\psi)$ of §2.2, which is the one other use of the letter E.

Four letters carry more than one meaning, and we flag them rather than pretend otherwise. N is a module in §2 and §14.3, the integer $k(2n-1-k)$ in §11, the saturated lattice $N(v)$ of §11, and a multiplicity N_q in §7. P is a projective P_j in §2 and an operator $P(x)$ in §13; the poset of §9 is $\mathcal{P}$. ℓ is the length of a reduced word or colour sequence, and a prime in §11; path length is written $|\cdot|$. Most often reused is p: a prime in §11, the block index of Theorem 3.6 and of the c-trace, a path in the proof of Lemma 2.8, a Bourbaki coordinate index in §4 and §8, and the number of positive signs in Lemma 4.4. Likewise c is $k-q-h$ in §7, $\langle \alpha_t^\vee, \mu' \rangle$ in §10, $\nu^{-1}(\kappa)$ in §4, and a c-trace entry c_p. In every case the meaning is fixed by the section together with whether the symbol carries a subscript, an argument or neither.

Lemma 4.2 (the profile of I_κ)**.** *The colour-t multiplicity sequence of I_κ is*

$$(1, \underbrace{2, \dots, 2}_{k-1}, 1) \quad \text{at the degrees} \quad t-k,\ t-k+2,\ \dots,\ t+k,$$

and $\dim(I_\kappa)_t[d] = 0$ *otherwise. A chain vertex $i \leq t$ carries multiplicity 2 in exactly* $\max(0, i+k-n+1)$ *degrees and multiplicity ≤ 2 throughout. The two leaves are multiplicity-free, with support*

$$t-k+1,\ t-k+3,\ \dots,\ t+k-1 \qquad (k \text{ degrees}),$$

and $a_d(n-1) = a_d(n)$ for every d. The top degree of I_κ is $2t$.

Proof. Since $\kappa = k \leq t$ is a chain vertex, Proposition 3.12(a) applies and gives, in the folded notation of Lemma 3.10,

$$\hat{a}_d(j) = A_d^k(j) + A_{2t-d}^k(j) \qquad (1 \leq j \leq L = t+1), \tag{6}$$

with top degree $2t$. Every assertion of the lemma is read off (6); we record the four readings. Throughout, degrees at j carry the parity of $j+k$, and by Lemma 3.9

$$A^k_d(j) = [\,|j-k| \le d \le j+k-2\,], \qquad A^k_{2t-d}(j) = [\,2t-(j+k-2) \le d \le 2t-|j-k|\,].$$

The node. At $j=t$ the two intervals are $[\,t-k,\,t+k-2\,]$ and $[\,t-k+2,\,t+k\,]$. Their union is $[\,t-k,\,t+k\,]$ and their intersection is $[\,t-k+2,\,t+k-2\,]$, which contains $\frac{(t+k-2)-(t-k+2)}{2}+1 = k-1$ degrees of the right parity. So the colour-t multiplicity sequence is $(1, 2^{k-1}, 1)$ at the degrees $t-k, t-k+2, \dots, t+k$, and 0 elsewhere.

Chain vertices. For $i \le t$ the two intervals are $[\,|i-k|,\ i+k-2\,]$ and $[\,2t+2-i-k,\ 2t-|i-k|\,]$, each carrying $\min(i,k)$ degrees by Lemma 3.9(iv), so $\hat{a}_d(i) \le 2$ throughout. As $i,k \le t$ we have $i+k-2 \le 2t-|i-k|$, so the intersection is $[\,\max(|i-k|,\ 2t+2-i-k),\ i+k-2\,]$; it is nonempty exactly when $2t+2-i-k \le i+k-2$ is not the binding constraint, and in all cases the number of degrees of the right parity in it is $\max\big(0,\ i+k-t-1\big) = \max(0,\ i+k-n+1)$.

The leaves. At $j = L = t+1$, using $k \le t$, both intervals are $[\,t+1-k,\ t+k-1\,]$, so $\hat{a}_d(L) = 2\,[\,t-k+1 \le d \le t+k-1\,]$. Since $\hat{a}_d(L) = \dim(I_\kappa)_{n-1}[d] + \dim(I_\kappa)_n[d]$ and τ exchanges the two leaves while fixing I_κ, the two summands are equal; hence $a_d(n-1) = a_d(n) = b_d = [\,t-k+1 \le d \le t+k-1\,]$, so the leaves are multiplicity-free with the displayed support of k degrees.

The top degree is $2t$ by Proposition 3.12(a). □

Remark 4.3. The predicted total is genuinely part of the argument, and it is *termwise non-negative* because (6) is a sum of two indicator functions rather than an alternating sum over the images of k under the infinite dihedral group. Its value, computed in the proof of Proposition 3.12(a), is $2\sum_{j=1}^{n-1}\min(j,k) = k(2n-1-k)$, which is exactly $\dim(\lambda - w_0\lambda)$ by Proposition 2.6.

Lemma 4.4. *With m and r as in Theorem 1.1 one has $v_t = r+2m$.*

Proof. Write $\lambda-\mu = \sum_j c_j e_j$. Comparing coefficients of e_j in $\lambda-\mu = \sum_i v_i\alpha_i$ gives $c_1 = v_1$ and $c_j = v_j - v_{j-1}$ for $2 \le j \le n-2$, so $v_t = \sum_{j\le n-2} c_j$. Since $k \le n-2$ the support of λ lies in the first $n-2$ coordinates, whence $\sum_{j\le n-2} c_j = k - \sum_{j\le n-2}\mu_j$. Writing p and m for the numbers of positive and negative signs of μ among the first $n-2$ coordinates, $\sum_{j\le n-2}\mu_j = p-m$ and $p+m = k-r$, so $v_t = k-(p-m) = r+2m$. □

4.1. Reduction to the trivalent node.

Lemma 4.5 (inward injectivity)**.** *Let $i<t$ be a chain vertex and let d satisfy $\dim(I_\kappa)_i[d] \ge 2$. Then the inward arrow $a_{i\to i+1} : (I_\kappa)_i[d] \to (I_\kappa)_{i+1}[d+1]$ is injective.*

Proof. Suppose $x \ne 0$ lies in the kernel. We first record the degree bound that drives the whole iteration. Since $\dim(I_\kappa)_i[d] \ge 2$, the slot (d,i) lies in the multiplicity-2 window of i, whose top degree is $i+k-2$ by Lemma 4.2; so

$$d \le i+k-2, \qquad \text{whence} \qquad d+(i-k) \ \le\ 2i-2 \ \le\ 2t-4 \ <\ 2t \tag{7}$$

(the last step because $i \le t-1$). Note also $d \le i+k-2 \le 2t-3 < 2t$.

Now argue as follows. For $i<t$ the vertex i has exactly the two neighbours $i-1$ and $i+1$. By Lemma 2.8 an element killed by every arrow out of i is either zero or the socle generator, which by Lemma 2.7 lives in the top degree $2t$ at the vertex κ; by (7) $d<2t$, so $y := a_{i\to i-1}(x) \ne 0$ (for $i=1$ there is no such y and we get the contradiction at once). The preprojective relation at i, $\varsigma\, a_{i+1\to i}a_{i\to i+1} + \varsigma'\, a_{i-1\to i}a_{i\to i-1} = 0$, applied to x, has vanishing first term, so $a_{i-1\to i}(y) = 0$. Thus $y \ne 0$ lies in $(I_\kappa)_{i-1}[d+1]$ and is killed by *its* inward arrow: the same situation one vertex further out and one degree higher.

At the general step we sit at $(d+p,\ i-p)$ with a nonzero element killed by its inward arrow, and we must know that the outward arrow does *not* kill it, so that the iteration continues. If both arrows out of $i-p$ killed it, the element would span a copy of S_{i-p} inside $\mathrm{soc}(I_\kappa)$, forcing $i-p = \kappa = k$ *and* $d+p = 2t$ by Lemma 2.7; but $i-p=k$ means $p = i-k$, and then (7) gives $d+p \le 2t-4 < 2t$. So the outward arrow survives at every step, and the iteration reaches vertex 1: it produces $0 \ne z \in (I_\kappa)_1[d+i-1]$ killed by $a_{1\to 2}$. Vertex 1 has 2 as its only neighbour, so $z \in \mathrm{soc}(I_\kappa) = S_\kappa$ and $\kappa = 1$ — contradicting $\kappa = k \ge 2$ of Convention 4.1. □

Theorem 4.6. *If $\dim V_t[d] \le 1$ for all d, then $\dim V_i[d] \le 1$ for all $i \in I$ and all d.*

Proof. The two leaves are multiplicity-free in I_κ by Lemma 4.2, so they never carry a repeat. Suppose $\dim V_i[d] \ge 2$ for a chain vertex $i<t$. Then $\dim(I_\kappa)_i[d] \ge 2$, so by Lemma 4.5 the inward arrow is injective on

$(I_\kappa)_i[d]$, hence on the subspace $V_i[d]$; as V is a submodule the image lies in $V_{i+1}[d+1]$, so $\dim V_{i+1}[d+1] \geq 2$. Iterating $t-i$ times gives $\dim V_t[d+t-i] \geq 2$. □

Remark 4.7. The more obvious argument — containment of multiplicity-2 windows — does not work. The multiplicity-2 window of vertex $i \leq t$ is $[2L-k-i,\ i+k)$ with $L = n-1$, and its endpoints do satisfy $2L-k-i \geq L-k+1$ and $i+k \leq (L-1)+k$, so it is contained in that of t *as an interval*. But the degrees carried by i obey the parity constraint $d \equiv i+k \pmod 2$, so the degree *sets* need not be nested: in D_6 with $\kappa = 3$ and $t = 4$, vertex 3 carries multiplicity 2 exactly at degree 4 while t carries it exactly at degrees 3 and 5 — disjoint sets. And even granting containment, nothing transports $\dim V_i[d] \geq 2$ to $\dim V_t[d'] \geq 2$. The degree shift built into Lemma 4.5 is what makes the D_6 numbers line up: $i = 3$ at $d = 4$ maps to t at $d = 5$.

4.2. The statement that carries the theorem.

Definition 4.8. Let $X \subseteq I_\kappa$ be a graded submodule and j a vertex. We say X is an *up-set at j* if there is a degree D with $X_j[d] = 0$ for $d < D$, $X_j[D] \subseteq (I_\kappa)_j[D]$ arbitrary, and $X_j[d] = (I_\kappa)_j[d]$ for $d > D$. At a vertex where $\dim(I_\kappa)_j[d] \leq 1$ this says exactly that X_j occupies a final segment of degrees; at t we also call it a *top truncation*.

The degree D above is not unique — it may be replaced by the next colour-j degree below whenever $X_j[D]$ happens to be full, or by the next one above whenever $X_j[D] = 0$. *We therefore fix once and for all the convention*

$$D_j(X) := \text{the largest degree at which } X_j \text{ is a proper subspace of } (I_\kappa)_j,$$

with $D_j(X) := -\infty$ when X_j is full in every degree. This is the only meaning the symbol $D_j(X)$ carries below, and it is the one all later arguments use: in particular $X_j[D_j(X)]$ is always proper, and $X_j[d] = (I_\kappa)_j[d]$ for every $d > D_j(X)$.

The following lemma reads off, from the profile $(1, 2^{k-1}, 1)$ of Lemma 4.2 alone, everything the induction of §10 needs to know about the node column of a top truncation. It is stated for an arbitrary graded submodule *that is a top truncation at t* — no reduced word, and no parameter of the nine families of §8: the integer h below is simply $\lfloor v'_t/2 \rfloor$, not the h of $\mu(h;x,y,z)$ — only the quantities v'_t and $D_t(X)$, which exist for any such X — and it is proved over an arbitrary field. The top-truncation hypothesis cannot be dropped from (a): if X_t is the line in the bottom one-dimensional slot, at degree $t-k$, then $v'_t = 1$ but $D_t(X) = t+k$, whereas the table for $v'_t = 1$ reads $D = t+k-2$. The lemma is placed here because it is used at $p = 2$, and because the standing hypothesis char $\neq 2$ of §§5–6 and §§8–9 is never in force for it.

Lemma 4.9 (the node column of a top truncation)**.** *Let $X \subseteq I_\kappa$ be a graded submodule which is a top truncation at t, put $v'_t = \dim X_t$ and let $D = D_t(X)$ be the threshold of Definition 4.8.*

(a) *If $v'_t < 2k$ then D is finite and*

$$D = \begin{cases} t+k, & v'_t = 0, \\ t+k-2h, & v'_t = 2h \text{ with } h \geq 1, \\ t+k-2h-2, & v'_t = 2h+1 \text{ with } h \geq 0, \end{cases}$$

and

$$X_t[D] \neq 0 \iff v'_t \text{ is even and } v'_t \geq 2,$$

in which case $X_t[D]$ is a line in the two-dimensional space $(I_\kappa)_t[D]$; while for v'_t odd, and for $v'_t = 0$, one has $X_t[D] = 0$.

(b) *If moreover $\underline{\dim} X$ is extremal and $c := \langle \alpha_t^\vee, \lambda - \underline{\dim} X \rangle \geq 1$, then $v'_t \leq 2k-1$, so the hypothesis $v'_t < 2k$ of* (a) *is automatic.*

Proof. (a). By Lemma 4.2 the colour-t column of I_κ consists of a one-dimensional slot at $t+k$, then $k-1$ two-dimensional slots at $t+k-2, t+k-4, \dots, t-k+2$, then a one-dimensional slot at $t-k$; its total dimension is $2k$, and there are no other colour-t degrees. Being a top truncation, X_t is full above D, proper at D and zero below D (Definition 4.8); so v'_t is the total dimension of the column strictly above D, plus $\dim X_t[D]$. Since $v'_t < 2k$ the column is not full, so some degree is proper and D is finite.

If $v'_t = 0$ then $X_t = 0$ and the largest proper degree is the top one, $D = t+k$; and $X_t[D] = 0$.

If $v'_t = 2h$ with $h \geq 1$, then $v'_t \leq 2k-2$, so $h-1 \leq k-2$ and the degrees $t+k, t+k-2, \dots, t+k-2(h-1)$ — the top slot and the top $h-1$ two-dimensional slots — carry $1+2(h-1) = 2h-1$ dimensions in all; one further dimension is taken at the next degree down, $t+k-2h$, which satisfies $t+k-2h \geq t-k+2$ and is

therefore two-dimensional. So $D = t + k - 2h$ and $X_t[D]$ is a line in a two-dimensional space: nonzero and proper.

If $v'_t = 2h+1$, then $v'_t \le 2k-1$, so $h \le k-1$ and the degrees $t+k, t+k-2, \dots, t+k-2h$ carry $1+2h = v'_t$ dimensions in all and are exactly the filled ones; the next degree down is $t+k-2h-2 \ge t-k$, which is a genuine colour-t degree and is untaken. So $D = t+k-2h-2$ and $X_t[D] = 0$.

This proves the displayed table, and the equivalence "$X_t[D] \neq 0 \iff v'_t$ even and ≥ 2" is read off it.

(b). Since $\underline{\dim} X$ is extremal, $\mu' := \lambda - \sum_i (\underline{\dim} X)_i \alpha_i$ lies in $W\lambda = W\omega_k$, and for $2 \le k \le n-2$ that orbit is $\{\sum_{i\in S} \eta_i e_i : |S| = k,\ \eta_i = \pm 1\}$ (see the first paragraph of the proof of Proposition 7.7); so we may write $\mu' = \sum_i \eta_i e_i$ with $\eta_i \in \{0, \pm 1\}$ and exactly k nonzero entries. By Lemma 4.4 applied to μ', $v'_t = r' + 2m'$ where $r' = |S' \cap \{n-1, n\}|$ and $m' = \#\{i \le n-2 : i \in S',\ \eta_i = -1\}$, S' being the support of μ'. Since $|S'| = k$ we have $m' \le k - r'$, hence

$$v'_t = r' + 2m' \ \le\ r' + 2(k - r') \ =\ 2k - r' \ \le\ 2k,$$

with equality throughout only if $r' = 0$ and $m' = k$. In that case every one of the k elements of S' is a chain coordinate $\le n-2$ carrying $\eta_i = -1$, and $r' = 0$ gives $\eta_{n-1} = \eta_n = 0$; so $\eta_{n-2} \in \{0, -1\}$ and

$$c = \langle \alpha_t^\vee, \mu' \rangle = \eta_{n-2} - \eta_{n-1} = \eta_{n-2} \le 0,$$

contradicting $c \ge 1$. Hence $v'_t \le 2k-1$. □

Remark 4.10. Lemma 4.9 is deliberately *not* a corollary of Lemma 8.9. The latter is stated inside §8, under the standing hypothesis char $\neq 2$, for a fixed representative $\mu = \mu(h; x, y, z)$, and its conclusions are written in the family parameters h, q, x, y, z, which do not exist for an arbitrary extremal submodule of I_κ. Its E_t is moreover a *different* quantity: it is the lowest *taken* colour-t degree, whereas the D of Definition 4.8 is the largest degree at which the column is *proper*. The two agree when v'_t is even *and positive* — there the lowest taken slot is the partial one — and differ by 2 when v'_t is odd, where the lowest taken slot is full and the threshold sits one slot below it. At $v'_t = 0$ there is no taken slot at all, so the E_t of Lemma 8.9 does not exist and no comparison is to be made, while the threshold of Definition 4.8 is $t+k$; this is the form recorded in §14. Lemma 4.9 is the statement in the D of Definition 4.8, and it is the one used in §10; part (b) supplies the boundary hypothesis $v'_t < 2k$, which Lemma 8.9 secures instead by the family computation $v_t = 2k - 2h - s$ with $2h + s \ge q \ge 1$.

Proposition 4.11 (top truncation)**.** *Let $\mu = w\lambda$ and $V = V(\mu)$. Then V is a top truncation at t: its colour-t multiplicities are the greedy top truncation of the sequence of Lemma 4.2 to total v_t.*

Proposition 4.11 is proved in §10. We record at once that it implies the theorem.

Proof of Theorem 1.1, assuming Proposition 4.11. By Lemma 4.2 the colour-t profile of I_κ is $(1, 2^{k-1}, 1)$. By Theorem 3.6 and Corollary 2.13, V is the unique submodule of I_κ with $\underline{\dim} V = v$, and Proposition 4.11 applies to it. The greedy top truncation of $(1, 2^{k-1}, 1)$ to total v_t is

$$v_t = 0 \mapsto (\,),\quad 1 \mapsto (1),\quad 2 \mapsto (1,1),\quad 3 \mapsto (1,2),\quad 4 \mapsto (1,2,1),\quad 5 \mapsto (1,2,2),\ \dots$$

so every colour-t multiplicity is ≤ 1 exactly when $v_t \le 2$ (for $v_t \ge 3$ the sequence contains a 2, since $k \ge 2$ guarantees a multiplicity-2 slot). By Theorem 4.6, simplicity at t implies simplicity everywhere, and conversely simplicity everywhere implies simplicity at t. Lemma 4.4 identifies $v_t = r + 2m$. Theorem 1.2 covers the spin weights. □

5. The strand decomposition

Throughout §5 and §6, and again throughout §8 and §9, we assume char $\mathbb{K} \neq 2$. The intervening §7 is *not* covered by this hypothesis and does not use it: Definition 7.1, Lemma 7.2, Corollary 7.4, Proposition 7.7, Definition 7.9 and (14) are proved over an arbitrary field, and they are quoted at $p = 2$ in §11; the hypothesis lapses at the head of §7 and resumes at the head of §8, as each of those sections says. The hypothesis is used to construct the witness submodule of (B); §11 transports (B), and with it Theorem 1.1, to every characteristic.

The automorphism τ fixes every chain vertex and exchanges the two leaves, and it acts on I_κ (Convention 4.1). Since 2 is invertible, each graded piece at a *chain* vertex splits into τ-eigenspaces. Write $a_d^{\pm}(j) = \dim\big((I_\kappa)_j[d]\big)^{\pm}$ for $j \le t$, and call the two eigen-parts the $+$ *strand* and the $-$ *strand*. Recall $t = n-2$ and that the top degree of I_κ is $2t$ (Lemma 4.2).

Proposition 5.1. *For every chain vertex j with $1 \le j \le t$ and every d,*

$$a_d^+(j) = \begin{cases} 1 & |j-k| \le d \le j+k-2 \text{ and } d \equiv j+k \ (2), \\ 0 & \text{otherwise,} \end{cases} \qquad a_d^-(j) = a_{2t-d}^+(j). \tag{8}$$

Consequently:

(W1) *each eigenspace is at most one dimensional in every chain slot, so a multiplicity-2 slot is exactly one $+$ line and one $-$ line;*

(W2) *the two strands are exchanged by the degree reversal $d \mapsto 2t-d$, that is $a_d^-(j) = a_{2t-d}^+(j)$ at every chain vertex — this is the second half of (8), and it is the statement referred to as* (W2) *elsewhere in the paper;*

(W3) *each strand occupies a contiguous run of degrees at every chain vertex, the $+$ strand the low ones and the $-$ strand the high ones, overlapping exactly in the multiplicity-2 window.*

We record the supports once and for all: for $1 \le j \le t$,

$$S^+(j) = \big[\, |j-k|,\ j+k-2 \,\big], \qquad S^-(j) = \big[\, 2t-(j+k-2),\ 2t-|j-k| \,\big], \tag{9}$$

both in steps of 2, degrees at j having the parity of $j+k$. We write $\mathrm{lo}_j^\varepsilon, \mathrm{hi}_j^\varepsilon$ for the endpoints, so

$$\mathrm{hi}_j^+ = j+k-2, \qquad \mathrm{hi}_j^- = 2t-|j-k|.$$

At the node, $S^+(t) \cup S^-(t) = [t-k,\, t+k]$, of dimension 2 on $[t-k+2,\, t+k-2]$ and of dimension 1 at exactly the two ends $d = t-k$ ($+$ alone) and $d = t+k$ ($-$ alone); this matches Lemma 4.2.

Proof of Proposition 5.1. Step 1: the eigen-refined recursion. The complex of Lemma 2.8 at a chain vertex j,

$$(I_\kappa)_j[d-1] \longrightarrow \bigoplus_{i \sim j} (I_\kappa)_i[d] \longrightarrow (I_\kappa)_j[d+1],$$

consists of τ-equivariant maps: the arrows between chain vertices commute with τ, and for $j = t$ the two leaf summands are exchanged by τ, so their sum $J[d] := (I_\kappa)_{n-1}[d] \oplus (I_\kappa)_n[d]$ is τ-stable. As $\mathrm{char} \neq 2$ the complex splits into eigen-parts, and exactness, together with the injectivity of the first map below the top degree (Lemma 2.8), is inherited by each summand. The pair $J[d]$ is the induced representation of $\mathbb{Z}/2$ on $(I_\kappa)_{n-1}[d]$, so each of its eigen-parts has dimension $b_d := \dim (I_\kappa)_{n-1}[d]$. Counting dimensions in the ε-part therefore gives, for every chain vertex j and every d with $0 \le d < 2t$ (the case $d = 0$ included: there the first map of the complex has source $(I_\kappa)_j[-1] = 0$ and the count is the trivial one, and it is needed to start the induction below),

$$a_{d+1}^\varepsilon(j) + a_{d-1}^\varepsilon(j) = \sum_{i \sim j,\ i \le t} a_d^\varepsilon(i) + [\, j = t \,]\, b_d, \tag{10}$$

while Lemma 2.8 at a leaf reads

$$b_{d+1} + b_{d-1} = a_d^+(t) + a_d^-(t). \tag{11}$$

The generator e_k spans $(I_\kappa)_k[0]$ and is τ-fixed, so $a_0^+(j) = \delta_{jk}$, $a_0^- \equiv 0$, $b_0 = 0$, and $a_{-1}^\pm = b_{-1} = 0$. Each equation expresses degree $d+1$ in terms of degrees d and $d-1$, so the system (10)–(11) with these initial data has *at most one* solution, determined by induction on d. It therefore suffices to check that (8) is a solution.

Step 2: the $+$ strand is the Dirichlet half-line wave. Let $A_d(j) := A_d^k(j)$ be the half-line wave of Lemma 3.9 with $\rho = k$; thus

$$A_d(j) = \big[\, |j-k| \le d \le j+k-2,\ d \equiv j+k \ (2) \,\big],$$

which is exactly the claimed $a_d^+(j)$ of (8), and by Lemma 3.9(i),(ii) it satisfies $A_{d+1}(j) + A_{d-1}(j) = A_d(j-1) + A_d(j+1)$ for every $j \ge 1$, with $A_d(0) = 0$, $A_0(j) = \delta_{jk}$ and $A_{-1} = 0$. Then (10) for $\varepsilon = +$ and $1 \le j \le t-1$ is the free recursion at j; and at $j = t$ it is the free recursion at t with the value $A_d(t+1)$ in the role of the leaf contribution. So the $+$ part of (10) holds, provided we show $b_d = A_d(t+1)$.

Step 3: the $-$ strand. Put $a_d^-(j) := A_{2t-d}(j)$. The recursion $X_{d+1}(j) + X_{d-1}(j) = X_d(j-1) + X_d(j+1)$ is invariant under $d \mapsto 2t-d$, so a^- satisfies (10) for $\varepsilon = -$ with the leaf value $A_{2t-d}(t+1)$. That this is the *same* leaf value as for the $+$ strand, i.e. that

$$A_d(t+1) = A_{2t-d}(t+1) \qquad \text{for every } d, \tag{12}$$

is the one symmetry the argument needs, and it is visible from the closed form: since $k \le t$, $|t+1-k| = t+1-k$ and $A_d(t+1) = [\, t+1-k \le d \le t+k-1 \,]$, an interval symmetric about t, so d lies in it if and only if $2t-d$ does; the parity condition $d \equiv t+1+k$ is likewise invariant under $d \mapsto 2t-d$. *Without (12) the $-$ part of (10) would not follow from the $+$ part.* Its initial data are right: $a_0^-(j) = A_{2t}(j)$ vanishes for $j \le t+1$ because

$A_{2t}(j) \neq 0$ would force $2t \leq j+k-2$, i.e. $j+k \geq 2t+2$, while $j \leq t+1$ and $k \leq t$ give $j+k \leq 2t+1$; and $a^-_{-1}(j) = A_{2t+1}(j) = 0$ likewise.

Step 4: the leaves. It remains to check that $A_d(t+1)$ satisfies (11), i.e. $A_{d+1}(t+1)+A_{d-1}(t+1) = a^+_d(t)+a^-_d(t)$. The free recursion at $j = t+1$ gives $A_{d+1}(t+1) + A_{d-1}(t+1) = A_d(t) + A_d(t+2)$, and $A_d(t) = a^+_d(t)$, so we must show $A_d(t+2) = a^-_d(t) = A_{2t-d}(t)$. Since $k \leq t$ we have $|t+2-k| = t+2-k$ and $|t-k| = t-k$, so

$$A_d(t+2) = [\,t+2-k \leq d \leq t+k\,], \qquad A_{2t-d}(t) = [\,t-k \leq 2t-d \leq t+k-2\,] = [\,t-k+2 \leq d \leq t+k\,],$$

which are the same interval; the parities agree too, both conditions reading $d \equiv t+k$ (2). Hence $A_\bullet(t+1)$ solves (11) and, by the uniqueness of Step 1, (8) holds.

Step 5: (W1) and (W3). Both are immediate from (8): each $a^\varepsilon_d(j) \in \{0,1\}$, and each strand occupies the interval (9). That $S^+(j)$ lies below $S^-(j)$ and that the overlap is the multiplicity-2 window is the statement $\mathrm{hi}^+_j = j+k-2 \leq \mathrm{hi}^-_j$ and $\mathrm{lo}^-_j = 2t+2-j-k \geq \mathrm{lo}^+_j = |j-k|$, both immediate for $j \leq t$, $k \leq t$. The overlap $W_j = [\,2t+2-j-k,\ j+k-2\,]$ has $o_j := \max(0,\, j+k-t-1)$ slots, in agreement with Lemma 4.2. □

Remark 5.2. This is the only place in the four sections §5, §6, §8, §9 where char $\neq 2$ is used — the rest of them use it only through Proposition 5.1 and its consequences — and it is used essentially: at char $= 2$ the operator τ is unipotent rather than semisimple and the splitting fails, so (W1)–(W4) are not available. They are also not repairable: see Remark 11.8. What §11 does instead is to dispense with them at $p = 2$ altogether.

We now prove that the arrows of I_κ transport each strand.

Lemma 5.3. *For either strand and every chain vertex $1 < j < t$, $\min\big(\mathrm{hi}^\varepsilon_{j-1}, \mathrm{hi}^\varepsilon_{j+1}\big) = \mathrm{hi}^\varepsilon_j - 1$.*

Proof. For $\varepsilon = +$, $\mathrm{hi}^+_j = j+k-2$ is affine in j, so the minimum is $\mathrm{hi}^+_{j-1} = \mathrm{hi}^+_j - 1$. For $\varepsilon = -$, $\mathrm{hi}^-_j = 2t-|j-k|$ and $\max(|j-1-k|, |j+1-k|) = |j-k|+1$. □

Lemma 5.4 (a death propagates)**.** *Let $1 < j < t$ and let x span the ε-line at (d,j). Suppose $a_{j\to i}x = 0$ for one chain neighbour i of j, the slot $(d+1,i)$ carrying ε. Write i' for the other chain neighbour. Then $a_{j\to i'}x$ spans the ε-line at $(d+1,i')$, the slot $(d+2,j)$ carries ε, and $a_{i'\to j}\big(a_{j\to i'}x\big) = 0$.*

Proof. As $j < t$ its only neighbours are i and i', so the ε-part of Lemma 2.8 (i.e. the injectivity of Y restricted to the ε-part) forces $a_{j\to i'}x \neq 0$; it lies in the ε-part of $(I_\kappa)_{i'}[d+1]$, which is at most a line by (W1), so that slot carries ε. Both slots $(d+1, j\pm1)$ therefore carry ε, whence $d+1 \leq \min(\mathrm{hi}^\varepsilon_{j-1}, \mathrm{hi}^\varepsilon_{j+1}) = \mathrm{hi}^\varepsilon_j - 1$ by Lemma 5.3, i.e. $d+2 \leq \mathrm{hi}^\varepsilon_j$; and $d+2 \geq d \geq \mathrm{lo}^\varepsilon_j$ with the parity unchanged, so $(d+2,j)$ carries ε. Finally the preprojective relation at j reads $\varsigma_1 a_{i\to j}a_{j\to i}x + \varsigma_2 a_{i'\to j}a_{j\to i'}x = 0$, whose first term vanishes. □

Proposition 5.5 (chain arrows towards the node)**.** *For either strand, $a_{j\to(j+1)}$ is nonzero on the ε-line at (d,j) whenever $(d+1, j+1)$ carries ε (here $1 \leq j \leq t-1$).*

Proof. Suppose $a_{j\to(j+1)}x = 0$. If $j = 1$ the only chain neighbour is 2 and the ε-part of Lemma 2.8 is contradicted at once. Otherwise Lemma 5.4 with $i = j+1$, $i' = j-1$ reproduces the same situation at $(d+1, j-1)$: its arrow towards the node dies and the target $(d+2,j)$ carries ε. Iterating lowers the vertex to 1. □

Proposition 5.6 (the folded leaf arrow)**.** *Let $J[d] = (I_\kappa)_{n-1}[d] \oplus (I_\kappa)_n[d]$. The induced map $J[d]^\varepsilon \to (I_\kappa)_t[d+1]^\varepsilon$ is nonzero whenever both sides are.*

Proof. The leaves are multiplicity-free (Lemma 4.2), so $(I_\kappa)_{n-1}[d]$ is a line, spanned by u say, and $J[d]^\varepsilon$ is spanned by $(u, \varepsilon\,\tau u)$, whose image is $(1+\varepsilon\tau)z$ with $z = a_{(n-1)\to t}u$. As $n-1$ has the single neighbour t, Lemma 2.8 gives $z \neq 0$, and the relation at $n-1$ reads $a_{t\to(n-1)}z = 0$. If $\dim(I_\kappa)_t[d+1] = 1$ the claim is exactly $z \neq 0$.

So let that slot be 2-dimensional and suppose z were a τ-eigenvector of eigenvalue ε. Applying τ to $a_{t\to(n-1)}z = 0$ gives $a_{t\to n}z = 0$, so both leaf arrows kill z; by Lemma 2.8 at t, $y := a_{t\to(t-1)}z \neq 0$, and y lies in the ε-part at $(d+2, t-1)$, which is therefore nonzero. The relation at t applied to z has both leaf terms zero, so $a_{(t-1)\to t}y = 0$. Finally $(d+3,t)$ carries ε: for $\varepsilon = +$ because $d+2 \leq \mathrm{hi}^+_{t-1} = \mathrm{hi}^+_t - 1$; for $\varepsilon = -$ because the slot $(d+1,t)$ then carries $+$ as well, so $d+1 \leq \mathrm{hi}^+_t = t+k-2$ and hence $d+3 \leq t+k = \mathrm{hi}^-_t$. Thus the arrow towards the node at $(d+2, t-1)$ dies with its target carrying ε, contradicting Proposition 5.5. So z is not an eigenvector and both of its eigen-components are nonzero. □

Proposition 5.7 (outward node arrows)**.** *Let z span the ε-eigenline of $(I_\kappa)_t[d]$ and let $l \in \{n-1, n\}$ satisfy $(I_\kappa)_l[d+1] \neq 0$. Then $a_{t\to l}z \neq 0$.*

Proof. Suppose $a_{t\to l}z = 0$.

(1) Both leaf arrows die. τ exchanges the leaves and the two arrows out of t, and $\tau z = \varepsilon z$; applying τ to $a_{t\to(n-1)}z = 0$ gives $\varepsilon\, a_{t\to n}z = 0$, so both leaf arrows kill z (and symmetrically).

(2) The chain arrow cannot die too. If $a_{t\to(t-1)}z = 0$ as well then all three arrows out of t kill z, so $\mathbb{K}z$ is a copy of S_t inside $\operatorname{soc}(I_\kappa) = S_\kappa$, which sits in the top degree $2t$; hence $d = 2t$. But the leaves have top degree $t + k - 1 \le 2t - 1$, so $(I_\kappa)_l[d+1] = 0$, against the hypothesis.

(3) So $y := a_{t\to(t-1)}z \neq 0$. It sits at $(d+1, t-1)$, and τ commutes with $a_{t\to(t-1)}$, so y is again an ε-eigenvector; by (W1) the slot $(d+1, t-1)$ carries ε. The relation at t,

$$\varsigma_1 a_{(t-1)\to t}a_{t\to(t-1)}z + \varsigma_2 a_{(n-1)\to t}a_{t\to(n-1)}z + \varsigma_3 a_{n\to t}a_{t\to n}z = 0,$$

has its last two terms zero by (1), so $a_{(t-1)\to t}y = 0$.

(4) By Proposition 5.5 applied to the arrow $(t-1)\to t$ at degree $d+1$, the slot $(d+2, t)$ does *not* carry ε, i.e. $d = \mathrm{hi}_t^{\varepsilon}$.

(5) Both values of ε are impossible. For $\varepsilon = +$, $\mathrm{hi}_t^{+} = t + k - 2$, so $d + 1 = t + k - 1 > t + k - 3 = \mathrm{hi}_{t-1}^{+}$ and the slot $(d+1, t-1)$ does not carry $+$, contradicting (3). For $\varepsilon = -$, $\mathrm{hi}_t^{-} = t + k$, so $d + 1 = t + k + 1$ exceeds the leaf top degree $t + k - 1$ and $(I_\kappa)_l[d+1] = 0$, contradicting the hypothesis. □

Corollary 5.8 ((W4))**.** *Every arrow of I_κ restricts to an isomorphism between consecutive nonzero slots of the same strand. Precisely: for every arrow $a_{j\to i}$ and every degree d such that the slots (d, j) and $(d+1, i)$ both carry ε (with the folded convention $J[\cdot]^{\varepsilon}$ at the leaves), the map $a_{j\to i}$ is nonzero, hence an isomorphism, on the ε-parts.*

Proof. There are four classes of arrow.

Chain inward (towards the node), $j \to j+1$ with $j \le t-1$: Proposition 5.5.

Leaf inward, $l \to t$: Proposition 5.6.

Node outward, $t \to l$: Proposition 5.7.

Chain outward (away from the node), $j \to j-1$ with $2 \le j \le t$: suppose the ε-line at (d, j) is killed, the slot $(d+1, j-1)$ carrying ε. If $j < t$, Lemma 5.4 (with $i = j-1$, $i' = j+1$) produces the same situation at $(d+1, j+1)$, again with a dying outward arrow; iterating raises the vertex to t. So we may assume $j = t$: the ε-line at (D, t) is killed by $a_{t\to(t-1)}$ and $(D+1, t-1)$ carries ε. The relation at t then makes the composite $J[D+1]^{\varepsilon} \to (I_\kappa)_t[D+2]^{\varepsilon}$ vanish on the image of the ε-line, while Lemma 2.8 makes that image nonzero in J^{ε}; by Proposition 5.6 this is a contradiction *provided* the slot $(D+2, t)$ carries ε. We check that it does, except in one case which we handle separately.

The slot $(D+1, t-1)$ carries ε, so $D + 1 \le \mathrm{hi}_{t-1}^{\varepsilon}$. For $\varepsilon = +$, $\mathrm{hi}_{t-1}^{+} = \mathrm{hi}_t^{+} - 1$, so $D + 2 \le \mathrm{hi}_t^{+}$ and $(D+2, t)$ carries $+$: no exception. For $\varepsilon = -$, $\mathrm{hi}_{t-1}^{-} = 2t - |t - 1 - k|$ and $\mathrm{hi}_t^{-} = 2t - (t - k) = t + k$; the condition $D + 1 \le \mathrm{hi}_{t-1}^{-}$ reads $D \le \mathrm{hi}_t^{-}$ (when $k \le t - 1$; when $k = t$ it reads $D \le \mathrm{hi}_t^{-} - 2$), while $(D+2, t)$ carries $-$ iff $D \le \mathrm{hi}_t^{-} - 2$. So the *only* uncovered arrival is $D = \mathrm{hi}_t^{-} = t + k$, and it occurs only for $k \le t - 1$.

At $D = t + k$ we argue directly and do not use Proposition 5.6 at all. The leaves are multiplicity-free with top degree $t + k - 1$ (Lemma 4.2), so $(I_\kappa)_{n-1}[D+1] = (I_\kappa)_n[D+1] = 0$, since $D + 1 = t + k + 1 > t + k - 1$. If in addition $a_{t\to(t-1)}z = 0$ for the vector z spanning the $-$line at (D, t), then all three arrows out of t kill z, so $\mathbb{K}z$ is a copy of S_t inside $\operatorname{soc}(I_\kappa) = S_\kappa$, which lies in the top degree $2t$. But $D = t + k < 2t$, because this configuration requires $k \le t - 1$. Contradiction. □

Remark 5.9. The last paragraph is the same socle argument that proves Lemma 2.8, and it is characteristic-free. It is needed: the target slot $(D+2, t)$ really is empty at $D = \mathrm{hi}_t^{-}$, so the appeal to Proposition 5.6 would be vacuous there.

6. Ranks of arrows between chain slots

By Corollary 5.8 the rank of an arrow $(d, j) \to (d+1, i)$ between chain slots is the number of strands carried by *both* slots. The following three consequences are what the closure arguments of §9 use. We call the $+$ line of a multiplicity-2 chain slot its *τ-eigenline* (it is the $\tau = +1$ eigenspace).

Proposition 6.1. *Let $i, j \le t$ be adjacent chain vertices and d a degree. Then:*

(T0) $\operatorname{rank}\big(a_{j\to i} : (I_\kappa)_j[d] \to (I_\kappa)_i[d+1]\big) = \min\big(\dim(I_\kappa)_j[d],\ \dim(I_\kappa)_i[d+1]\big)$;

(T1) *if* $\dim(I_\kappa)_j[d] = 1$ *and* $\dim(I_\kappa)_i[d+1] = 2$ *then the image of $a_{j\to i}$ is the τ-eigenline of $(I_\kappa)_i[d+1]$;*

(T2) *if* $\dim(I_\kappa)_j[d] = 2$ *and* $\dim(I_\kappa)_i[d+1] = 1$ *then $a_{j\to i}$ annihilates the τ-eigenline of $(I_\kappa)_j[d]$.*

Moreover the τ-eigenline of a multiplicity-2 chain slot is not *annihilated by an arrow into a 2-dimensional chain slot (Corollary 5.8), nor by an arrow into a nonzero leaf slot (Proposition 5.7).*

Proof. By Corollary 5.8 the rank is $|\Sigma(d,j)\cap\Sigma(d+1,i)|$, where Σ denotes the set of strands present. So (T0) says exactly that the two sets $\Sigma(d,j),\Sigma(d+1,i)$ are *nested*; and (T1), (T2) say that in the two mixed cases the single strand present at the 1-dimensional slot is $+$, respectively $-$. All three are therefore statements about the intervals (9), and we verify them. Throughout $i,j,k\le t$ and $|i-j|=1$; degrees at j have the parity of $j+k$.

(T0). Suppose the sets are not nested; then one slot carries only $+$ and the other only $-$.

Case A: $+$ only at (d,j), $-$ only at $(d+1,i)$. From $+$ at (d,j), $d\ge|j-k|\ge|i-k|-1$, so $d+1\ge|i-k|$; hence the failure of $+$ at $(d+1,i)$ must be $d+1>i+k-2$, i.e. $d\ge i+k-2$, while $+$ at (d,j) gives $d\le j+k-2$. If $i=j+1$ this reads $d\le i+k-3<i+k-2\le d$, absurd. If $i=j-1$ then $d\in\{i+k-2,i+k-1\}$ and the parity $d\equiv j+k=i+k+1$ forces $d=i+k-1=j+k-2$; now $-$ at $(d+1,i)=(i+k,i)$ needs $i+k\ge 2t-i-k+2$, i.e. $i+k\ge t+1$, while the absence of $-$ at (d,j) needs $j+k-2<2t-j-k+2$, i.e. $j+k\le t+1$ — impossible for $j=i+1$.

Case B: $-$ only at (d,j), $+$ only at $(d+1,i)$. From $+$ at $(d+1,i)$, $d\le i+k-3$. The absence of $+$ at (d,j) is either $d>j+k-2$, which with $d\le i+k-3$ forces $j<i-1$, impossible; or $d<|j-k|$, in which case $-$ at (d,j) gives $2t-j-k+2\le d<|j-k|$, reading $2t+2<2j$ if $j\ge k$ and $2t+2<2k$ if $j<k$, both impossible.

(T1). Here the source carries one strand and the target two. Suppose the source strand were $-$. This is *not* the configuration of Case B: there the target carried $+$ only, whereas here it carries both strands. But the proof of Case B never uses the absence of $-$ at $(d+1,i)$. Reading it again, it uses exactly three facts — that $+$ is present at $(d+1,i)$, that $+$ is absent at (d,j), and that $-$ is present at (d,j) — and derives a contradiction from those alone. All three hold in the present situation: the target is 2-dimensional, so it carries $+$; and the source is 1-dimensional carrying $-$, so it carries $-$ and not $+$. The same contradiction therefore applies, and the source strand is $+$. By Corollary 5.8 the image is then the $+$ line of the target.

(T2). Here the source carries both strands and the target one; suppose the target strand were $+$. The absence of $-$ at $(d+1,i)$ is either $d\le 2t-i-k$, which against $d\ge 2t-j-k+2$ (from $-$ at (d,j)) gives $i\le j-2$; or $d\ge 2t-|i-k|$, which against $d\le j+k-2$ (from $+$ at (d,j)) gives $2t+2\le i+j$ when $i\ge k$ and $2t+2\le j-i+2k\le 2t+1$ when $i<k$. All are impossible. So the target carries $-$ only, and the $+$ line of the source dies. □

Remark 6.2. Proposition 6.1 is stated and proved only for arrows *between chain vertices*; its proof is entirely the interval combinatorics of $S^\varepsilon(i),S^\varepsilon(j)$ with $i,j\le t$, and it says nothing about an arrow out of, or into, a leaf. The one leaf arrow needed later, namely the injectivity of $a_{l\to t}$ on a leaf slot, is Lemma 2.8 verbatim: at a leaf l the only neighbour is t, so the map Y of Lemma 2.8 *is* $a_{l\to t}$, and the leaf support tops out at $t+k-1\le 2t-1$, strictly below the top degree $2t$, so the hypothesis of Lemma 2.8 holds at every leaf slot without exception. Since a leaf slot is at most a line (Lemma 4.2), the image of $a_{l\to t}$ on a nonzero leaf slot is a line.

7. Removing the word: the statement $(*^\circ)$ and its representatives

This section is characteristic-free. The hypothesis char $\neq 2$ declared at the head of §5 is suspended here and is used nowhere in this section: Lemma 7.2 rests on Theorem 3.6, Corollary 7.4 on Lemma 7.2, Proposition 7.7 is sign-vector combinatorics in the Weyl orbit, and (14) is Corollary 2.13 — all proved over an arbitrary field. This matters: §11 quotes Definition 7.1, Corollary 7.4, Proposition 7.7, Definition 7.9 and (14) at $p=2$. The hypothesis resumes at the head of §8.

The colour-t step of the induction of §10 needs to know that the partial module is an up-set at the chain neighbour $t-1$. Since the partial module is itself extremal (Corollary 3.8), the requirement can be stated with no reference to w or to any order of construction.

Definition 7.1. $(*^\circ)$ is the statement: *every extremal submodule $M\subseteq I_\kappa$ with $\langle\alpha_t^\vee,\lambda-\underline{\dim}\,M\rangle\ge 1$ is an up-set at $t-1$.*

The side condition is essential and cannot be dropped: there exist extremal modules that are not up-sets at $t-1$, and every one of them has $\langle\alpha_t^\vee,\lambda-\underline{\dim}\,M\rangle=0$ (Proposition 7.5; the existence is exhibited in §14). So any reduction of $(*^\circ)$ to special weights must preserve the side condition.

7.1. The parabolic. For $\mu\in W\lambda$ write $c_t(\mu)=\langle\alpha_t^\vee,\mu\rangle$. In D_n the vertices adjacent to $t=n-2$ are exactly $t-1$, $n-1$ and n. Put

$$J' \;=\; I\setminus\{t-1,\,t,\,n-1,\,n\} \;=\; \{1,2,\dots,t-2\}.$$

Lemma 7.2. *Let $\mu \in W\lambda$.*
(a) *For $j \in J'$ we have $c_t(s_j\mu) = c_t(\mu)$.*
(b) *For any $j \in I$ that is a descent of μ (i.e. $\langle \alpha_j^\vee, \mu\rangle \le -1$), the modules $V(\mu)$ and $V(s_j\mu)$ have the same graded profile at every vertex other than j. In particular, for $j \in J'$ they agree at $t-1$ and at t, and for $j = t$ they agree at $t-1$.*

Proof. (a) For $j \in J'$ the simple roots α_j and α_t are orthogonal, so $s_j\alpha_t^\vee = \alpha_t^\vee$ and $c_t(s_j\mu) = \langle s_j\alpha_t^\vee, \mu\rangle = c_t(\mu)$. (b) Let $\nu = s_j\mu$; since j is a descent, $\langle \alpha_j^\vee, \nu\rangle \ge 1$ and ν is shorter, and by Theorem 3.6 $V(\mu) = \operatorname{soc}_{(j)} V(\nu)$. A socle step of colour j enlarges a submodule only at vertex j. This argument is independent of j. For $j \in J'$ one has $t-1 \notin J'$ and $t \notin J'$, so $j \ne t-1, t$; and $t \ne t-1$. □

Remark 7.3. This is the only choice of parabolic that works. For the larger parabolic $J = I \setminus \{t-1, t\}$, which also leaves the profile at $t-1$ unchanged, the side condition is *not* invariant: for a descent $j \in \{n-1, n\}$, which is adjacent to t, $c_t(s_j\mu) = c_t(\mu) + \langle \alpha_j^\vee, \mu\rangle \le c_t(\mu) - 1$, so the W_J-minimal representative of an orbit can fall outside the hypothesis set — and does so for the large majority of instances (§14). With J' the side condition is literally invariant and the reduction is closed by construction.

Corollary 7.4. $(*^\circ)$ *holds for all μ if and only if it holds for every $W_{J'}$-minimal $\mu \in W\lambda$ with $c_t(\mu) \ge 1$.*

Proof. Let μ satisfy $c_t(\mu) \ge 1$ and let μ_0 be the $W_{J'}$-minimal element of $W_{J'}\mu$. Any chain $\mu \to s_{j_1}\mu \to \cdots \to \mu_0$ of descents $j_i \in J'$ keeps $c_t \ge 1$ and keeps the profile at $t-1$ unchanged, by Lemma 7.2. Being an up-set at $t-1$ depends only on that profile. □

7.2. Sharpness of the side condition. $(*^\circ)$ says that $c_t \ge 1$ forces the up-set property at $t-1$. The converse-shaped statement — that the side condition is exactly right, i.e. that the extremal modules which fail to be up-sets at $t-1$ are confined to the single value $c_t = 0$ — is also true, and is a short consequence of $(*^\circ)$ itself together with Lemma 7.2(b).

Proposition 7.5 (sharpness)**.** *Granted* $(*^\circ)$ (*proved in §9*)*: every extremal submodule $M \subseteq I_\kappa$ that is* not *an up-set at $t-1$ satisfies $\langle \alpha_t^\vee, \lambda - \underline{\dim}\, M\rangle = 0$.*

Proof. Put $\mu = \lambda - \underline{\dim}\, M \in W\lambda$; by Corollary 2.13 the extremal submodule with that dimension vector is unique, so $M = V(\mu)$, and $c_t(M) = c_t(\mu)$. If $c_t(\mu) \ge 1$ then $(*^\circ)$ makes M an up-set at $t-1$, contrary to hypothesis; so $c_t(\mu) \le 0$.

Suppose $c_t(\mu) \le -1$. Then t is a descent of μ, so Lemma 7.2(b) applies with $j = t$: writing $\nu = s_t\mu \in W\lambda$, the modules $V(\mu)$ and $V(\nu)$ have the same graded profile at every vertex other than t, in particular at $t-1$. But $c_t(\nu) = \langle \alpha_t^\vee, s_t\mu\rangle = -\langle \alpha_t^\vee, \mu\rangle = -c_t(\mu) \ge 1$, and $V(\nu)$ is extremal, so $(*^\circ)$ makes $V(\nu)$ an up-set at $t-1$. Being an up-set at $t-1$ depends only on the graded profile at $t-1$, so $V(\mu) = M$ is an up-set at $t-1$ — again contrary to hypothesis.

Hence $c_t(\mu) = 0$. □

Remark 7.6. The value $c_t \le -1$ is not vacuous: since $c_t(\mu) = \eta_{n-2} - \eta_{n-1}$ in the coordinates of Proposition 7.7, c_t takes all five values $-2, \dots, 2$ on $W\omega_k$, and over D_4–D_8 exactly 2565 of the 7464 extremal modules have $c_t \le -1$ (§14.2). So Proposition 7.5 is not the triviality "$c_t \ge 0$ always" — that statement is *false*, and $(*^\circ)$ alone gives only $c_t \le 0$. The work is done by the reflection s_t, which converts a negative c_t into a positive one at no cost at $t-1$ precisely because the socle step of colour t moves nothing at $t-1$.

7.3. The representatives.

Proposition 7.7. *Let $2 \le k \le n-2$. The $W_{J'}$-minimal $\mu \in W\omega_k$ with $c_t(\mu) \ge 1$ are exactly the weights*

$$\mu(h; x, y, z) \;=\; e_1 + \cdots + e_h \;-\; (e_{n-2-c} + \cdots + e_{n-3}) \;+\; x\, e_{n-2} + y\, e_{n-1} + z\, e_n, \tag{13}$$

where $(x, y, z) \in \{0, \pm 1\}^3$ satisfies $x - y \ge 1$, and $h, c \ge 0$ with $h + c = k - q$, $q := \#\{\text{nonzero among } x, y, z\}$. The admissible triples are the nine

$$\begin{array}{ccc} (1,0,0) & (1,0,1) & (1,0,-1) \\ (1,-1,0) & (1,-1,1) & (1,-1,-1) \\ (0,-1,0) & (0,-1,1) & (0,-1,-1), \end{array}$$

with $q = 1$ for two of them, $q = 2$ for five and $q = 3$ for two, and their number is

$$\sum_q N_q\,(k - q + 1) = 2k + 5(k-1) + 2(k-2) = 9(k-1),$$

independently of n, *and with no exception at* $k = n-2$. *The automorphism* τ *acts by* $z \mapsto -z$, *pairing the nine triples into six* τ*-classes.*

Proof. For $k \le n-2$ one has $W\omega_k = \{\sum_{i\in S} \eta_i e_i : |S| = k,\ \eta_i = \pm 1\}$, so a weight is a vector $(\eta_1, \dots, \eta_n)$ with $\eta_i \in \{0, \pm 1\}$ and exactly k nonzero entries. The pairings are $\langle \alpha_j^\vee, \mu\rangle = \eta_j - \eta_{j+1}$ for $j \le n-1$ and $\langle \alpha_n^\vee, \mu\rangle = \eta_{n-1} + \eta_n$.

$W_{J'}$ is generated by $s_1, \dots, s_{t-2} = s_{n-4}$, hence is the symmetric group permuting the coordinates $\eta_1, \dots, \eta_{n-3}$ and fixing $\eta_{n-2}, \eta_{n-1}, \eta_n$. Minimality in the orbit means no descent in J', i.e. $\eta_j \ge \eta_{j+1}$ for $1 \le j \le n-4$: the sequence $(\eta_1, \dots, \eta_{n-3})$ is weakly decreasing, hence of the form $1^h 0^b (-1)^c$ with $h+b+c = n-3$. That is precisely the displayed shape, with $x = \eta_{n-2}$, $y = \eta_{n-1}$, $z = \eta_n$. The side condition is $c_t(\mu) = \eta_{n-2} - \eta_{n-1} = x - y \ge 1$, which with $x, y \in \{0, \pm 1\}$ leaves $(x, y) \in \{(1, 0), (1, -1), (0, -1)\}$ and z free, i.e. the nine listed triples. Counting nonzero entries gives $h + c + q = k$, so $h + c = k - q$ and h ranges over $0, \dots, k-q$, which is $k - q + 1$ values; and $b = n - 3 - h - c = n - 3 - k + q \ge 0$ since $k \le n-2$ and $q \ge 1$. Summing over the $N_1 = 2$, $N_2 = 5$, $N_3 = 2$ triples gives $9(k-1)$, uniformly in n. When $k = n-2$ the value $b = 0$ is attained and the triple $(h; x, y, z) = (k-1; 1, 0, 0)$ gives $\mu = e_1 + \cdots + e_k = \lambda$; this is a *genuine* member of the list and nothing is to be subtracted for it, since λ is $W_{J'}$-minimal (its Bourbaki coordinates $1^k 0^{n-k}$ are weakly decreasing) and satisfies the side condition, $c_t(\lambda) = \eta_{n-2} - \eta_{n-1} = [\,k = n-2\,] - 0 = 1 \ge 1$; so nothing is subtracted at $k = n-2$. Finally τ fixes e_i for $i \le n-1$ and negates e_n, so it sends $\mu(h; x, y, z)$ to $\mu(h; x, y, -z)$. □

Remark 7.8. The τ-fixed family $(x, y, z) = (1, 0, 0)$ consists of the k weights $\mu_h = e_1 + \cdots + e_h - (e_{n-2-c} + \cdots + e_{n-3}) + e_{n-2}$, $h + c = k - 1$. With one exception these are exactly the weights whose *only* descent is $t-1$; the exception is $k = n-2$, $h = k-1$, where $e_{n-2} = e_k$ and $\mu_{k-1} = e_1 + \cdots + e_k = \lambda$, which is dominant and has no descent at all. (Proposition 7.7 states the list correctly in either case: the condition there is $W_{J'}$-minimality together with $c_t(\mu) \ge 1$, and λ satisfies both when $k = n-2$.) They are the W_J-minimal weights for the larger parabolic $J = I \setminus \{t-1, t\}$, and they are not enough: the reduction through J is not closed, and the other $8k - 9$ representatives are genuinely new.

The statement we shall prove at the representatives is stronger than $(*^\circ)$ requires.

Definition 7.9. (B) is the statement: *for every* μ *of Proposition 7.7, the graded multiplicities of* $V(\mu)$ *are the greedy top truncation of those of* I_κ *at every vertex.*

A greedy top truncation is an up-set, so (B) gives $(*^\circ)$ at the representatives, and hence, by Corollary 7.4, in general. By Corollary 2.13, (B) is equivalent to:

for each representative μ, *some* submodule of I_κ has the greedy top truncation as its graded profile. (14)

Indeed, such a submodule has dimension vector $v = \lambda - \mu$, and by rigidity it is then the unique one, namely $V(\mu)$. *This removes the socle construction from the problem completely*: what remains is to exhibit a subspace of I_κ, closed under all arrows, whose graded profile is greedy from the top. That is done in §9.

8. The nine families

From here to the end of §9 we again assume $\operatorname{char}\mathbb{K} \ne 2$, as declared at the head of §5: everything below rests on the strand decomposition.

Fix a representative $\mu = \mu(h; x, y, z)$ as in (13), write $q = \#\{\text{nonzero among } x, y, z\}$, $b = n - 3 - k + q$ (the number of zero coordinates among $\eta_1, \dots, \eta_{n-3}$), $c = k - q - h$, and $v = \lambda - \mu$ in root coordinates. Everything below is controlled by the two integers $q + x \bmod 2$ and z.

8.1. The dimension vector.

Lemma 8.1. *With the above notation,*

$$v_j = \big(\min(j, k) - h\big)^+ + \big(j - h - b\big)^+ \quad (1 \le j \le t-1), \qquad v_t = 2k - 2h - q - x, \tag{15}$$

and

$$v_{n-1} = \tfrac{v_t - y + z}{2}, \qquad v_n = \tfrac{v_t - y - z}{2}. \tag{16}$$

In particular $v(h; x, y, -z)$ *is* $v(h; x, y, z)$ *with the two leaf entries exchanged, and every chain entry is independent of* z.

Proof. Write $X_p = \lambda_p - \mu_p$. From (13),

$$X_p = 0 \ (p \le h), \quad X_p = [p \le k] \ (h < p \le h+b), \quad X_p = [p \le k] + 1 \ (h+b < p \le n-3),$$

and $X_{n-2} = [k = t] - x$, $X_{n-1} = -y$, $X_n = -z$. Comparing coefficients of e_p in $\sum_j v_j \alpha_j$ gives $v_j = X_1 + \cdots + X_j$ for $j \le t$, whence the first formula of (15) for $j \le n-3 = t-1$. At $j = t$,

$$v_t = v_{t-1} + [k = t] - x.$$

If $k \le t-1$ then $v_{t-1} = (k-h) + (t-1-h-b) = (k-h) + c = 2k - 2h - q$ (using $t-1-h-b = k-h-q = c$), and $v_t = 2k - 2h - q - x$. If $k = t$ then $v_{t-1} = (t-1-h) + c = 2k - 2h - q - 1$ and $v_t = v_{t-1} + 1 - x = 2k - 2h - q - x$. For the leaves, the coefficient of e_{n-1} in $\sum_j v_j \alpha_j$ is $-v_t + v_{n-1} + v_n$ and that of e_n is $-v_{n-1} + v_n$; equating with $X_{n-1} = -y$, $X_n = -z$ gives $v_{n-1} + v_n = v_t - y$ and $v_{n-1} - v_n = z$, i.e. (16). □

8.2. Occupation numbers. For $1 \le j \le t$ put

$$\mathrm{m}_j = \min(j, k), \qquad o_j = \max(0,\ j + k - t - 1) = \max(0,\ j - B), \quad B := n - 1 - k, \qquad u_j = v_j - \mathrm{m}_j.$$

By Proposition 5.1, each strand has m_j slots at j, of which o_j lie in the multiplicity-2 window W_j; above W_j there are $\mathrm{m}_j - o_j$ minus-only slots and below it $\mathrm{m}_j - o_j$ plus-only slots. Taking v_j dimensions greedily from the top exhausts the minus-only slots first, then fills window slots; writing $r_j := v_j - (\mathrm{m}_j - o_j) = u_j + o_j$, it fills $\lfloor r_j/2 \rfloor$ window slots completely and, if r_j is odd and $r_j < 2o_j$, leaves exactly one window slot half filled. Consequently the numbers of occupied slots in the two strands are

$$s_j^+ = \max\Big(0,\ u_j,\ \Big\lceil \frac{u_j + o_j}{2} \Big\rceil\Big), \qquad s_j^- = v_j - s_j^+, \tag{17}$$

and in each strand the occupied slots are the top ones. We also set

$$g_j := s_j^+ - u_j = \max\Big(0,\ -u_j,\ \Big\lceil \frac{o_j - u_j}{2} \Big\rceil\Big), \tag{18}$$

the *dual occupation*, obtained from s^+ by $u \mapsto -u$. Then identically $s_j^- = \mathrm{m}_j - g_j$.

Lemma 8.2. *Define, for every chain vertex j and either strand,*

$$E_j^\varepsilon := \mathrm{hi}_j^\varepsilon - 2s_j^\varepsilon + 2.$$

When $s_j^\varepsilon > 0$ this is the lowest occupied degree of strand ε at j; when $s_j^\varepsilon = 0$ it equals $\mathrm{hi}_j^\varepsilon + 2$, which lies above the whole support $S^\varepsilon(j)$. In either case, a slot (d, j) of strand ε is occupied if and only if $d \ge E_j^\varepsilon$, *and*

$$E_j^+ = j + k - 2u_j - 2g_j, \qquad E_j^- = 2t + 2 - j - k + 2g_j.$$

Proof. The occupied slots of strand ε at j are its top s_j^ε slots, i.e. the degrees $d \in S^\varepsilon(j)$ with $d \ge \mathrm{hi}_j^\varepsilon - 2(s_j^\varepsilon - 1) = E_j^\varepsilon$; for $s_j^\varepsilon = 0$ this condition is $d \ge \mathrm{hi}_j^\varepsilon + 2$, which no slot satisfies, as required. The formulas: $E_j^+ = (j + k - 2) - 2s_j^+ + 2 = j + k - 2s_j^+ = j + k - 2u_j - 2g_j$ since $s_j^+ = u_j + g_j$. For the minus strand, $s_j^- = \mathrm{m}_j - g_j$ and $E_j^- = 2t - |j - k| - 2(\mathrm{m}_j - g_j) + 2$; if $j \le k$ this is $2t - k + j - 2j + 2g_j + 2 = 2t + 2 - j - k + 2g_j$, and if $j \ge k$ it is $2t - j + k - 2k + 2g_j + 2 = 2t + 2 - j - k + 2g_j$. Note that no case distinction on s_j^ε was made, so both formulas are valid unconditionally. □

Proposition 8.3 (the closure criterion, strandwise)**.** *Let occupations $s_j^\pm$ ($0 \le s_j^\varepsilon \le \mathrm{m}_j$) be given at every chain vertex j, defining an occupied set of strand slots as in Lemma 8.2, and suppose that at every chain edge $j \to j+1$*

$$\Delta s_j^+ = \Delta u_j + \Delta g_j \in \{0, 1\} \qquad \textit{and} \qquad \Delta g_j \in \{0, 1\}, \tag{19}$$

where $\Delta f_j := f_{j+1} - f_j$. Then, for either strand ε, either direction of the edge, and every degree d:

if the ε-slot at (d, j) is occupied and the slot $(d+1, i)$ $(i = j \pm 1)$ carries ε, then that slot is occupied too.

Consequently the graded subspace which at each chain vertex is the direct sum of the top s_j^+ slots of the $+$ strand and the top s_j^- slots of the $-$ strand is carried into itself by every chain arrow; and no obligation runs from a taken *chain slot to an* untaken *nonzero chain slot, except possibly from a slot in which only one strand is occupied to a slot carrying only the other strand.*

Proof. By Lemma 8.2 the two bounds (19) say exactly

$$E^+_{j+1} - E^+_j = 1 - 2\Delta s^+_j \in \{-1, 1\}, \qquad E^-_{j+1} - E^-_j = -1 + 2\Delta g_j \in \{-1, 1\},$$

so $|E^\varepsilon_{j+1} - E^\varepsilon_j| \le 1$ for both strands. Now let the ε-slot at (d, j) be occupied, so $d \ge E^\varepsilon_j$, and let $(d+1, i)$ carry ε with $i = j \pm 1$. Then $d + 1 \ge E^\varepsilon_j + 1 \ge E^\varepsilon_i$, so the target is occupied. The same argument with the roles of j and i exchanged covers the other direction.

For the two consequences: by Corollary 5.8 a chain arrow maps the ε-line of (d, j) isomorphically onto that of $(d + 1, i)$ when both slots carry ε, and to 0 otherwise, so the displayed statement is precisely closure under chain arrows. Finally, if (d, j) is taken in strand ε and $(d + 1, i)$ is a nonzero chain slot that carries ε, the displayed statement makes it taken; so an untaken nonzero target forces $(d + 1, i)$ not to carry ε, which at a chain slot means it carries only $-\varepsilon$. If the source is full it carries $-\varepsilon$ as well and is occupied there, so the target is taken after all. □

Remark 8.4 (why this form, and why it applies to T^l)**.** Two points. First, E^ε_j is now *defined at every chain vertex*, including those where $s^\varepsilon_j = 0$, for which reading E^ε_j as "the lowest occupied degree" is ill-posed on an unoccupied strand, and the statement of Proposition 8.3 is asserted at every chain edge. Lemma 8.2 shows that the two closed formulas hold unconditionally, which is what makes the uniform statement legitimate. Second, only the *sufficiency* of (19) is used anywhere in this paper, and the conclusion as stated is a statement about *which slots are occupied*, not about which line sits in a partial slot. This matters because Theorem 9.4 applies the conclusion to $T^l(v)$, which is *not* a sum of two strand top-segments; see Lemma 9.2.

Remark 8.5. It is tempting, and wrong, to phrase the minus condition as $|\Delta s^-_j| \le 1$. Since $\mathrm{hi}^-_j = 2t - |j - k|$ is a tent rather than a ramp, the correct condition is one-sided and changes side at $j = k$: $\Delta s^- \in \{0, 1\}$ for $j < k$ and $\Delta s^- \in \{-1, 0\}$ for $j \ge k$. The symmetric bound $|\Delta s^-| \le 1$ is strictly weaker on the wrong side of the tent, and is not equivalent to up-closedness. The formulation of Proposition 8.3 in terms of the thresholds E^ε_j of Lemma 8.2 — equivalently, of Δs^+ and Δg — is uniform in ε and has no case split; the asymmetry lives entirely inside hi^ε.

8.3. The two increment bounds.

Lemma 8.6. *For every one of the nine families:*
(i) $\Delta u_j \in \{-1, 0, 1\}$ *and* $\Delta o_j \in \{0, 1\}$ *at every chain edge* $j \to j + 1$, $1 \le j \le t - 1$*;*
(ii) $\Delta u_j = -1$ *occurs only for* $j + 1 \le h$*, or for* $j + 1 = t$ *with* $x = 1$*;*
(iii) $\Delta u_j = +1$ *forces* $j \ge h + b$*;*
(iv) $u_{t-1} \le o_{t-1} + 1$.

Proof. For $j + 1 \le t - 1$, (15) gives $v_{j+1} - v_j = [h < j + 1 \le k] + [j + 1 > h + b] \in \{0, 1, 2\}$ and $\mathrm{m}_{j+1} - \mathrm{m}_j = [j + 1 \le k]$, so $\Delta u_j = [h < j + 1 \le k] + [j + 1 > h + b] - [j + 1 \le k] \in \{-1, 0, 1\}$; it is -1 only when $[h < j + 1 \le k] = 0 = [j + 1 > h + b]$ and $[j + 1 \le k] = 1$, i.e. $j + 1 \le h$. It is $+1$ only when $[j + 1 > h + b] = 1$, i.e. $j \ge h + b$. At the last edge $j + 1 = t$, Lemma 8.1 gives $v_t - v_{t-1} = [k = t] - x$ and $\mathrm{m}_t - \mathrm{m}_{t-1} = [t \le k] = [k = t]$, so $\Delta u_{t-1} = -x$. That proves (i)–(iii); $o_j = \max(0, j - B)$ is nondecreasing with steps in $\{0, 1\}$.

For (iv): $o_{t-1} = \max(0, t - 1 + k - t - 1) = k - 2$ since $k \ge 2$. If $k \le t - 1$ then $u_{t-1} = v_{t-1} - k = (2k - 2h - q) - k = k - 2h - q$; if $k = t$ then $u_{t-1} = v_{t-1} - (t - 1) = (2k - 2h - q - 1) - (k - 1) = k - 2h - q$ as well. So $u_{t-1} - o_{t-1} = 2 - 2h - q \le 1$ because $q \ge 1$. □

Proposition 8.7 (increasing bound)**.** $\Delta u_j + \Delta g_j = \Delta s^+_j \in \{0, 1\}$ *at every chain edge, for every one of the nine families.*

Proof. Upper bound. In (17) the three arguments are 0 (constant), u_j (rising by $\Delta u \le 1$) and $\lceil (u_j + o_j)/2 \rceil$ (rising by $\lceil (\Delta u + \Delta o)/2 \rceil \le 1$, since $\Delta u + \Delta o \le 2$). A maximum of finitely many quantities each rising by at most 1 rises by at most 1.

Lower bound. A maximum can only fall if some argument falls, so by Lemma 8.6 we need only examine the two exceptional steps.

(a) $j + 1 \le h$. Then $j < h \le k - q < k$, so $\mathrm{m}_j = j$ and $v_j = 0$, giving $u_j = -j$. Also $j \le h - 1 \le k - q - 1$ and $B = n - 1 - k \ge 1$, so $j - B \le k - q - 1 - (n - 1 - k) = 2k - q - n \le n - 4 - q < 0$ and $o_j = 0$. Hence $u_j + o_j = -j \le 0$, so $s^+_j = 0$; the same computation gives $s^+_{j+1} = 0$. No drop.

(b) $j + 1 = t$ *with* $x = 1$. Here $\Delta u = -1$ and, since $k \ge 2$, $\Delta o = o_t - o_{t-1} = (k - 1) - (k - 2) = 1$. So $u_j + o_j$ is unchanged and the third argument of (17) is unchanged; the first is constant; only u falls. Therefore s^+ can

drop only if the maximum at $j = t-1$ was attained *strictly* by u_{t-1}, i.e. $u_{t-1} > \lceil (u_{t-1}+o_{t-1})/2 \rceil$, which forces $u_{t-1} \geq o_{t-1}+2$. This contradicts Lemma 8.6(iv). No drop.

(c) $j+1 = t$ with $x = 0$. Then $\Delta u = 0$ and $\Delta o = 1$, so no argument falls. □

Proposition 8.8 (decreasing bound)**.** *$\Delta g_j \in \{0, 1\}$ at every chain edge, for every one of the nine families.*

Proof. Write $g_j^1 = -u_j$ and $g_j^2 = \lceil (o_j - u_j)/2 \rceil$, so $g_j = \max(0,\, g_j^1,\, g_j^2)$, and put $\delta := \Delta o - \Delta u$.

Upper bound. g^1 rises by $-\Delta u \leq 1$; g^2 rises by $\lceil \delta/2 \rceil \leq 1$ since $\delta \leq 1-(-1) = 2$; the third argument is the constant 0.

Lower bound. If $\Delta u \leq 0$ then g^1 does not fall and $\delta \geq 0$, so g^2 does not fall: g does not fall. So assume $\Delta u = +1$; by Lemma 8.6(iii), $j \geq h+b$, and then (15) gives $u_j = j-2h-b$ (indeed $j \geq h+b \geq h$ and $h \leq k-q < k$, so $(\min(j,k)-h)^+ - \min(j,k) = -h$ and $(j-h-b)^+ = j-h-b$).

Case $\Delta o = 1$, i.e. $\delta = 0$. Then g^2 is unchanged and the third argument is constant, so g can only fall if $g_j = g_j^1$ strictly exceeds $\max(0, g_j^2)$. Putting $U = -u_j$, this requires $U \geq 1$ and $U - 1 \geq \lceil (o_j + U)/2 \rceil$, i.e. $U \geq o_j + 2$, i.e. $u_j + o_j \leq -2$. We show this is impossible. $\Delta o = 1$ means $j+1 > B$, so $j \geq B$ and $o_j = j - B \geq 0$. Since $B = n-1-k$ and $b = n-3-k+q$ we have $B = b+2-q$, and therefore

$$u_j + o_j = (j-2h-b) + (j-B) = 2j-2h-b-B \;\geq\; 2(h+b)-2h-b-B = b-B = q-2 \;\geq\; -1,$$

using $j \geq h+b$ and $q \geq 1$. So $u_j + o_j \geq -1 > -2$.

Case $\Delta o = 0$. Since o is nondecreasing with $o_{j+1} = \max(0, j+1-B)$, $\Delta o = 0$ forces $o_{j+1} = 0$, hence $o_j = 0$ and $j+1 \leq B$. Combined with $j \geq h+b = h+B+q-2$ this gives $h+q \leq 1$, so $q = 1$, $h = 0$ and $j = B-1 = b$. Then $u_j = j-2h-b = 0 = o_j$, so $g_j = 0$; and $u_{j+1} = 1$, $o_{j+1} = 0$, so $g_{j+1} = \max(0, -1, \lceil -1/2 \rceil) = 0$. No fall.

The last edge $j = t-1 \to t$. There $\Delta u = -x \leq 0$, so the first case of the lower bound applies. □

8.4. The node and the leaves.

Lemma 8.9. *Let E_t denote the lowest* taken *degree at t in the greedy top truncation to total $v_t > 0$, and $E_l = t+k+1-2v_l$ the lowest taken degree at a leaf l with $v_l > 0$. Put $s := q+x$. Then*

$$E_t = t-k+2h+q+x+[\,q+x \text{ odd}\,], \tag{20}$$

$$E_{n-1} - E_t = y-z+[\,q+x \text{ even}\,], \qquad E_n - E_t = y+z+[\,q+x \text{ even}\,], \tag{21}$$

and the slot (E_t, t) is partial (*taken to dimension 1 out of 2*) *if and only if v_t is even, if and only if $q+x$ is even. Moreover the two offsets in (21) are always $+1$ or -1, never anything else, and a leaf offset -1 occurs exactly when $z \neq 0$, or when $y = -1$ and $q+x$ is odd.*

Proof. By Lemma 4.2 the node column has one 1-dimensional slot at degree $t+k$, then $k-1$ two-dimensional slots at $t+k-2, \ldots, t-k+2$, then one 1-dimensional slot at $t-k$. Taking v_t dimensions from the top, the lowest taken degree is $t+k-2\lceil (v_t-1)/2 \rceil$, and the last slot reached is partial exactly when v_t is even and $v_t < 2k$, i.e. when the slot reached is two-dimensional. By Lemma 8.1, $v_t = 2k-2h-s$, so $v_t \equiv s \pmod 2$, and $v_t < 2k$ because $2h+s \geq q \geq 1$. If s is even then $\lceil (v_t-1)/2 \rceil = v_t/2$ and $E_t = t+k-v_t = t-k+2h+s$; if s is odd then $\lceil (v_t-1)/2 \rceil = (v_t-1)/2$ and $E_t = t+k-v_t+1 = t-k+2h+s+1$. That is (20).

For the leaves, Lemma 4.2 gives the leaf support $t-k+1, \ldots, t+k-1$, whence $E_l = t+k+1-2v_l$, and by (16)

$$E_{n-1} = t+k+1-(v_t-y+z) = t-k+2h+s+1+y-z,$$

so $E_{n-1} - E_t = 1+y-z-[\,s \text{ odd}\,] = y-z+[\,s \text{ even}\,]$; the formula for E_n is the same with $z \mapsto -z$.

Finally, the nine families satisfy $x-y \geq 1$ with $x \leq 1$, so $y \leq 0$ and $y \in \{0, -1\}$, and $z \in \{0, \pm 1\}$. Enumerating the nine triples:

$$\begin{array}{llll} (1,0,0):\ s=2, & (+1,+1); & (1,0,\pm1):\ s=3, & (\mp1,\pm1);\\ (1,-1,0):\ s=3, & (-1,-1); & (1,-1,\pm1):\ s=4, & (\mp1,\pm1);\\ (0,-1,0):\ s=1, & (-1,-1); & (0,-1,\pm1):\ s=2, & (\mp1,\pm1). \end{array}$$

All eighteen values are ± 1, and the pattern of -1's is as stated. □

Proposition 8.10 (the partial-slot closed form)**.** *At every chain vertex j with $1 \leq j \leq t$ the greedy top truncation has* at most one *partial slot. It exists if and only if*

$$|u_j| < o_j \quad \text{and} \quad u_j + o_j \text{ is odd},$$

and then its degree is

$$d_j \;=\; t - u_j. \tag{22}$$

No partial slot occurs at a leaf. This statement is characteristic-free, being a statement about the greedy truncation of the multiplicity profile alone.

Proof. The window $W_j = [\,2t+2-j-k,\ j+k-2\,]$ has o_j slots, sitting below the $\mathrm{m}_j - o_j$ minus-only slots and above the $\mathrm{m}_j - o_j$ plus-only slots. Greedy from the top exhausts the minus-only slots first; writing $r = v_j - (\mathrm{m}_j - o_j) = u_j + o_j$ for what is left, it fills $\lfloor r/2 \rfloor$ window slots completely, and leaves one window slot half filled precisely when r is odd and $0 < r < 2o_j$, i.e. $u_j > -o_j$ and $u_j < o_j$. That slot is the $\frac{r+1}{2}$-th from the top of W_j, of degree

$$\max(W_j) - 2\Big(\frac{r+1}{2} - 1\Big) = (j+k-2) - (r-1) = j+k-1-r.$$

Existence forces $o_j > 0$, so $o_j = j+k-t-1$ and $j+k-1-r = j+k-1-u_j-(j+k-t-1) = t-u_j$. A leaf slot is at most a line by Lemma 4.2, hence taken wholly or not at all. □

Corollary 8.11. *Along the region $j \geq \max(B, h+b)$ one has $u_j = j-2h-b$ and $o_j = j-B$, so $u_j + o_j = 2j-2h-b-B \equiv b+B \equiv q \pmod 2$. Hence the parity of r_j is constant and equal to that of q: if q is even there is no partial slot at any chain vertex $j \leq t-1$, and if q is odd the partial chain vertices form a contiguous interval ending at $t-1$. Outside that region no partial slot can occur: for $j < h+b$ one would need $B < b$, i.e. $q > 2$, and then $h+B = h+b-1$ leaves no integer j with $h+B < j < h+b$; and for $j < h$ the column is empty.*

9. Closure of the truncation

We must exhibit, for each representative μ, a graded submodule of I_κ whose profile is the greedy top truncation; by (14) this proves (B) and hence $(*^\circ)$.

A greedy top truncation is not determined by its multiplicities: at a partial slot a *line* in a two-dimensional space must be chosen. We therefore define two candidate subspaces.

Definition 9.1. Let v be the dimension vector of a representative. The *canonical top truncation* $T(v)$ is the graded subspace of I_κ which at every vertex is full above the greedy threshold, zero below it, and, at a partial slot, is the τ-eigenline of that slot. When some leaf l has $E_l = E_t - 1$, the *leaf-pinned truncation* $T^l(v)$ is defined in the same way except that at (E_t, t) it carries the line $\mathrm{im}(a_{l\to t})$, and at each further partial slot it carries the transport of that line along the path of Proposition 9.7 below. (This second clause presupposes that (E_t, t) is itself partial; by Lemma 8.9 that holds, together with the existence of such a leaf, exactly in the four families named below, and $T^l(v)$ is defined only there.) In both cases $\underline{\dim}\, T = v$ by construction.

By Lemma 8.9 a leaf with $E_l = E_t - 1$ exists, *and the node slot (E_t, t) is partial*, exactly in the four families with $q+x$ even and $z \neq 0$, namely $(0, -1, \pm 1)$ and $(1, -1, \pm 1)$. Note that $\mathrm{im}(a_{l\to t})$ is a line, by Lemma 2.8 at the leaf (see the Remark after Proposition 6.1), and that it is *not* a τ-eigenvector: by Proposition 5.6 the folded leaf map is nonzero on both eigen-parts, so the image of the one-dimensional $(I_\kappa)_l[E_t - 1]$ has nonzero component in each eigenline. In those four families $T(v)$ is therefore *not* a submodule and the pinned version is forced.

Lemma 9.2 (T and T^l have the same taken pattern)**.** *$T(v)$ and $T^l(v)$ have the same graded dimension at every slot, namely the greedy top truncation of the profile of I_κ to total v_j at each vertex j. In particular a slot is full, partial or untaken for $T(v)$ if and only if it is so for $T^l(v)$; the two subspaces differ only in* which *line occupies each partial slot.*

Proof. Immediate from Definition 9.1: both are "full above the greedy threshold, zero below it, one line at a partial slot", and the threshold and the set of partial slots depend only on the multiplicity profile of I_κ and on v, not on the choice of line. A line has dimension 1 in either case. □

Consequently every statement about the *taken/untaken* pattern of $T(v)$ — in particular every consequence of Proposition 8.3, which by Remark 8.4 is such a statement — holds verbatim for $T^l(v)$. We use this silently below, and it is the only way Proposition 8.3 is ever applied in the four pinned families, where the hypothesis "a sum of two strand top-segments" fails for $T^l(v)$ itself but holds for the auxiliary $T(v)$ with the same occupations $s_j^\pm$.

9.1. The obligation census. Call an *obligation* a pair consisting of a taken slot (d, j) and a slot $(d+1, i)$ with $i \sim j$; closure of T means that $a_{j\to i}$ maps the taken part of (d, j) into the taken part of $(d+1, i)$. We classify obligations by the state of source and target: *full*, *line* (a partial slot), or *empty*.

Proposition 9.3. *For the nine families the partial slots are as follows.* (*The node entry* (E_t, t) *presupposes* $v_t > 0$*, so that the* E_t *of Lemma 8.9 exists; this holds at every representative except the single degenerate one* $(x, y, z) = (1, 0, 0)$ *with* $h = k-1$*, which is treated at the end of the proof.*)

family	q	$q+x$	*partial slots*
$(1,0,0)$	1	*even*	*a chain interval ending at* $t-1$*, and* (E_t, t)
$(0,-1,0)$	1	*odd*	*a chain interval ending at* $t-1$*;* (E_t, t) *is full*
$(1,-1,0)$	2	*odd*	*none*
$(1,0,\pm1)$	2	*odd*	*none*
$(0,-1,\pm1)$	2	*even*	*exactly one:* (E_t, t)
$(1,-1,\pm1)$	3	*even*	*a chain interval ending at* $t-1$*, and* (E_t, t)

Proof. Combine Corollary 8.11 (chain partials occur iff q is odd, and then form an interval ending at $t-1$) with Lemma 8.9 (the node slot is partial iff $q+x$ is even).

For $(1,0,0)$ and $(0,-1,0)$, where $q=1$, the interval may be empty, and we record exactly when. Here $b = n-2-k$ and $B = b+1$, and for $j \geq \max(B, h+b)$ Corollary 8.11 gives $u_j = j-2h-b$ and $o_j = j-B = j-b-1$, so $r_j = 2(j-h-b)-1$. Thus $r_j > 0$ iff $j > h+b$, and $r_j < 2o_j$ reads $-2h < 1$, i.e. $h \geq 1$. So the partial vertices are exactly the j with $h+b+1 \leq j \leq t-1$ when $h \geq 1$, and there are none when $h = 0$; and since $h+b+1 = h+n-1-k$ and $t-1 = n-3$, the range is empty also when $h \geq k-1$. As $h \leq k-q = k-1$, the conclusion is:

the chain interval is empty exactly when $h = 0$ *or* $h = k-1$*; in particular it is always empty when* $k = 2$*.*

(The statement in the table is that the partial slots form a chain interval ending at $t-1$, possibly empty, and that is what is used below; no later argument depends on the interval being nonempty in these two families.)

The node entry of the table needs the same care, and only in the family $(1,0,0)$. By Lemma 8.1, $v_t = 2k-2h-q-x$ is decreasing in h, so its least value over the range $0 \leq h \leq k-q$ of Proposition 7.7 is attained at $h = k-q$ and equals $q-x$; running over the nine triples, $q-x$ is 0 for $(1,0,0)$, 1 for $(0,-1,0)$, $(1,0,\pm1)$ and $(1,-1,0)$, and 2 for $(0,-1,\pm1)$ and $(1,-1,\pm1)$. Hence $v_t > 0$ at every representative save one: the family $(1,0,0)$ at the top of its range, $h = k-1$, where $v_t = 0$. There the hypothesis $v_t > 0$ of Lemma 8.9 fails — the colour-t column is untaken in every degree, E_t does not exist, and $\mu = e_1 + \cdots + e_{k-1} + e_{n-2}$ has $v = \lambda - \mu = \alpha_k + \cdots + \alpha_{t-1}$ supported on the chain alone, with $v_t = v_{n-1} = v_n = 0$. The table row is to be read accordingly: at $h = k-1$ there is no partial slot at all, neither on the chain (previous paragraph) nor at the node. Nothing below is affected. The only consumer of this proposition is Proposition 9.10(a), which concerns $(0,-1,\pm1)$, where $h \leq k-2$ and so $v_t = 2k-2h-2 \geq 2$; and in Theorem 9.4 the family $(1,0,0)$ is discharged entirely through Proposition 8.3 (classes (ii), (iii) and (vi)) and Proposition 9.5, neither of which mentions E_t — the latter derives $v_l > 0$, respectively $v_t > 0$, from the source of the obligation being taken, so that at $v_t = 0$ its two cases are vacuous — while class (v) requires $z \neq 0$. □

Theorem 9.4. (B) *holds: for every representative* $\mu(h; x, y, z)$ *of Proposition 7.7 there is a graded submodule of* I_κ *of dimension vector* $v = \lambda - \mu$ *whose profile at every vertex is the greedy top truncation, namely* $T(v)$ *in the five families with* $q+x$ *odd or* $z = 0$*, and* $T^l(v)$ *in the four families with* $q+x$ *even and* $z \neq 0$*. Consequently* $V(\mu)$ *is that module, and* $(*^\circ)$ *holds.*

Proof. Every obligation of T falls into one of six classes, and each class has its own discharge.

(i) Target full. Nothing to check. This covers all full→full and line→full obligations, at chain arrows and at leaf arrows alike.

(ii) full→partial on the chain, in the five families with $q+x$ odd or $z = 0$. In these five families the module built is $T(v)$ and every partial slot carries the τ-eigenline. Here the source slot is 1-dimensional and taken, the target 2-dimensional and half taken. By (T0) the arrow has rank 1, and by (T1) its image is the τ-eigenline of the target — which is what $T(v)$ carries there. That the thresholds line up so that no other configuration arises is the strandwise statement of Proposition 8.3, i.e. the two increment bounds (19), established in Propositions 8.7 and 8.8.

In the remaining four families — those with $q+x$ even and $z \neq 0$, where the module built is $T^l(v)$ and the partial slots carry the pinned line $\mathrm{im}(a_{l\to t})$ *and its transports, which by* (N6) *is provably* not *a τ-eigenvector — the discharge just given is unavailable, and it is the negation of what $T^l(v)$ carries. In those four families this class is instead* empty*: no full→partial obligation occurs along a chain arrow at all. That is Proposition 9.10 below, and it is what makes Theorem 9.4 true there.*

(iii) line→partial on the chain, for the eigenline families $(1,0,0)$ and $(0,-1,0)$. Here every partial slot carries the $+$ line, so $T(v)$ is at every chain vertex the direct sum of a top segment of the $+$ strand and a top segment of the $-$ strand, and Proposition 8.3 applies verbatim: closure follows from (19), which holds by Propositions 8.7 and 8.8.

(iv) line→partial on the chain, for $(1,-1,\pm1)$. Here the partial slots carry the transported line, which is not an eigenvector, so $T^l(v)$ is *not* a sum of two strand parts and (iii) does not apply. This class is discharged by Propositions 9.7, 9.8 and 9.9 below: the partial slots form a path with (E_t, t) an endpoint, these obligations are exactly its edges, and each edge has rank 2 by (T0), hence is an isomorphism, so the pinned line propagates uniquely along the path.

(v) the single full→partial leaf obligation. This is $a_{l\to t} : (E_t - 1, l) \to (E_t, t)$ and occurs exactly in the four families with $q+x$ even and $z \neq 0$ (Lemma 8.9). Its image is a line by Lemma 2.8 at the leaf, and $T^l(v)$ is defined to carry precisely that line; so the obligation holds by construction.

For $(0,-1,\pm1)$ that is the only partial slot at all, so no further propagation is needed. The line at (E_t, t) has *three* obligations out of it, one for each neighbour of t: to $(E_t+1, t-1)$, to (E_t+1, l) and to (E_t+1, l'), where l' is the other leaf. All three targets are full or zero, hence all three fall in class (i). Indeed, by Proposition 9.10(a) the lowest taken degree at $t-1$ is E_t+1 and $t-1$ carries no partial slot, so $(E_t+1, t-1)$ is full or zero; and at each leaf $E_l \leq E_t+1$ by Lemma 8.9, so a nonzero leaf slot at E_t+1 is taken, hence full (a leaf slot is at most a line). (For the leaf l itself one may also see the obligation vanish outright: $a_{t\to l}a_{l\to t} = 0$ is the preprojective relation at l, whose only neighbour is t.)

(vi) line→empty on the chain. Let the source be the partial slot (d, j), carrying by Proposition 8.10 the $+$ line (the lowest occupied slot of the $+$ strand, by Lemma 8.2), and let the target $(d+1, i)$, $i = j \pm 1$, be a nonzero untaken chain slot.

We first show the target is 1-dimensional. Suppose it were 2-dimensional; then it carries $+$. If $s_i^+ > 0$ then, by the proof of Proposition 8.3, $|E_i^+ - E_j^+| = 1$ and $E_j^+ = d$, so $E_i^+ \leq d+1$ and the target is occupied in the $+$ strand, hence taken — a contradiction. So $s_i^+ = 0$. If $i = j+1$ this gives $\Delta s_j^+ = -s_j^+ \leq -1$, contradicting Proposition 8.7. If $i = j-1$, use Proposition 8.10: the partial slot at j has degree $d = t - u_j$ and lies in the window, so $o_j = j+k-t-1$ and $s_j^+ = \lceil r_j/2 \rceil$ with $r_j = u_j + o_j$; the hypothesis that $(d+1, i)$ carries $+$ reads $d+1 \leq \mathrm{hi}_{j-1}^+ = j+k-3$, i.e. $t - u_j + 1 \leq o_j + t - 2$, i.e. $r_j \geq 3$, whence $s_j^+ \geq 2$ and $\Delta s_{j-1}^+ = s_j^+ - s_{j-1}^+ = s_j^+ \geq 2$, again contradicting Proposition 8.7. So the target is 1-dimensional.

Now (T2) applies: the source is a 2-dimensional chain slot, the target a 1-dimensional one, and (T2) says the arrow annihilates the τ-eigenline of the source, which is the line $T(v)$ carries. So the obligation is discharged. (This class occurs only in the two families $(1,0,0)$ and $(0,-1,0)$, where the module is $T(v)$; for $(1,-1,\pm1)$ there is no line→empty obligation at all, by Proposition 9.9.)

It remains to show that *no other class occurs*: no full→empty obligation anywhere, and no line→partial or line→empty obligation at a leaf. On the chain, full→empty is the last consequence recorded in Proposition 8.3: a full source slot has both strands occupied, so whichever strand the nonzero target carries, that strand is occupied there too. This is a statement about occupations only, so by Lemma 9.2 it holds for $T^l(v)$ as well as for $T(v)$; the occupations satisfy (19) by Propositions 8.7 and 8.8 in all nine families. For $(1,-1,\pm1)$ the corresponding conclusion for the transported partial slots is Proposition 9.9, and the absence of full→partial chain obligations in all four pinned families is Proposition 9.10. The three leaf classes are Proposition 9.5.

Thus $T(v)$, respectively $T^l(v)$, is closed under every arrow, i.e. is a graded submodule, with $\underline{\dim}\, T = v$. By Corollary 2.13 it is the unique submodule of that dimension vector, hence equals $V(\mu)$; and its profile is greedy from the top at every vertex, which is (B). By Corollary 7.4, $(*^\circ)$ follows. □

9.2. The leaf classes.

Proposition 9.5. *For every family instance $(n,k,h;x,y,z)$ the greedy truncation has* (F1) *no line→partial obligation at a leaf,* (F2) *no full→empty obligation at a leaf, and* (F3) *no line→empty obligation at a leaf.*

Proof. An obligation is "at a leaf" if one end of its arrow is a leaf slot.

(F1). A partial target has dimension 2, so by Lemma 4.2 it is not a leaf slot; hence the target is the node and the *source* is a leaf slot, which has dimension ≤ 1 and is therefore full or empty, never a line.

(F2), (F3). Two directions.

Source at the leaf, $a_{l\to t} : (d,l) \to (d+1,t)$ with the source taken. Then $v_l > 0$ and $d \ge E_l$; by Lemma 8.9 $E_l \ge E_t - 1$, so $d+1 \ge E_t$. The node column takes every nonzero slot of degree $\ge E_t$, so the target is taken: full or partial, never empty.

Source at the node, $a_{t\to l} : (d,t) \to (d+1,l)$ with the source taken, so $v_t > 0$ and $d \ge E_t$. If $v_l > 0$, the leaf column takes every nonzero slot of degree $\ge E_l$, and $E_l \le E_t + 1 \le d+1$ by Lemma 8.9; so the target is taken, hence full, never empty. If $v_l = 0$, then by (16) either $v_t = y - z$ or $v_t = y + z$; the nine families have $x - y \ge 1$ with $x \le 1$, so $y \le 0$, and $z \in \{0, \pm 1\}$, whence $v_t \le 1$. Since the source is taken, $v_t = 1$, so $E_t = t + k$ is the top node degree, the only one taken, and its leaf target sits at degree $t+k+1$, beyond the leaf top degree $t+k-1$: there is no obligation at all. □

9.3. The path of partial slots for $(1,-1,\pm 1)$. Throughout this subsection $(x,y,z) = (1,-1,z)$ with $z = \pm 1$, so $q = 3$, $x = 1$, $b = n-k$, $B = b-1$, $h \le k-3$, and by Lemma 8.1 $u_j = \max(-\min(j,h),\ j-2h-b)$ for $j \le t-1$.

Lemma 9.6. *The partial slots of the greedy truncation are exactly*

$$\mathcal{P} = \big\{(d_j,\, j)\ :\ h+b \le j \le t-1\big\} \ \cup\ \big\{(E_t,\, t)\big\}, \qquad d_j = t + 2h + b - j,$$

and $E_t = d_{t-1} + 1 = 2h + b + 2$. The chain part is never empty.

Proof. By Proposition 8.10 a vertex $j \le t-1$ is partial iff $r_j = u_j + o_j$ is odd and $0 < r_j < 2o_j$, where $o_j = \max(0, j - B) = \max(0, j - b + 1)$.

If $j < b$ then $o_j = 0$ and there is no partial slot. If $b \le j < h + b$ then $u_j = -\min(j,h)$ and $r_j = (j - b + 1) - \min(j,h)$; for $j \ge h$ this is $j - h - b + 1 \le 0$, and for $j < h$ it is $1 - b \le -1$ since $b = n - k \ge 2$. Either way $r_j \le 0$: no partial slot. If $j \ge h + b$ then $u_j = j - 2h - b$ and $o_j = j - b + 1$, so $r_j = 2(j - h - b) + 1$ is odd and positive, and $r_j < 2o_j$ reads $-2h < 1$, always true. So every $j \in [h+b, t-1]$ is partial, at degree $d_j = t - u_j = t + 2h + b - j$ by (22).

At the node, $q + x = 4$ is even, so (E_t, t) is partial and, by (20), $E_t = t - k + 2h + 4$. Since $b = n - k = t + 2 - k$ we get $E_t = 2h + b + 2 = d_{t-1} + 1$. Finally $h \le k - q = k - 3$ gives $h + b \le k - 3 + n - k = n - 3 = t - 1$, so the chain part is nonempty. □

Proposition 9.7 ((L1)). *$\mathcal{P}$, with the slot adjacency of I_κ, is a path, and (E_t, t) is one of its two endpoints.*

Proof. Two slots (d,j) and (d',i) are joined by an arrow iff $i \sim j$ and $|d - d'| = 1$. By Lemma 9.6 the vertices carrying a partial slot are the consecutive $h+b, h+b+1, \dots, t-1, t$, and the degrees $d_{h+b} = t + h$, $d_{h+b+1} = t + h - 1$, $\dots$, $d_{t-1} = 2h + b + 1$ decrease by exactly one per step, with $E_t = d_{t-1} + 1$. So consecutive vertices give adjacent slots, and non-consecutive chain vertices i, j have $|d_i - d_j| = |i - j| \ge 2$, hence no arrow; the leaves carry no partial slot by Proposition 8.10. Therefore $\mathcal{P}$ is the path

$$(d_{h+b}, h+b) \text{ — } (d_{h+b+1}, h+b+1) \text{ —} \cdots \text{— } (d_{t-1}, t-1) \text{ — } (E_t, t),$$

whose endpoints are $(t+h, h+b)$ and (E_t, t). □

Proposition 9.8 ((L3)). *The line→partial obligations of $T^l(v)$ are exactly the edges of the path $\mathcal{P}$, each traversed once.*

Proof. An obligation is line→partial iff its source is the partial slot of its vertex j and its target is the partial slot of a neighbour i; by Proposition 8.10 each vertex has at most one partial slot, so this says $d_i = d_j + 1$ with $i \sim j$. With $d_j = t + 2h + b - j$ on the chain and $E_t = d_{t-1} + 1$, the possibilities are $i = j - 1$ (requiring $j - 1 \ge h + b$) and $(j,i) = (t-1, t)$; whereas $i = j + 1$ gives $d_i = d_j - 1$, the node-to-chain step $(j,i) = (t, t-1)$ gives $d_{t-1} = E_t - 1$, and the leaves carry no partial slot. Those are exactly the edges of $\mathcal{P}$, each traversed once, in the direction away from vertex $t - 1$. □

Proposition 9.9 ((L4)). *No line→empty obligation occurs for $T^l(v)$ in the family $(1,-1,\pm 1)$.*

Proof. Such an obligation would be $\big((d_j, j), (d_j + 1, i)\big)$ with $(d_j + 1, i)$ nonzero in I_κ but untaken, i.e. $d_j + 1 < E_i$, where E_i is the lowest taken degree at i. At a partial vertex $E_i = d_i$. We run over the neighbours of a partial vertex j, i.e. $h + b \le j \le t$.

- $i = j - 1$ with $j - 1 \ge h + b$: $E_i = d_i = d_j + 1$, so the target is taken (partial): a path edge.

- $i = j+1 \le t-1$: $E_i = d_i = d_j - 1 \le d_j + 1$: taken, and full.
- $(j,i) = (t-1,t)$: $E_t = d_{t-1} + 1$: a path edge.
- $(j,i) = (t,t-1)$: $E_{t-1} = d_{t-1} = E_t - 1 \le E_t + 1$: taken.
- $j = t$, i a leaf: by Lemma 8.9 the offsets are $E_l - E_t = \mp z$ or $\pm z$, i.e. ± 1, so $E_l \le E_t + 1$ and the target $(E_t + 1, l)$ is taken whenever it exists.
- $j = h+b$, $i = h+b-1 \ge 1$: the only nontrivial case. Here $d_j = t+h$ and the target is $(t+h+1,\, h+b-1)$. At $i = h+b-1$ we have $o_i = \max(0, i-B) = \max(0,h) = h$ and, since $i = h+b-1 \ge h$ (as $b \ge 2$), $u_i = -\min(i,h) = -h$; so $r_i = u_i + o_i = 0$ and i carries no partial slot, and $v_i = \mathrm{m}_i - o_i$ is exactly the number of minus-only slots at i. Hence the greedy stops at the bottom minus-only slot, of degree
$$E_i = \big(2t - |i-k|\big) - 2\big(\mathrm{m}_i - h\big) + 2.$$
Using $k = t+2-b$ and $i = h+b-1 = h+t+1-k$, this equals $t+h+1$ whether $i \le k$ or $i > k$: for $i \le k$ it is $2t-k+i-2i+2h+2 = 2t-k-i+2h+2 = t+h+1$, and for $i > k$ it is $2t-i+k-2k+2h+2 = 2t-i-k+2h+2 = t+h+1$. So the obligation lands exactly on E_i: the target is taken, and full.

Hence no line→empty obligation occurs. □

9.4. No full→partial chain obligation in the four pinned families. This subsection isolates and proves the statement that class (ii) of Theorem 9.4 cannot supply in the four families with $q+x$ even and $z \neq 0$. It is a genuine gap in the class list otherwise: in those families the partial slots of $T^l(v)$ carry $\mathrm{im}(a_{l\to t})$ and its transports, which by Proposition 5.6 (see (N6)) is *not* a τ-eigenvector, so the discharge "(T1) makes the image the τ-eigenline, which is what the module carries" asserts exactly the opposite of the truth. A single full→partial chain obligation in those families would make $T^l(v)$ fail to be a submodule and Theorem 9.4 — hence Theorem 1.1 — false. We prove that the class is empty. Nothing below is a census: the argument is the closed-form threshold arithmetic of §8, run once for $(0,-1,\pm1)$ and once for $(1,-1,\pm1)$.

Throughout, for a vertex j we write E_j for the lowest taken degree of the greedy top truncation at j (so $E_j = d_j$ at a partial vertex, the partial slot being the last one the greedy reaches).

Proposition 9.10 ((L5)). *Let $\mu = \mu(h;x,y,z)$ belong to one of the four families with $q+x$ even and $z \neq 0$, i.e. $(x,y,z) \in \{(0,-1,\pm1),(1,-1,\pm1)\}$. Then:*
(a) *for $(x,y,z) = (0,-1,\pm1)$ the unique partial slot is (E_t,t), and $E_{t-1} = E_t + 1$;*
(b) *for $(x,y,z) = (1,-1,\pm1)$ every chain obligation whose target is a partial slot has a partial slot as its source, i.e. is an edge of the path $\mathcal{P}$ of Proposition 9.7;*
(c) *in all four families,* no full→partial obligation occurs along a chain arrow. *Equivalently, every obligation of $T^l(v)$ whose target is a partial slot is either the leaf obligation $a_{l\to t} : (E_t - 1, l) \to (E_t, t)$ of class* (v) *or an edge of $\mathcal{P}$.*

Proof. (a). Here $q = 2$, $x = 0$, $y = -1$, $z = \pm 1$. Since q is even, Corollary 8.11 gives no partial slot at any chain vertex $j \le t-1$; and since $q+x = 2$ is even, Lemma 8.9 makes (E_t,t) partial. So (E_t,t) is the unique partial slot, as recorded in Proposition 9.3, and by (20)
$$E_t = t - k + 2h + q + x + [\,q+x \text{ odd}\,] = t - k + 2h + 2.$$
We compute E_{t-1}. By Lemma 8.1 (and its proof) $v_{t-1} = 2k - 2h - q - [k=t] = 2k - 2h - 2 - [k=t]$, while $\mathrm{m}_{t-1} = \min(t-1,k)$ and $o_{t-1} = k-2$ (Lemma 8.6(iv)), so the number of minus-only slots at $t-1$ is
$$\mathrm{m}_{t-1} - o_{t-1} = \begin{cases} k-(k-2) = 2, & k \le t-1, \\ (t-1)-(k-2) = 1, & k = t, \end{cases} \qquad \text{i.e.} \qquad 2 - [k=t].$$
Greedy from the top exhausts these first, leaving $r_{t-1} = v_{t-1} - (\mathrm{m}_{t-1} - o_{t-1}) = 2k - 2h - 4$ dimensions, which is even and non-negative (as $h \le k - q = k-2$) and at most $2o_{t-1} = 2k-4$. So the greedy fills exactly $k-h-2$ window slots *completely* and leaves none partial — confirming Corollary 8.11 — and occupies $(2-[k=t]) + (k-h-2)$ degrees in all. Since $\mathrm{hi}^-_{t-1} = 2t - |t-1-k| = t+k+1-2[k=t]$, the lowest of them is
$$E_{t-1} = \mathrm{hi}^-_{t-1} - 2\big((2-[k=t]) + (k-h-2) - 1\big) = \big(t+k+1-2[k=t]\big) - \big(2k-2h-2-2[k=t]\big) = t-k+2h+3,$$
the terms in $[k=t]$ cancelling. Hence $E_{t-1} = E_t + 1$, as claimed.

(b). Here $q = 3$, $x = 1$, $b = n-k$, $B = b-1$, $h \le k-3$. By Lemma 9.6 the partial slots are (d_j, j) for $h+b \le j \le t-1$, with $d_j = t+2h+b-j$, together with (E_t,t) where $E_t = d_{t-1}+1$; and the leaves carry none (Proposition 8.10). An obligation with target a partial slot (d_i, i) has source $(d_i - 1, j)$ for some neighbour j of

i. We run over all of them. This is exactly the bullet list in the proof of Proposition 9.9, read in the opposite direction: there we asked, for each partial vertex, what lies one degree *above* at each neighbour; here we ask what lies one degree *below*. The same four threshold computations answer both questions.

- $i \le t-1$, $j = i+1 \le t-1$. Then $d_{i+1} = d_i - 1$, so the source slot $(d_i - 1, i+1)$ *is* the partial slot at $i+1$: a line. This is a path edge.
- $i = t-1$, $j = t$. The source is $(d_{t-1} - 1, t) = (E_t - 2, t)$, and $E_t = E_t > E_t - 2$, so the source is untaken: there is no obligation.
- $i \le t-1$, $j = i-1 \ge h+b$. Then $E_{i-1} = d_{i-1} = d_i + 1 > d_i - 1$, so the source is untaken: no obligation.
- $i = h+b$, $j = h+b-1$. This vertex carries no partial slot, and the last bullet of the proof of Proposition 9.9 computes $E_{h+b-1} = t+h+1 = d_{h+b}+1 = d_i + 1 > d_i - 1$: the source is untaken, so again there is no obligation. (The vertex exists, since $b = n-k \ge 2$ gives $h+b-1 \ge 1$.)
- $i = t$, $j = t-1$. The source is $(E_t - 1, t-1) = (d_{t-1}, t-1)$, the partial slot at $t-1$: a line. This is the remaining path edge.
- $i = t$, j *a leaf.* These are the two leaf obligations into (E_t, t); by Lemma 8.9 exactly one of them is nonzero, namely the one with $E_l = E_t - 1$, and it is the class (v) obligation. Neither is a chain arrow.

So every chain obligation into a partial slot has a partial source, which is (b).

(c). In the two families of (a) the only partial slot is (E_t, t), so any full→partial chain obligation would have source $(E_t - 1, t-1)$; but $E_{t-1} = E_t + 1 > E_t - 1$, so that slot is untaken and no obligation starts there. (The other two neighbours of t are leaves, whose obligations are not chain arrows.) In the two families of (b), every chain obligation into a partial slot has a *line* source, never a full one. In all four families, therefore, the class is empty. □

Remark 9.11 (the hypotheses actually used)**.** Proposition 9.10 draws on exactly four earlier facts, and each holds in the stated range. (1) Corollary 8.11, that chain partial slots occur iff q is odd — used for $(0, -1, \pm 1)$, where $q = 2$. (2) Lemma 8.9, that (E_t, t) is partial iff $q + x$ is even and that the leaf offsets are ± 1 — both families have $q + x$ even. (3) Lemma 8.1 and Lemma 8.6(iv), for v_{t-1} and $o_{t-1} = k - 2$; the latter needs $k \ge 2$, which is Convention 4.1. (4) Lemma 9.6 and the last bullet of Proposition 9.9, for $(1, -1, \pm 1)$; these need $h \le k - 3$ and $b = n - k \ge 2$, both guaranteed by Proposition 7.7 with $q = 3$ and $k \le n - 2$. In particular the proof never assumes that the chain interval of partial slots is nonempty in the two families of (a) — it is empty there, by Corollary 8.11 — nor that it is nonempty in (b), where Lemma 9.6 shows that it is.

Remark 9.12 (what a census would and would not give)**.** An exhaustive census over D_4–D_{44}, all $2 \le k \le n-2$, all nine families and all h — some 4.8×10^7 obligations — confirms Proposition 9.10: the class "full→partial along a chain arrow" is populated only in the family $(1, 0, 0)$, never in the four pinned families, and in the pinned families every chain obligation into a partial slot has a line source (§14). That census is a check, not the proof; the proof is the threshold arithmetic above, which is uniform in n, k and h.

10. The induction along σ

We now prove Proposition 4.11. Fix $\mu = w\lambda$ and a reduced word for w, and let $0 = X_0 \subseteq X_1 \subseteq \cdots \subseteq X_\ell = V$ be the filtration of Theorem 3.6, indexed by blocks. Each X_p is extremal (Corollary 3.8). We carry along the hypothesis

$$(H) \qquad X \text{ is a top truncation at } t \text{ and an up-set at each of the two leaves,}$$

which holds for $X_0 = 0$. Steps at colours $j \notin \{t, n-1, n\}$ change X only at j, so (H) is untouched by them. We write $v' = \underline{\dim}\, X$, $\mu' = \lambda - \sum_i v'_i \alpha_i = \sum_i \eta_i e_i$ with $\eta_i \in \{0, \pm 1\}$, and $D = D_t(X)$ for the threshold of Definition 4.8.

10.1. Weight arithmetic at the node.

Lemma 10.1. *For any graded submodule $X \subseteq I_\kappa$ with $\underline{\dim}\, X$ extremal,*

$$v'_{n-1} + v'_n = v'_t - \eta_{n-1}, \qquad v'_{n-1} - v'_n = \eta_n,$$

and, writing $c := \langle \alpha_t^\vee, \mu' \rangle$,

$$c = [k = t] - 2v'_t + v'_{t-1} + v'_{n-1} + v'_n.$$

If $c \ge 1$ then $\eta_{n-1} \le 0$, and

$$v'_{t-1} \;\ge\; v'_t + 1 + \eta_{n-1} - [k = t] \;\ge\; v'_t - [k = t]. \tag{23}$$

Proof. The coefficient of e_{n-1} in $\sum_j v'_j\alpha_j$ is $-v'_t+v'_{n-1}+v'_n$ and that of e_n is $-v'_{n-1}+v'_n$; since λ has support in the first $n-2$ coordinates, these equal $-\eta_{n-1}$ and $-\eta_n$. The formula for c is $\langle\alpha_t^\vee,\lambda\rangle-\sum_j a_{tj}v'_j$. Next, $c=\langle\alpha_t^\vee,\mu'\rangle=\eta_{n-2}-\eta_{n-1}$ with $\eta_i\in\{0,\pm1\}$, so $c\geq1$ forces $\eta_{n-1}\leq0$. Substituting $v'_{n-1}+v'_n=v'_t-\eta_{n-1}$ into $c\geq1$ gives (23). □

Lemma 10.2. *Suppose X is an up-set at $t-1$ and put* $\mathrm{cum}:=\sum_{d\geq D+1}\dim(I_\kappa)_{t-1}[d]$. *If $v'_{t-1}\geq\mathrm{cum}$ then $X_{t-1}[D+1]$ is full; and if $v'_{t-1}\geq\mathrm{cum}-1$ then $X_{t-1}[D+1]$ is deficient by at most one dimension.*

Proof. Let D' be the threshold of X at $t-1$. If $D'\geq D+1$ then X_{t-1} is supported in degrees $\geq D'$ and is proper at D', so $v'_{t-1}\leq\mathrm{cum}-1$; contrapositively $v'_{t-1}\geq\mathrm{cum}$ forces $D'<D+1$, i.e. $X_{t-1}[D+1]$ full. If $D'=D+1$ the deficiency at $D+1$ is $\dim(I_\kappa)_{t-1}[D+1]-\dim X_{t-1}[D+1]$ and $v'_{t-1}\leq\mathrm{cum}-$(that deficiency), so $v'_{t-1}\geq\mathrm{cum}-1$ bounds it by 1; if $D'>D+1$ then $v'_{t-1}\leq\mathrm{cum}-1-\dim(I_\kappa)_{t-1}[D+1]$, which combined with $v'_{t-1}\geq\mathrm{cum}-1$ forces $(I_\kappa)_{t-1}[D+1]=0$ and the slot is full vacuously. □

Lemma 10.3 (the profile bound at $t-1$)**.** *Assume (H), and let $h\geq0$.*

(i) *If $v'_t=2h$, so that $D=t+k-2h$, then* $\mathrm{cum}\leq\max(0,h-1)+\max(0,h+1-[k=t])$, *which equals $2h-[k=t]$ when $h\geq1$ and $1-[k=t]$ when $h=0$.*

(ii) *If $v'_t=2h+1$, so that $D=t+k-2h-2$, then* $\mathrm{cum}\leq2h+2-[k=t]$.

Proof. By (6) the multiplicity profile at $t-1$ is the sum of two indicator intervals: the *low* one on $[\,|t-1-k|,\ t+k-3\,]$ and the *high* one on $[\,t-k+3,\ 2t-|t-1-k|\,]$, each with $\min(t-1,k)$ slots, in steps of 2. Degrees at $t-1$ have the parity of $t-1+k$. (When char $\neq2$ these two intervals are the supports of the $+$ and $-$ strands of Proposition 5.1, but the counting below uses only the profile, so this lemma is valid over every field.)

(i) $D+1=t+k-2h+1\equiv t-1+k$. The low interval contributes the slots $d\leq t+k-3$ with $d\geq D+1$, of which there are $\lfloor((t+k-3)-(t+k-2h+1))/2\rfloor+1=h-1$ when $h\geq1$ and none otherwise, capped at $\min(t-1,k)$: that is $\max(0,h-1)$. The high interval tops out at $2t-|t-1-k|$, which is $t+k+1$ if $k\leq t-1$ and $2t-1$ if $k=t$; the number of its slots $\geq D+1$ is $h+1$, respectively h, when nonnegative: that is $\max(0,h+1-[k=t])$, again capped. Summing gives the bound, and for $h\geq1$ the two terms are $h-1$ and $h+1-[k=t]$.

(ii) $D+1=t+k-2h-1$. The same count gives h from the low interval and $h+2-[k=t]$ from the high interval, whence $\mathrm{cum}\leq2h+2-[k=t]$. □

10.2. Leaf steps.

Lemma 10.4. *Assume (H) and let the step have colour $j\in\{n-1,n\}$. Then (H) is preserved.*

Proof. The leaf column is multiplicity-free with degrees $t+k-1>t+k-3>\cdots>t-k+1$ (Lemma 4.2), and by (H) the module occupies the top v'_j of them, so the largest unfilled leaf degree is $D_j=t+k-1-2v'_j$. We have $c_j=\langle\alpha_j^\vee,\mu'\rangle=v'_t-2v'_j\geq1$ because the word is reduced (Lemma 3.5(i)). If $v'_t=2h$ this forces $v'_j\leq h-1$ and $D_j\geq t+k-2h+1>D$; if $v'_t=2h+1$ it forces $v'_j\leq h$ and $D_j\geq t+k-2h-1>D$. Either way $D_j>D$.

The socle step at j adds exactly the unfilled leaf slots (e,j) for which $a_{j\to t}\big((I_\kappa)_j[e]\big)\subseteq X_t[e+1]$. If $e\geq D$ then $e+1>D$ and $X_t[e+1]$ is full by (H), so the condition holds. If $e<D-1$ then $X_t[e+1]=0$ by (H), while $a_{j\to t}$ is injective on the leaf slot by Lemma 2.8, so the condition fails. The degree $e=D-1$ is not decided by this dichotomy: there $X_t[e+1]=X_t[D]$ is *proper* by Definition 4.8, so the inclusion $a_{j\to t}\big((I_\kappa)_j[D-1]\big)\subseteq X_t[D]$ may hold or fail, and $D-1$ need not be added. Hence, the unfilled degrees being exactly those $\leq D_j$, the added set *contains* $[D,\,D_j]$ and *is contained in* $[\max(D-1,\ \text{bottom}),\,D_j]$. Both of these are top segments of the unfilled block; and since the only leaf degree in which they differ is $D-1$, the added set is one of the two, hence itself a top segment of the unfilled block. Adding a top segment of the unfilled block to a final segment of degrees leaves a final segment. So X is still an up-set at j, and it is unchanged at t and at the other leaf. □

10.3. Colour-t steps: Claim G and Claim F.

Proposition 10.5 (Claim G)**.** *Assume (H), that $\underline{\dim}\,X$ is extremal, that $c=\langle\alpha_t^\vee,\mu'\rangle\geq1$, and $(*^\circ)$ for X (whose hypotheses X meets, being extremal with $c\geq1$; $(*^\circ)$ itself is available from Corollary 7.4 — which is characteristic-free — together with (B), and (B) comes from Theorem 9.4 when* char $\neq2$ *and from Step 4 of Theorem 11.1 at $p=2$, where Theorem 9.4 is not claimed). Then:*

(G1) *if $X_t[D]\neq0$ then every neighbour of t is full at degree $D+1$;*

(G2) *if $X_t[D] = 0$ then at most one neighbour of t is deficient at degree $D+1$, and that deficiency is one dimensional; moreover if some neighbour is deficient then* $\dim(I_\kappa)_t[D] = 2$.

Proof. Under (H) the column at t is a top truncation, and X is extremal with $c \geq 1$, so Lemma 4.9(b) gives $v'_t < 2k$ and Lemma 4.9(a) then gives

$$X_t[D] \neq 0 \iff v'_t \text{ is even and } \geq 2,$$

and $D = t+k-2h$ when $v'_t = 2h$ with $h \geq 1$, while $D = t+k-2h-2$ when $v'_t = 2h+1$, and $D = t+k$ when $v'_t = 0$. (Lemma 8.9 is *not* what is being used here: it is stated only for the nine representatives and only for char $\neq 2$, and its E_t is the lowest taken degree rather than the threshold of Definition 4.8; see Remark 4.10.)

(G1): $v'_t = 2h$, $h \geq 1$. The leaves. By Lemma 10.1, $\eta_{n-1} \leq 0$ and $v'_{n-1} + v'_n = 2h - \eta_{n-1} \in \{2h, 2h+1\}$ with difference η_n. If the sum is $2h$ its parity forces $\eta_n = 0$ and both leaves equal h; if the sum is $2h+1$ then $\eta_n = \pm 1$ and the leaves are $\{h, h+1\}$. Either way $\min(v'_{n-1}, v'_n) \geq h$. Now $D+1 = t+k-2h+1$ is the h-th leaf degree from the top, and an up-set at a leaf is full there exactly when $v'_l \geq h$. So both leaves are full at $D+1$. *The chain.* By (23), $v'_{t-1} \geq 2h - [k=t]$, and by Lemma 10.3(i) with $h \geq 1$, cum $\leq 2h - [k=t]$. By $(*^\circ)$, X is an up-set at $t-1$, so Lemma 10.2 applies and $X_{t-1}[D+1]$ is full.

(G2), first branch: $v'_t = 2h+1$. Here $D+1 = t+k-2h-1$ is the $(h+1)$-st leaf degree from the top, so a leaf is full at $D+1$ iff $v'_l \geq h+1$. By Lemma 10.1, $v'_{n-1} + v'_n = 2h+1-\eta_{n-1}$ with difference η_n, and $\eta_{n-1} \leq 0$.

- If $\eta_{n-1} = 0$ the sum is $2h+1$, odd, so $\eta_n = \pm 1$ and the leaves are $\{h, h+1\}$: exactly one leaf is deficient, by exactly one dimension. But then (23) reads $v'_{t-1} \geq 2h+2-[k=t] \geq$ cum by Lemma 10.3(ii), so the chain neighbour is *full*.
- If $\eta_{n-1} = -1$ the sum is $2h+2$, even, so $\eta_n = 0$ and both leaves equal $h+1$: both leaves are full. Only the chain neighbour can be deficient, and (23) gives $v'_{t-1} \geq 2h+1-[k=t] \geq$ cum -1, so by Lemma 10.2 it is deficient by at most one dimension.

Either way at most one neighbour is deficient, by at most one dimension.

For the last clause, suppose a neighbour is deficient. We have $\dim(I_\kappa)_t[D] = 1$ only for $D \in \{t-k, t+k\}$, and $D = t+k$ is excluded since $v'_t \geq 1$. So suppose $D = t-k$, i.e. $v'_t = 2k-1$. Then $v'_t = r' + 2m'$ (Lemma 4.4 applied to μ') is odd, so $r' = 1$ and $m' = k-1$; thus all $k-1$ chain entries of $\operatorname{supp}\mu'$ carry sign -1, whence $\eta_{n-2} \leq 0$, and $c = \eta_{n-2} - \eta_{n-1} \geq 1$ forces $\eta_{n-2} = 0$ and $\eta_{n-1} = -1$, hence $\eta_n = 0$ and $r' = 1$ is accounted for by $n-1$. So both leaves have $v'_l = k$ and are full (the leaf column has exactly k slots). For the chain, cum is the whole column at $t-1$ above degree $D+1 = t-k+1$; the column starts at $|t-1-k|$, which is $t-1-k$ when $k \leq t-1$ and 1 when $k = t$, so cum $= 2k-1$ when $k \leq t-1$ (the single bottom $+$ slot is excluded) and cum $= 2k-2$ when $k = t$. Now (23) gives $v'_{t-1} \geq 2k-1-[k=t]$, which is $\geq$ cum in both cases, so by Lemma 10.2 the chain neighbour is full too. No neighbour is deficient after all — contradiction. Hence $\dim(I_\kappa)_t[D] = 2$.

(G2), second branch: $v'_t = 0$. Here $X_t = 0$ and $D = t+k$, the multiplicity-one top slot. We show that no neighbour is deficient at all.

- *The leaves are empty.* If $0 \neq u \in X_{n-1}[e]$, its only arrow lands in $X_t[e+1] = 0$, so u spans a copy of the leaf simple inside $\operatorname{soc}(I_\kappa) = S_\kappa$; but $\kappa = k \leq n-2$ is a chain vertex. So $X_{n-1} = 0$ and likewise $X_n = 0$: $v'_{n-1} = v'_n = 0$.
- *The chain neighbour is not empty.* By Lemma 10.1, $c = [k=t] + v'_{t-1} \geq 1$, so $v'_{t-1} \geq 1 - [k=t]$.
- cum ≤ 1. Here $D+1 = t+k+1$. At $t-1$ the low interval tops out at $t+k-3$ and the high interval at $2t - |t-1-k|$, which is $t+k+1$ if $k \leq t-1$ and $t+k-1$ if $k = t$. So cum $= 1 - [k=t]$ exactly.
- *No deficiency.* By Lemma 4.2 the leaf top degree is $t+k-1 < t+k+1 = D+1$, so $(I_\kappa)_l[D+1] = 0$ for both leaves and they are full vacuously. For the chain: if $k = t$ then cum $= 0$ and $(I_\kappa)_{t-1}[D+1] = 0$, again full vacuously; if $k \leq t-1$ then $v'_{t-1} \geq 1 =$ cum and Lemma 10.2 gives fullness.

So (G2) holds vacuously in this branch, and the last clause is vacuous too. □

Corollary 10.6 (Claim F)**.** *Assume (H), $c \geq 1$,* and that $\underline{\dim}\, X$ is extremal. *If $X_t[D]$ is a proper subspace of $(I_\kappa)_t[D]$ then $\operatorname{soc}_{(t)}(I_\kappa/X)$ has a nonzero component in degree D.*

Remark 10.7. The extremality hypothesis is not decorative and cannot be removed: by (N3) of §14.3, Claim F is *false* for arbitrary graded submodules satisfying (H) and $c \geq 1$. It enters through Proposition 10.5, whose proof invokes $(*^\circ)$ for X, and $(*^\circ)$ is a statement about *extremal* modules only (Definition 7.1). At the one place where Corollary 10.6 is used — the proof of Proposition 4.11 — the module $X = X_p$ is a prefix of the filtration of Theorem 3.6 and is therefore extremal by Corollary 3.8, so the hypothesis holds.

Proof. An element $z \in (I_\kappa)_t[D]$ lies in $\mathrm{soc}_{(t)}(I_\kappa/X)$ iff all three images $a_{t\to i}z$ ($i \in \{t-1, n-1, n\}$) lie in X. If no neighbour is deficient at $D+1$ — which is (G1), and also the case $v'_t = 0$ of (G2) — then all three conditions are automatic and the whole of $(I_\kappa)_t[D]/X_t[D] \neq 0$ lies in the socle. Otherwise, by (G2), exactly one neighbour is deficient, by exactly one dimension, and $\dim(I_\kappa)_t[D] = 2$. So membership in the socle is cut out by one linear condition on a two-dimensional space, and a nonzero solution exists; it is not contained in $X_t[D]$ for dimension reasons unless $X_t[D]$ is the whole solution space, which is impossible because $X_t[D] = 0$ in the (G2) branch. □

10.4. Preservation of (H) at colour-t steps.

Lemma 10.8. *Assume (H). Then*
(a) $\mathrm{soc}_{(t)}(I_\kappa/X)$ *vanishes in every degree* $< D-2$*;*
(b) *if* $\mathrm{soc}_{(t)}(I_\kappa/X)$ *is nonzero in degree* $D-2$ *then* $\dim X_t[D] = 1$ *and* $\dim(I_\kappa)_t[D] = 2$.

Proof. For (a), let $0 \neq x \in (I_\kappa)_t[d]$ represent an element of $\mathrm{soc}_{(t)}(I_\kappa/X)$ with $d < D-2$. Then $X_t[d+2] = 0$ by (H), so every double path $a_{j\to t}a_{t\to j}x$ vanishes. Not all $a_{t\to j}x$ can vanish: otherwise x spans a copy of S_t inside $\mathrm{soc}(I_\kappa) = S_\kappa$, forcing $d = 2t$, whereas $d < D \le t+k \le 2t$. So some $y = a_{t\to j}x \neq 0$ satisfies $a_{j\to t}y = 0$. If j is a leaf, y is killed by its only arrow and hence lies in $\mathrm{soc}(I_\kappa) = S_\kappa$, forcing κ to be a leaf — excluded, as $\kappa = k \le n-2$. So $j = t-1$. Before propagating we record the degree bound. By (H) and Definition 4.8, $D \le t+k$ (the top colour-t degree, Lemma 4.2), so $d < D-2 \le t+k-2$; the element $y = a_{t\to(t-1)}x$ sits at degree $d+1 \le t+k-2$, at the vertex $t-1$. Since $k \le t$,

$$d+1+p \;\le\; t+k-2+p \;<\; 2t \qquad \text{for every } 0 \le p \le t-k-1, \tag{24}$$

and more crudely $d+1 \le t+k-2 \le 2t-2 < 2t$. (Note that the relevant bound is this one and not the leaf top degree $t+k-1$: y sits at the chain vertex $t-1$, and the bound that applies comes from $d < D-2$ and $D \le t+k$.)

Now propagate outward exactly as in Lemma 4.5: the relation at $t-1$ applied to y has its t-term zero, so $a_{(t-2)\to(t-1)}\big(a_{(t-1)\to(t-2)}y\big) = 0$, and $a_{(t-1)\to(t-2)}y \neq 0$ because otherwise y would be killed by both arrows out of $t-1$ and hence lie in $\mathrm{soc}(I_\kappa)$, which by Lemma 2.7 forces the vertex to be $\kappa = k$ and the degree to be $2t$; the degree is $d+1 < 2t$. At the general step we sit at $(d+1+p,\, t-1-p)$, and the same alternative arises: the element is killed by both its arrows only if $t-1-p = k$ and $d+1+p = 2t$. The first gives $p = t-1-k$, and then (24) gives $d+1+p \le 2t-3 < 2t$. So the iteration never stops early, reaches vertex 1, whose only neighbour is 2, and lands in $\mathrm{soc}(I_\kappa) = S_\kappa$, forcing $\kappa = 1$ — excluded, as $k \ge 2$.

For (b), let x be nonzero in $\mathrm{soc}_{(t)}(I_\kappa/X)$ at degree $D-2$. If $X_t[D]$ were 0, the double paths of x would all land in $X_t[D] = 0$ and the argument just given applies verbatim with $d = D-2$, a contradiction. So $X_t[D] \neq 0$; it is proper by the definition of D, hence a line in a two-dimensional space. □

Proof of Proposition 4.11. We show (H) is preserved by every socle step. Steps at colours $\notin \{t, n-1, n\}$ do not change X at t or the leaves; leaf steps are Lemma 10.4. Let the step have colour t, write Y for the result, and put $s := \dim\big(\mathrm{soc}_{(t)}(I_\kappa/X)\big)[D-2] \ge 0$.

Case 0 (no-op). If $\mathrm{soc}_{(t)}(I_\kappa/X) = 0$ then $Y = X$ and there is nothing to prove. This occurs exactly at the repetitions inside a block of σ: by Remark 3.3, a colour-t socle step is idempotent, so only the first application in a block can contribute.

Effective steps. Assume $Y \neq X$. Then $c = \langle \alpha_t^\vee, \mu' \rangle \ge 1$ by Lemma 3.5(i); moreover $\underline{\dim}\, X = \underline{\dim}\, X_p$ is extremal by Corollary 3.8, so Corollary 10.6 applies. We record the new profile degree by degree:

- degrees $< D-2$ are untouched by Lemma 10.8(a) and were 0 by (H), so stay 0;
- degrees $> D$ were full and remain full;
- at degree $D-2$: $X_t[D-2] = 0$ by (H), so $\dim Y_t[D-2] = s$;
- at degree D: $X_t[D]$ is proper by the definition of D, so by Corollary 10.6 the step contributes something there, and $Y_t[D] \neq 0$.

Now split on s.

- $s > 0$. The step reaches $D-2$, so Lemma 10.8(b) gives $\dim X_t[D] = 1$ and $\dim(I_\kappa)_t[D] = 2$; hence $(I_\kappa)_t[D]/X_t[D]$ is a line and $Y_t[D] = (I_\kappa)_t[D]$ is full. The new profile is 0 below $D-2$, of dimension s at $D-2$, and full from D up: a top truncation with threshold $D-2$, or, if s is maximal there, with the next colour-t degree below.

- $s = 0$. The new profile is 0 at every colour-t degree $< D$, nonzero at D, and full above D. Whether $Y_t[D]$ is proper or full, this is again "zero below one degree, proper or full at it, full above", i.e. a top truncation, with threshold D or the next colour-t degree below.

In every case Y is a top truncation at t, and it is unchanged at the two leaves. So (H) is preserved, and by induction $V = X_\ell$ satisfies (H). In particular the colour-t multiplicities of V are "zero below one degree, proper at it, full above", i.e. the greedy top truncation of the sequence of Lemma 4.2 to total v_t. □

Remark 10.9. The proof of Proposition 4.11 uses $(*^\circ)$, whose proof (§§7–9) uses only rigidity and the strand apparatus, never Proposition 4.11. There is no circularity.

Remark 10.10. Both $(*^\circ)$ and Claim F genuinely use extremality. The corresponding statement for arbitrary graded submodules of I_κ satisfying (H) is false; see §14.

11. All characteristics: specialisation from the generic fibre

The proof just completed assumed $\operatorname{char}\mathbb{K} \neq 2$ at exactly one place, the strand decomposition of §5, and used it for exactly one purpose, the construction in Theorem 9.4 of a witness submodule for (B). We now remove the hypothesis. Nothing in this section repairs the strand apparatus at $p = 2$; it makes it unnecessary there.

Theorem 11.1. *Let $\Gamma = D_n$, $\lambda = \omega_k$ with $2 \le k \le n-2$, and let $\mathbb{K}$ be an arbitrary field. Then (B), $(*^\circ)$, Proposition 4.11 and Theorem 1.1 all hold over $\mathbb{K}$. More precisely, for every $\mu \in W\lambda$ the graded dimensions $\dim_{\mathbb{K}} V(\mu)_i[d]$ are independent of $\mathbb{K}$.*

Proof. Let p be a prime and let $R = \mathbb{Z}_{(p)}$, a discrete valuation ring with fraction field $\mathbb{Q}$ and residue field $\mathbb{F}_p$. It suffices to treat $\mathbb{K} = \mathbb{F}_p$ and $\mathbb{K} = \mathbb{Q}$: a field of characteristic 0 or p is a flat — indeed free — extension of its prime field, and all the dimensions below are computed slotwise, so they do not change under field extension.

Step 1: a $\mathbb{Z}$-form, free in every graded slot. The sign convention $\varsigma(i,j) = \operatorname{sign}(j-i)$ of §2 has coefficients ± 1, so Π is defined by a presentation over $\mathbb{Z}$: writing $\Pi_{\mathbb{Z}}$ for the quotient of $\mathbb{Z}\overline{Q}$ by those relations, we have $\Pi_{\mathbb{Z}} \otimes_{\mathbb{Z}} \mathbb{K} = \Pi_{\mathbb{K}}$ for every field $\mathbb{K}$, both sides being the same presented quotient and $- \otimes \mathbb{K}$ being right exact. Put $\Lambda := \Pi_{\mathbb{Z}} e_k$, so that $\Lambda \otimes \mathbb{K} = \Pi_{\mathbb{K}} e_k = I_\kappa(\mathbb{K})$, compatibly with the path-length grading, which is defined over $\mathbb{Z}$.

Λ *is a finitely generated $\mathbb{Z}$-module.* This needs an argument, because $\mathbb{Z}\overline{Q}$ is not. The path-length grading is defined over $\mathbb{Z}$, so $\Lambda = \bigoplus_{d \ge 0} \Lambda[d]$, and $\Lambda[d]$ is spanned over $\mathbb{Z}$ by the images of the paths of length d starting at k — a finite set — so each $\Lambda[d]$ is finitely generated. It remains to see that $\Lambda[d] = 0$ for $d > 2t$. Now $\Lambda[d] \otimes \mathbb{K} = (I_\kappa(\mathbb{K}))[d]$, which vanishes for $d > 2t$ over every field $\mathbb{K}$, by Lemma 4.2 (proved characteristic-free). A finitely generated abelian group A with $A \otimes \mathbb{Q} = 0$ is finite, and a nonzero finite abelian group has $A \otimes \mathbb{F}_\ell \neq 0$ for some prime ℓ; so $\Lambda[d] = 0$ for $d > 2t$, and $\Lambda = \bigoplus_{d \le 2t} \Lambda[d]$ is finitely generated.

Λ *is free.* Write $\Lambda \cong \mathbb{Z}^N \oplus T$ with T finite. By Corollary 3.13 — which is proved over an arbitrary field, from Lemma 2.8, Lemma 2.7 and Proposition 2.4, and so in particular holds in characteristic 2 — we have $\dim_{\mathbb{K}} \Lambda \otimes \mathbb{K} = k(2n-1-k)$ for *every* $\mathbb{K}$. Taking $\mathbb{K} = \mathbb{Q}$ gives $N = k(2n-1-k)$; taking $\mathbb{K} = \mathbb{F}_\ell$, *for each prime ℓ in turn*, gives $N + \#\{\text{cyclic summands of } T \text{ of } \ell\text{-power order}\} = N$, so T has no ℓ-torsion, for every ℓ. Hence $T = 0$ and Λ is free. (Only the absence of p-torsion is needed for the fixed p at hand, and that is what the single prime p would give; we range over all primes because it costs nothing and makes the conclusion the clean one.) Each graded slot $\Lambda_i[d]$ is a direct summand of Λ as a $\mathbb{Z}$-module, hence projective, hence free; and its rank is $\dim_{\mathbb{K}}(I_\kappa(\mathbb{K}))_i[d]$ for every $\mathbb{K}$. (In particular this re-proves Lemma 4.2 in every characteristic from the characteristic-0 case, although Lemma 4.2 was in any case proved characteristic-free.)

Step 2: the witness in the generic fibre. Let μ be one of the representatives of Proposition 7.7 and $v = \lambda - \mu$. Since $\operatorname{char}\mathbb{Q} = 0 \neq 2$, Theorem 9.4 applies over $\mathbb{Q}$ and produces a graded submodule $V_{\mathbb{Q}} \subseteq \Lambda_{\mathbb{Q}} := \Lambda \otimes \mathbb{Q}$ with $\underline{\dim}\, V_{\mathbb{Q}} = v$ whose graded profile is the greedy top truncation at every vertex. (It is $T(v)$ or $T^l(v)$, built explicitly inside $I_\kappa(\mathbb{Q})$; no descent from an algebraic closure is needed, and by Corollary 2.13 it is the unique submodule of its dimension vector.)

Step 3: saturate over R. Put $\Lambda_R := \Lambda \otimes R$ and

$$N(v) \;:=\; V_{\mathbb{Q}} \cap \Lambda_R \;\subseteq\; \Lambda_{\mathbb{Q}}.$$

This is a Π_R-submodule of Λ_R, since Π_R preserves $V_{\mathbb{Q}}$ and Λ_R separately. It is graded, because both $V_{\mathbb{Q}}$ and Λ_R are: Λ_R carries the path-length grading, which is defined over $\mathbb{Z}$, and $V_{\mathbb{Q}}$ is graded *by construction* — in Step 2 it is $T(v)$ or $T^l(v)$, which Definition 9.1 defines slotwise as a graded subspace of $I_\kappa(\mathbb{Q})$, and in the run of

Step 5 below for an arbitrary μ it is $\mathrm{soc}_\sigma(I_\kappa(\mathbb{Q}))$, which is graded because a socle step is degree-homogeneous (see the proof of Theorem 1.2). No appeal to Proposition 2.16 — a statement over $\mathbb{C}$ — is made or needed. So $N(v) = \bigoplus_{i,d} N(v)_i[d]$ with $N(v)_i[d] = V_{\mathbb{Q},i}[d] \cap \Lambda_{R,i}[d]$.

Each $N(v)_i[d]$ is *saturated* in the free R-module $\Lambda_{R,i}[d]$: if $x \in \Lambda_{R,i}[d]$ and $px \in N(v)_i[d]$ then $x \in V_{\mathbb{Q},i}[d]$, that being a $\mathbb{Q}$-subspace, so $x \in N(v)_i[d]$. Over a principal ideal domain a saturated submodule of a finitely generated free module is a direct summand. Hence $N(v)_i[d]$ is a free direct summand of $\Lambda_{R,i}[d]$ of rank $\dim_{\mathbb{Q}} V_{\mathbb{Q},i}[d]$.

Step 4: reduce. Being a direct summand in each slot, $N(v)$ stays a direct summand after $-\otimes_R \mathbb{F}_p$; in particular

$$N(v) \otimes_R \mathbb{F}_p \;\longrightarrow\; \Lambda_R \otimes_R \mathbb{F}_p = I_\kappa(\mathbb{F}_p)$$

is injective, its image is a graded $\Pi_{\mathbb{F}_p}$-submodule, and

$$\dim_{\mathbb{F}_p} (\text{image})_i[d] = \dim_{\mathbb{Q}} V_{\mathbb{Q},i}[d] \qquad \text{for all } i, d. \tag{25}$$

In particular the image has dimension vector v and its graded profile is the greedy top truncation at every vertex. This is exactly the assertion (B) over $\mathbb{F}_p$, in the form (14); by Corollary 2.13, which holds over every field, the image is the unique submodule of $I_\kappa(\mathbb{F}_p)$ of dimension vector v, i.e. it *is* $V_{\mathbb{F}_p}(\mu)$.

Step 5: conclude. With (B) available over $\mathbb{F}_p$, Corollary 7.4 gives $(*^\circ)$ over $\mathbb{F}_p$, and the induction of §10 gives Proposition 4.11 over $\mathbb{F}_p$. We list exactly what that induction consumes — the list is the transitive closure of the citations in §10, not only its immediate ones — since the point of this theorem is that none of it carries a characteristic hypothesis.

- $(*^\circ)$, and with it from §7: Definition 7.1, Lemma 7.2, Corollary 7.4, Proposition 7.7, Definition 7.9 and (14). These are characteristic-free, as the head of §7 records, which is why quoting them at $p = 2$ is legitimate. (That is everything the induction consumes from §7. The one further result of that section, Proposition 7.5, is a consequence of $(*^\circ)$ rather than an input to it, and is not consumed here.)
- From §4: Lemma 4.2, Lemma 4.4, and Lemma 4.9, the last supplying the node column of a top truncation in the parameters v_t' and $D_t(X)$ and replacing Lemma 8.9 — a statement of §8, made for the nine representatives and for char $\neq 2$ — at every point of use. Also Definition 4.8 (the threshold $D_j(X)$), Theorem 4.6 and Lemma 4.5.
- Through the proof of Lemma 4.2: the display (6), which the proof of Lemma 10.3 quotes directly, and with it Proposition 3.12(a), Lemma 3.10, Lemma 3.9 and Lemma 3.11. All are proved from Lemma 2.8, Lemma 2.7 and Proposition 2.4 by integer combinatorics, over an arbitrary field.
- Remark 3.3, which is what Case 0 of the induction in the proof of Proposition 4.11 rests on: a colour-t socle step is idempotent. It is a consequence of Lemma 3.1 alone.
- From §§2–3: Convention 4.1, the sign convention and the relation (2) of §2, Definition 3.4, Lemma 2.7, Lemma 2.8, Lemma 2.3 and Proposition 2.4, Lemma 3.1, Corollary 3.2, Corollary 2.13, Lemma 3.5, Theorem 3.6 with its equation (4), and Corollary 3.8.
- Proposition 2.6 — equivalently Corollary 3.13, which is where it is proved — and behind it Proposition 2.1(b). These enter at four places: through Lemma 4.2 and Proposition 3.12; through Theorem 3.6; through Corollary 2.13 and Lemma 3.1 (whose proofs use the injectivity of I_κ and $\mathrm{soc}\, I_\kappa = S_\kappa$ — both definitional, and not part of the citation); and through Lemma 10.8 and Lemma 2.7. Consequently the only input of the present theorem that is quoted rather than proved is (b), a citation to [1] consumed only through the three items (b1)–(b3) itemised in Remark 2.2.

Nothing from §5, §6, §8 or §9 enters — those four sections reach $p = 2$ only through their consequence (B), supplied by Step 4 — and neither does Proposition 2.9 or Proposition 2.16. §7, which lies between them, *does* enter, and carries no characteristic hypothesis.

Theorem 1.1 follows from Proposition 4.11 exactly as in §4: the greedy top truncation of $(1, 2^{k-1}, 1)$ to total v_t has all multiplicities ≤ 1 iff $v_t \leq 2$, and $v_t = r + 2m$ by Lemma 4.4. Finally, running Steps 2–4 for an arbitrary extremal μ rather than a representative gives (25) for every $\mu \in W\lambda$, which is the last assertion. For such a μ, Step 2 is supplied not by Theorem 9.4 but by Theorem 3.6: over $\mathbb{Q}$ it constructs the submodule $V_{\mathbb{Q}}(\mu) = \mathrm{soc}_\sigma(I_\kappa(\mathbb{Q}))$ of dimension vector $\lambda - \mu$, graded, with no hypothesis on the field; Corollary 2.13 then says only that it is the *unique* such submodule, which is all that is asked of it. (The existence half of Corollary 2.13 is the clause "over $\mathbb{C}$ there is exactly one", which is not available over $\mathbb{Q}$ without the geometry, and is not used here — exactly as the paragraph following Corollary 2.13 records.) □

11.1. The $\mathbb{Z}$-form of a chain slot. Step 1 of the proof just given produced the $\mathbb{Z}$-form $\Lambda = \Pi_{\mathbb{Z}} e_k$ and showed that every graded slot $\Lambda_i[d]$ is a free $\mathbb{Z}$-module whose rank is $\dim_{\mathbb{K}}(I_\kappa(\mathbb{K}))_i[d]$ for *every* field $\mathbb{K}$. We now determine $\Lambda_i[d]$ at a chain vertex as a module over $\mathbb{Z}[\tau] = \mathbb{Z}[C_2]$. Nothing in the rest of the paper depends on this subsection; it is here because the answer is exactly what governs which shadows of §5 survive at $p = 2$, and because it is not visible in the rational data.

The point at issue is the following. A rank-2 $\mathbb{Z}[C_2]$-lattice whose rational character is $\varepsilon_+ \oplus \varepsilon_-$ is either the regular representation $\mathbb{Z}[C_2]$, which has a $\mathbb{Z}$-basis $\{e, f\}$ with $\tau e = f$, or the split lattice $\mathbb{Z}_+ \oplus \mathbb{Z}_-$; and these are not distinguished by the character. They are distinguished mod 2 — τ is a single Jordan block in the first case and the identity in the second — and equivalently by the index of the sublattice spanned by the two primitive rational eigenvectors, which is 2 for $\mathbb{Z}[C_2]$ and 1 for $\mathbb{Z}_+ \oplus \mathbb{Z}_-$. Only in the regular case do the two rational eigenlines have the *same* reduction mod 2.

Two integral facts drive everything, and neither survives passage to $\mathbb{Q}$.

Remark 11.2 (τ carries no sign twist)**.** With the convention $\varsigma(i, j) = \operatorname{sign}(j - i)$ of §2 we have $\varsigma(t, n-1) = \varsigma(t, n) = +1$ and $\varsigma(n-1, t) = \varsigma(n, t) = -1$, while ς is unchanged on chain arrows; so $\varsigma(\tau i, \tau j) = \varsigma(i, j)$ for every arrow. Hence τ carries the relation at i to the relation at τi on the nose, and is an automorphism of $\Pi_{\mathbb{Z}}$ over $\mathbb{Z}$ fixing e_κ. In particular τ *permutes paths*, and $\tau a_{j\to i} = a_{\tau j \to \tau i}\tau$ on Λ with no intervening sign.

Lemma 11.3 (generation)**.** *For every vertex i and every $d \geq 1$, $\Lambda_i[d] = \sum_{j\sim i} a_{j\to i}\Lambda_j[d-1]$.*

Proof. $\Lambda = \Pi_{\mathbb{Z}} e_\kappa$ is generated by e_κ and graded by path length, so $\Lambda_i[d]$ is spanned by the images of the paths of length d from κ to i, and every such path is an arrow $a_{j\to i}$ applied to a path of length $d-1$. □

Lemma 11.4 (the arrows toward the node are integrally surjective)**.** (i) *Let $1 \leq j \leq t-1$ and $d \geq 1$ with $d \neq j - k$. Then*

$$a_{j+1\to j}\big(\Lambda_{j+1}[d-1]\big) = \Lambda_j[d].$$

(ii) *Let $d \geq 1$ with $d \neq t - k$. Then*

$$\Lambda_t[d] = a_{n-1\to t}\big(\Lambda_{n-1}[d-1]\big) + a_{n\to t}\big(\Lambda_n[d-1]\big),$$

that is, the node slot is generated by the two leaves alone.

Proof. For $1 \leq j \leq t-1$ and $d \in \mathbb{Z}$ write $Q(j, d)$ for the assertion of (i), and for $d \in \mathbb{Z}$ write $N(d)$ for the assertion of (ii). Only the preprojective relations and Lemma 11.3 are used; no exactness and no hypothesis on a field.

Trivial cases. $Q(j, d)$ holds whenever $\Lambda_j[d] = 0$: in particular for $d < 0$, and for $d = 0$ with $j \neq k$, since $\Lambda_j[0] = \mathbb{Z}e_\kappa$ or 0 according as $j = k$ or not.

Base. $Q(1, d)$ holds for every $d \geq 1$: the vertex 1 has 2 as its only neighbour, so Lemma 11.3 reads $\Lambda_1[d] = a_{2\to 1}\Lambda_2[d-1]$.

Step. Let $2 \leq j \leq t-1$ and $d \geq 1$. Then $Q(j-1, d-1)$ implies $Q(j, d)$. Indeed by Lemma 11.3 $\Lambda_j[d] = a_{j-1\to j}\Lambda_{j-1}[d-1] + a_{j+1\to j}\Lambda_{j+1}[d-1]$, so it suffices to see that the first summand lies in the second. Let $w \in \Lambda_{j-1}[d-1]$. By $Q(j-1, d-1)$ we may write $w = a_{j\to j-1}v$ with $v \in \Lambda_j[d-2]$. The preprojective relation at the chain vertex j, applied to v, reads

$$\varsigma(j, j-1)\, a_{j-1\to j}a_{j\to j-1}v + \varsigma(j, j+1)\, a_{j+1\to j}a_{j\to j+1}v = 0,$$

whence $a_{j-1\to j}w = -\varsigma(j, j-1)\varsigma(j, j+1)\, a_{j+1\to j}\big(a_{j\to j+1}v\big) \in a_{j+1\to j}\Lambda_{j+1}[d-1]$.

Step at the node. Likewise $Q(t-1, d-1)$ implies $N(d)$ for $d \geq 1$: by Lemma 11.3, $\Lambda_t[d]$ is the sum of the images of $a_{t-1\to t}$, $a_{n-1\to t}$ and $a_{n\to t}$; writing $w \in \Lambda_{t-1}[d-1]$ as $a_{t\to t-1}v$ and applying the relation at t,

$$-a_{t-1\to t}a_{t\to t-1}v + a_{n-1\to t}a_{t\to n-1}v + a_{n\to t}a_{t\to n}v = 0,$$

puts $a_{t-1\to t}w$ in the sum of the two leaf images.

Descent. Iterating the Step from (j, d) gives $Q(j-m, d-m) \Rightarrow Q(j, d)$ for every $m \leq \min(j-1, d)$ — the m-th application of the Step is at degree $d - m + 1$, which must be ≥ 1. If $d \geq j$ we reach $(1, d-j+1)$ with $d - j + 1 \geq 1$ and invoke the Base. If $d < j$ we reach $(j-d, 0)$, and $\Lambda_{j-d}[0] = 0$ unless $j - d = k$; so $Q(j, d)$ holds unless $d = j - k$. This is (i). For (ii), $N(d)$ follows from $Q(t-1, d-1)$, which by the same count holds unless $d - 1 = (t-1) - k$, i.e. unless $d = t - k$. □

The excluded diagonal is harmless: $d = j - k$ is the bottom degree $\mathrm{lo}_j^+ = |j - k|$ of the $+$ strand when $j \geq k$, a slot that is fed from $j - 1$ and not from $j + 1$, and it never carries multiplicity 2.

Proposition 11.5 (the $\mathbb{Z}$-form at a chain slot)**.** *Let $1 \le j \le t$ and let d be a degree.*
(1) If $\operatorname{rk}\Lambda_j[d] = 2$ then $\Lambda_j[d] \cong \mathbb{Z}[C_2]$ as a $\mathbb{Z}[\tau]$-module: it has a $\mathbb{Z}$-basis $\{x, \tau x\}$.
(2) If $\operatorname{rk}\Lambda_j[d] = 1$ then $\Lambda_j[d] \cong \mathbb{Z}_\varepsilon$, where ε is the unique strand carried by the slot.
Consequently τ reduces mod 2 to a single Jordan block at every multiplicity-2 chain slot and to the identity at every 1-dimensional one; at a multiplicity-2 slot the two rational eigenlines $\langle x + \tau x\rangle$ and $\langle x - \tau x\rangle$ have the same reduction mod 2, namely $\ker(\tau - 1) = \operatorname{im}(\tau - 1)$, and the sublattice they span has index 2.

Proof. (2) is immediate: an involution of $\mathbb{Z}$ is ± 1, and the sign is the eigenvalue of τ on the slot over $\mathbb{Q}$, which is ε by Proposition 5.1.

For (1), recall from Proposition 5.1 that the multiplicity-2 degrees at the chain vertex j are exactly

$$d \in W_j = [\,2t + 2 - j - k,\ j + k - 2\,], \qquad d \equiv j + k\ (2), \tag{26}$$

and that by Step 1 of the proof of Theorem 11.1 these are also the d with $\operatorname{rk}\Lambda_j[d] = 2$.

(a) The window moves inward. If $d \in W_j$ and $j \le t-1$ then $d-1 \in W_{j+1}$: the lower bound $2t+2-(j+1)-k \le d-1$ is $d \ge 2t+2-j-k$, and the upper bound $d-1 \le (j+1)+k-2$ is $d \le j+k$, both implied by (26); the parity is right because $d-1 \equiv (j+1)+k$. Iterating, $(t,\, d^\circ)$ with $d^\circ := d-(t-j)$ is a multiplicity-2 node slot.

(b) The chain arrows are surjective along the way. For $j \le j' \le t-1$ put $d' = d-(j'-j)$. Lemma 11.4(i) applies at (j', d'): its degree hypothesis holds because $d' \ge d^\circ \ge t+2-k \ge 2$, and its exclusion $d' = j'-k$ would read $2j' = d+j+k \ge 2t+2$ by (26), forcing $j' \ge t+1$. Hence

$$\Lambda_j[d] = \big(a_{j+1\to j} \circ \cdots \circ a_{t\to t-1}\big)\big(\Lambda_t[d^\circ]\big),$$

a surjection of $\mathbb{Z}[\tau]$-modules, τ fixing every chain vertex.

(c) The node slot is generated by one τ-orbit. Since $d^\circ \in W_t = [\,t+2-k,\ t+k-2\,]$ we have $d^\circ \ne t-k$, so Lemma 11.4(ii) gives $\Lambda_t[d^\circ] = a_{n-1\to t}\Lambda_{n-1}[d^\circ - 1] + a_{n\to t}\Lambda_n[d^\circ - 1]$. By Lemma 4.2 the two leaves are multiplicity-free with support $t-k+1, t-k+3, \dots, t+k-1$, and $d^\circ - 1$ lies in that range with the right parity; so $\Lambda_{n-1}[d^\circ-1] = \mathbb{Z}u$ and $\Lambda_n[d^\circ-1] = \mathbb{Z}\tau u$ are of rank 1, and $J := \Lambda_{n-1}[d^\circ-1] \oplus \Lambda_n[d^\circ-1]$ has the $\mathbb{Z}$-basis $\{u, \tau u\}$ permuted by τ, i.e. $J \cong \mathbb{Z}[C_2]$. By Remark 11.2 the map $\varphi := a_{n-1\to t} + a_{n\to t} : J \to \Lambda_t[d^\circ]$ is τ-equivariant, and it is onto. So $\Lambda_t[d^\circ]$ is a quotient of $\mathbb{Z}[C_2]$; both have rank 2, so the kernel of φ has rank 0 and, sitting inside the torsion-free J, is zero. Hence $\Lambda_t[d^\circ] \cong \mathbb{Z}[C_2]$, with basis $\{\varphi(u), \tau\varphi(u)\}$.

(d) Descent along the chain. By (b), $\Lambda_j[d]$ is a τ-equivariant quotient of $\Lambda_t[d^\circ] \cong \mathbb{Z}[C_2]$ of rank 2; by the same argument the quotient map is an isomorphism. This proves (1).

The final sentence is the standard dictionary recalled above: in the basis $\{x, \tau x\}$ the matrix of τ is $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, which mod 2 is a single Jordan block with $\ker(\tau-1) = \operatorname{im}(\tau-1) = \langle \overline{x+\tau x}\rangle$; and $\det\left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right) = -2$. □

Remark 11.6. The proof uses no exactness and no field: only that Λ is generated by e_κ in degree 0, that the preprojective relations have coefficients ± 1 with $\varsigma(\tau i, \tau j) = \varsigma(i,j)$, that the slot ranks are those of Proposition 5.1 and Lemma 4.2, and that the slots are $\mathbb{Z}$-free (Step 1 of Theorem 11.1). The two places where integrality — as opposed to the rational character — enters are Remark 11.2 and the surjectivity of Lemma 11.4, which is a statement about $\mathbb{Z}$-modules and not merely about ranks. Proposition 11.5 was also verified directly, by an independent integral construction of Λ with Smith normal forms in place of row reduction; see §14.2.

Remark 11.7 (what the argument does and does not give)**.** It gives the theorem at $p = 2$; it gives no new information there. Being a specialisation argument, it shows only that the $p = 2$ answer is the shadow of the characteristic-0 one. It does *not* make (W1)–(W4), (T0)–(T2) or the canonical truncation $T(v)$ true at $p = 2$ intrinsically.

Some of them do survive as shadows, and we record the mechanism. By Proposition 11.5 the $\mathbb{Z}$-form of a multiplicity-2 chain slot is the regular representation $\mathbb{Z}[C_2] = \langle e, f\rangle$ with τ swapping e and f; hence the two rational eigenlines $\langle e+f\rangle$ and $\langle e-f\rangle$ have the *same* reduction mod 2, namely $\ker(\tau-1) = \operatorname{im}(\tau-1)$. So $T(v)$ remains definable at $p = 2$ with "τ-eigenline" read as $\ker(\tau-1)$, and (T0)–(T2) then hold verbatim.

This is not formal. A rank-2 $\mathbb{Z}[C_2]$-lattice whose rational character is $\varepsilon_+ \oplus \varepsilon_-$ is either the regular representation $\mathbb{Z}[C_2]$, whose reduction mod 2 is a single Jordan block, or the split lattice $\mathbb{Z}_+ \oplus \mathbb{Z}_-$, whose reduction is the identity, and the rational data do not distinguish them; in the split case the two eigenlines would have *different* reductions and the previous paragraph would fail. Which one occurs is decided in §11.1, by an integral argument — the surjectivity of Lemma 11.4 and the absence of a sign twist in τ (Remark 11.2) — and not by any rational input. Nothing in this paper depends on the outcome: Theorem 11.1 goes through the $\mathbb{Z}$-form slotwise and never asks what τ does; and, as the next remark shows, even with Proposition 11.5 in hand one cannot rebuild §5.

Remark 11.8 (the filtration route does not work)**.** It is natural to try to replace the eigen-splitting at $p = 2$ by the filtration $0 \le \ker(\tau - 1) \le V$, which is available because $(\tau - 1)^2 = 2(1 - \tau) = 0$ there, and to read the associated graded as a substitute for the two strands. This fails, and it fails at the level of (W2), so it is worth recording so that no one repeats the attempt.

The obstruction sits at the 1-dimensional slots, and it does not depend on any unproved input. At a 1-dimensional slot the only unipotent operator on a line is the identity, so the filtration grading is $(1, 0)$ *always*. (At a multiplicity-2 slot the filtration grading is $(1, 1)$, by Proposition 11.5; the argument below does not use this.) Over $\mathbb{Q}$ a 1-dimensional chain slot carries one strand only, with eigen-dimensions $(1, 0)$ or $(0, 1)$ according to which. The filtration grading is therefore wrong at every "$-$ only" slot, whatever it does at the multiplicity-2 slots. These are not rare: over D_4–D_{14} and all k there are 3575 nonzero chain slots, of which 1001 have multiplicity 2, 1287 are "$+$ only" and 1287 are "$-$ only" — and the filtration misreads all 1287 of the last kind.

Consequently Proposition 5.1 is false for the filtration grading in either assignment of sub- and quotient to $+$ and $-$. Writing $f^0_d(j)$ and $f^1_d(j)$ for the two graded pieces, f^0 is supported on $S^+(j) \cup S^-(j)$ and f^1 on the window $W_j = S^+(j) \cap S^-(j)$; both are symmetric under $d \mapsto 2t - d$, since $S^+(j) \cup S^-(j)$ and W_j are, whereas a^+ is not, so the reversal symmetry $a^-_d(j) = a^+_{2t-d}(j)$ of (W2) fails. The casualty downstream is (T2), whose proof works by showing that a 1-dimensional target carries $-$ and not $+$: over $\mathbb{F}_2$ the filtration cannot see that label at all, since τ acts trivially on every line, and (T2) is not even statable in filtration language. This is a loss of information, not a gap in an argument, and it is why Theorem 11.1 goes through the $\mathbb{Z}$-form instead.

12. A necessary condition at the trivalent node

Theorem 1.1 is a condition on μ. It has a consequence that is a condition on the *reduced word*, and a sharp one: simple spectrum forbids the c-trace from ever exceeding 1 at the trivalent node. Since λ-minuscule means that no c-trace value anywhere exceeds 1 (Definition 3.4 and [5, Def. 5.1]), this isolates the one colour at which λ-minuscularity may not be relaxed. Nothing later depends on this section.

We first record the bookkeeping identity behind it.

Lemma 12.1. *Let w be of minimal length in wW_λ, with reduced word $s_{i_1} \cdots s_{i_\ell}$ and c-trace (c_p). Then*

$$v_i \;=\; \sum_{p\,:\,i_p = i} c_p \qquad (i \in I). \tag{27}$$

Proof. In the proof of Theorem 3.6, $\mu[p] = \mu[p-1] - c(p)\alpha_{i(p)}$ with $\mu[0] = \lambda$ and $\mu[\ell] = w\lambda$. Telescoping gives $\lambda - w\lambda = \sum_p c(p)\alpha_{i(p)}$, and comparing coefficients of α_i against $\lambda - w\lambda = \sum_i v_i\alpha_i$ gives (27). □

The point of the next lemma is that deleting the trivalent node from D_n leaves no room to build a heavy step at it.

Lemma 12.2 (a lone node step is light)**.** *Let $\lambda = \omega_k$ with $1 \le k \le n-2$, let w be of minimal length in wW_λ, and let $s_{i_1} \cdots s_{i_\ell}$ be a reduced word for w in which the trivalent node $t = n-2$ occurs exactly once, say $i_p = t$. Then $c_p = 1$.*

Proof. By Definition 3.4, $c_p = \langle \alpha_t^\vee, u\lambda \rangle$ with $u = s_{i_{p+1}} \cdots s_{i_\ell}$. By hypothesis none of $i_{p+1}, \dots, i_\ell$ is t, so $u \in W_{I\setminus\{t\}}$. Deleting t from D_n leaves the chain $\{1, \dots, n-3\}$ and the two isolated leaves $n-1$ and n, so

$$W_{I\setminus\{t\}} = \langle s_1, \dots, s_{n-3} \rangle \times \langle s_{n-1} \rangle \times \langle s_n \rangle,$$

and $\langle s_1, \dots, s_{n-3} \rangle$, generated by the reflections in $e_j - e_{j+1}$ for $j \le n-3$, is the symmetric group permuting $e_1, \dots, e_{n-2}$. Now $\alpha_{n-1} = e_{n-1} - e_n$ and $\alpha_n = e_{n-1} + e_n$ are supported on the coordinates $n-1, n$, where $\lambda = e_1 + \cdots + e_k$ vanishes because $k \le n-2$; hence s_{n-1} and s_n fix λ, and $u\lambda = a\lambda$ for the permutation part a of u. Therefore $a\lambda = \sum_{j \in A} e_j$ for some k-element $A \subseteq \{1, \dots, n-2\}$, and

$$c_p = \Big\langle e_{n-2} - e_{n-1},\; \sum_{j \in A} e_j \Big\rangle = [\, n-2 \in A \,] \;\le\; 1.$$

Since $c_p \ge 1$ by Lemma 3.5(i), $c_p = 1$. □

Corollary 12.3 (no heavy step at the node)**.** *Let $\lambda = \omega_\kappa$ be* any *fundamental weight of D_n and let w be of minimal length in wW_λ. If some c-trace value at the trivalent node is ≥ 2, then T does not act on V with simple spectrum. Equivalently, simple spectrum forces $c_p = 1$ at* every *occurrence of t.*

Proof. For the two spin weights $\kappa \in \{n-1, n\}$ there is nothing to prove: by Theorem 1.2 every such w is λ-minuscule, so no c-trace value anywhere is ≥ 2 and the hypothesis is vacuous. So let $\kappa = k \leq n-2$ and suppose $c_p \geq 2$ with $i_p = t$. By Lemma 12.2 the colour t occurs at least twice in the word, so (27) together with $c_q \geq 1$ for the other occurrences gives $v_t \geq c_p + 1 \geq 3$. For $k \geq 2$ Theorem 1.1 now denies simple spectrum; and $k = 1$ cannot occur at all, since there $|S| = 1$ forces $(m, r) \in \{(0,0), (0,1), (1,0)\}$ and hence $v_t = r + 2m \leq 2$ by Lemma 4.4, contradicting $v_t \geq 3$. □

Remark 12.4. The converse fails, and must. By (27), $c_p = 1$ at every occurrence of t says exactly that $v_t = \#\{p : i_p = t\}$, and that number is unbounded; the shortest witness is D_4, $\lambda = \omega_2$, $w = s_2 s_4 s_2 s_3 s_1 s_2$, with $\mu = -e_2 - e_4$ and $v = (1, 3, 1, 2)$, where $t = 2$ occurs three times with $c_p = 1$ each, so $v_t = 3$ and the spectrum is not simple — consistently with Theorem 1.1, since $m = r = 1$. Corollary 12.3 is thus strictly weaker than Theorem 1.1, as it must be: it bounds how heavy a node step may be, while v_t also counts how many there are. Example 1.3 is not a witness but an instance: its c-trace is $(1, 1, 1, 1, 2)$, heavy at t.

Remark 12.5 (outside type D)**.** Corollary 12.3 is proved only for D_n, since it quotes Theorem 1.1. Its proof splits, however, into a part local at t (Lemma 12.2, which uses only that deleting t disconnects the diagram and that λ vanishes off the component carrying κ) and the appeal to Theorem 1.1, which is not. We record, as a computation and not as a claim of this paper, that its conclusion also holds in E_6: of the 1272 extremal weights of the six fundamental λ, the 831 whose c-trace exceeds 1 at the trivalent node all fail to have simple spectrum. We do not know a proof, and nothing here uses it.

13. A word-free complement: the staircase property

Proposition 4.11 says that the *extremal* module V is a top truncation at t. There is a stronger, word-free statement, which we prove here because it explains why extremality is the right hypothesis and because it is the form in which the result is easiest to check. Nothing in §§2–11 depends on this section.

Definition 13.1. A graded submodule $M \subseteq I_\kappa$ is a *staircase* if it is a top truncation at t and an up-set at each of the two leaves.

Theorem 13.2 (staircase property)**.** *Over an arbitrary field: if* $\underline{\dim}\, M$ *is* w*-extremal then* M *is a staircase. Equivalently, a graded submodule that is not a staircase satisfies* $\delta(\underline{\dim}\, M) > 0$ *and is therefore never the unique submodule of its dimension vector.*

Proof of Theorem 13.2. Let $v = \underline{\dim}\, M$ be extremal, say $v = \lambda - w\lambda$. By Corollary 2.13 the submodule of I_κ with dimension vector v is unique, so $M = V$; and by Proposition 4.11 together with Lemma 10.4 (applied along the filtration of Theorem 3.6, which ends at V and preserves (H)), V is a top truncation at t and an up-set at each leaf. The equivalent form follows from Corollary 2.13. Both inputs are available over every field — Corollary 2.13 directly and Proposition 4.11 by Theorem 11.1 — so no hypothesis on the characteristic is needed here. □

14. What is verified, what is proved, and what fails

14.1. Statements asserted on computational evidence. Every statement in §§2–13 is proved here or attributed to a cited reference. *No* statement of this paper, and in particular no step in the proof of Theorem 1.1, rests on computational evidence; the counts of §14.2 are checks on proved statements, and nothing depends on them. Earlier versions of this paper asserted three statements on computational evidence — the conclusion at char = 2, the Jordan type of τ at a multiplicity-2 chain slot, and the sharpness of the side condition in $(*^\circ)$. They are now Theorem 11.1, Proposition 11.5 and Proposition 7.5 respectively, and their computations are retained in §14.2 as checks.

Exactly one assertion in this paper does rest on a computation, and is flagged where it appears: the final remark of §12, that the conclusion of Corollary 12.3 also holds in E_6. It concerns a type this paper does not treat, and nothing here depends on it.

14.2. Ranges of the independent checks. The statements below are all proved above. Each was additionally confirmed by computation, as a check and not as a step: no statement of §§2–13 rests on any entry of this table, and no proof here cites this section. "closed form" means the combinatorial statement was evaluated symbolically in n and k rather than on modules. Each range is the one the corresponding script actually covers, and every entry was checked without exception.

statement	range	instances
Theorem 1.1, against a direct solver	full orbits D_4–D_{11}, $p = 2, 3, 5$; D_4–D_{10} over $\mathbb{Q}$ and $p = 7$; the bands $m \leq 2$ for D_{12}, D_{13} and $m \leq 1$ for D_{14}	241008 per p; 77196; 156663, 370221, 257339
Theorem 11.1, slotwise over $\mathbb{Q}$ against $\mathbb{F}_2$	full orbits D_4–D_{10}, all k	77196 weights
Proposition 7.5, the up-set property at $t-1$	full orbits D_4–D_{10}, all k	77196 modules, 2778 non-up-sets, all with $c_t = 0$
Proposition 11.5, the $\mathbb{Z}$-form	D_4–D_{22}, all k, via Smith normal form	7315 multiplicity-2 chain slots
$\mathrm{soc}(\Pi e_j)$ one dimensional in the top degree $2t$, i.e. Proposition 2.1(b) as consumed here	D_4–D_{10}, every vertex, rebuilt from scratch over $\mathbb{Q}$ and $\mathbb{F}_{2,3,5,7}$	245 cases
the block induction of Theorem 3.6, $\delta_p = c(p)$	full orbits D_4–D_8, all k, $p \in \{0, 2, 3, 5\}$	519184 blocks
Lemma 2.3 and Proposition 2.4	module pairs over D_4–D_6, $p \in \{2, 3, 5\}$	1641 pairs
Lemma 4.9 and Remark 4.10	D_4–D_{40}, all k	19684 greedy truncations
Propositions 7.7, 8.7, 8.8, 8.10, 9.7, 9.8, 9.9, Lemmas 8.1, 8.9 and the obligation census of Theorem 9.4	closed form, D_6–D_{30}, all k, all nine families, every h	every fatal obligation class empty
the census behind Proposition 9.10	closed form, D_4–D_{44}, all k, all families, every h	4.8×10^7 obligations
Proposition 6.1	closed form to D_{100}, all k	2.4×10^7 arrow/degree pairs
Proposition 3.12 and Corollary 3.14	closed form D_4–D_{80} resp. D_4–D_{150}; against I_κ over D_4–D_8	all k, all three spin κ
Theorem 13.2, Lemma 10.8, Corollary 10.6, against *every* graded submodule of I_κ	D_7, D_8, all k, over $\mathbb{F}_{2,3,5}$	counts are field-dependent

Two points about the table are worth stating rather than tabulating. First, Proposition 7.5 has content: on extremal modules c_t takes all five values $-2, \dots, 2$, so "$c_t \geq 0$ always" is false, and of the 77196 extremal modules over D_4–D_{10} the 26181 with $c_t \leq -1$ are all up-sets at $t-1$. Second, the check of Theorem 1.1 was produced by a from-scratch implementation sharing no code with the one used elsewhere in this project; the two agree on 7020 graded dimensions and 11880 arrow ranks over D_4–D_{12}, and the verdict is independent of the reduced word (checked on a second word for 1685 weights of D_{11}).

14.3. Negative results worth recording. These delimit the argument and explain why several shorter routes do not exist.

(N1) *Full commutativity is neither necessary nor sufficient.* In D_4, $w = s_2 s_4 s_3 s_1 s_2$ is fully commutative and does not have simple spectrum (Example 1.3); [5, Ex. 5.12] gives a w with simple spectrum that is not fully commutative.

(N2) *The greedy top truncation at every vertex is false for general extremal μ.* It holds at the representatives of Proposition 7.7 — that is (B), Theorem 9.4 — but not in general. The smallest failure is D_5, $\lambda = \omega_2$, $\mu = (1, -1, 0, 0, 0)$, where V_2 occupies degrees 2 and 6 while the top truncation would give 6 and 4; here $t - 1 = 2$ is the chain neighbour of the node. That element has $r + 2m = 2$ and so *does* have simple spectrum, so the failure is not confined to degenerate cases. It is precisely why the theorem is proved at t only, and why the reduction to a bounded list of weights is necessary.

(N3) *Extremality cannot be dropped from $(*^\circ)$.* The localised statement — for every graded $N \subseteq I_\kappa$, (H) together with $\langle \alpha_t^\vee, \lambda - \underline{\dim}\, N \rangle \geq 1$ implies N is an up-set at $t-1$ — holds for every graded submodule when $\kappa \leq 3$ and fails at $\kappa = 4$, where $t - 1$ first carries multiplicity 2 in two different degrees. The smallest counterexample

is D_6, $\kappa = 4$, $t = 4$, with graded profile ($\dim N / \dim I_\kappa$ by degree)

1 :	0/1	1/1			4 :	0/1	0/2	1/2	2/2	1/1
2 :	0/1	1/2	1/1		5 :	0/1	1/1	1/1	1/1	
3 :	0/1	1/2	1/2	1/1	6 :	0/1	0/1	1/1	1/1	

which satisfies (H), has $\langle \alpha_t^\vee, \lambda - \underline{\dim}\, N\rangle = 1$, and is partial at two degrees at $t-1$. Likewise Claim F (Corollary 10.6) is false for arbitrary graded submodules.

(N4) *The staircase property needs extremality.* Graded submodules of I_κ that are not staircases exist, and in the enumeration described in §14 every one of them has $\delta(\underline{\dim}\, M) > 0$, in accordance with Theorem 13.2. We quote no count here: the number of graded submodules depends on the ground field (a two-dimensional slot has $|\mathbb{F}_p| + 1$ lines in it), so a bare count is not a well-posed statistic.

(N5) *The larger parabolic does not work.* With $J = I \setminus \{t-1, t\}$, which contains the two leaves, the side condition $\langle \alpha_t^\vee, \mu\rangle \geq 1$ is not W_J-invariant, and the W_J-minimal representative of the orbit of a μ satisfying it usually does not satisfy it: over D_4–D_7, all $2 \leq k \leq n-2$, there are 764 weights $\mu \in W\omega_k$ with $c_t(\mu) \geq 1$, and 648 of them have W_J-minimal representative outside the hypothesis set; over D_4–D_8 the counts are 2565 and 2207. Since the W_J-minimal element of an orbit is unique, there is no alternative descent path, and $(*^\circ)$ is false without its side condition — the 64 respectively 244 extremal non-up-sets recorded in §14.2 are exactly the counterexamples. This is why Proposition 7.7 uses $J' = \{1, \ldots, t-2\}$ and produces $9(k-1)$ representatives rather than k.

(N6) *The canonical truncation is not always the right module.* In the four families with $q + x$ even and $z \neq 0$, the leaf pin forces the line at (E_t, t) to be $\mathrm{im}(a_{l\to t})$, which by Proposition 5.6 is *not* a τ-eigenvector. So $T(v)$ fails to be a submodule there and must be replaced by $T^l(v)$ (Theorem 9.4).

(N7) *Containment of multiplicity-2 windows does not prove Theorem 4.6*; see Remark 4.7. Nor is the naive mechanism for (G1) available: when $X_t[D]$ is a line it is always one of the two special lines, so one leaf arrow annihilates it and that leaf's fullness at $D+1$ must come from elsewhere in X.

References

[1] S. Brenner, M. C. R. Butler and A. D. King, *Periodic algebras which are almost Koszul*, Algebr. Represent. Theory **5** (2002), 331–368.

[2] P. Baumann and J. Kamnitzer, *Preprojective algebras and MV polytopes*, Represent. Theory **16** (2012), 152–188.

[3] P. Baumann, J. Kamnitzer and P. Tingley, *Affine Mirković–Vilonen polytopes*, Publ. Math. Inst. Hautes Études Sci. **120** (2014), 113–205.

[4] W. Crawley-Boevey, *On the exceptional fibres of Kleinian singularities*, Amer. J. Math. **122** (2000), 1027–1037.

[5] H. Dinkins, I. Karpov and V. Krylov, *The quantum Hikita conjecture via quasimaps*, arXiv:2608.16746.

[6] A. Dranowski, B. Elek, J. Kamnitzer and C. Morton-Ferguson, *Heaps, crystals, and preprojective algebra modules*, arXiv:2202.02490.

[7] C. Geiss, B. Leclerc and J. Schröer, *Kac–Moody groups and cluster algebras*, Adv. Math. **228** (2011), 329–433.

[8] H. Nakajima, *Quiver varieties and Kac–Moody algebras*, Duke Math. J. **91** (1998), 515–560.

[9] A. Savage and P. Tingley, *Quiver Grassmannians, quiver varieties and the preprojective algebra*, Pacific J. Math. **251** (2011), 393–429.

[10] J. Stembridge, *Minuscule elements of Weyl groups*, J. Algebra **235** (2001), 722–743.

Department of Mathematics, Purdue University, West Lafayette, IN 47907

Email address: jpadhi@purdue.edu